PAPERPORT III

PRINTER / SMART OPTIONS INTERFACE SPECIFICATION

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INTRODUCTION

1.1 PURPOSE

The purpose of this document is to describe the interface between a laser printer and its paper handling "Smart Options" that permits the printer to control and monitor the status of the "Smart Options".

Three specific aspects of the Printer / Smart Option interface will be addressed in this document: link communications, software communications, and the electrical interface.

Any one working on a paper handling device should verify with Lexmark International that support for the specific device and its characteristics are, or will be implemented in the Printer to which the paper handling device is to be attached.

1.2 OVERVIEW

The opening chapter of this document will present several philosophical ideas and/or guidelines that were used during the development of this interface.

Following this section will be several detailed chapters describing the specific interface (ie, communication commands, responses, and protocols; as well as, the electrical interface).

1.3 RELATED DOCUMENTS

General documents listed here as applicable.

Device-Specific documents provided as necessary

INTERFACE PHILOSOPHY

1.4 PRINTER / DEVICE RELATIONSHIP

The Printer and Smart Option(s) possess a ***Master / Slave*** relationship:

- The Printer is the Master, and the Smart Option (ie, device) is the Slave.

- There exists a communication path (ie, physical interface) between the master and the slave that permits the exchange of data (ie, commands and responses).
- The Master is required to initiate ***ALL*** communications by sending a command to the slave (ie, the slave doesn't talk until first spoken to by the master).
- The master will not issue a second command until the slave has responded to the previous command.
- Some of the commands and responses are multiple bytes in length. When the printer issues a multi-byte command, the slave must respond to each byte sent by the printer (master) by sending an *acknowledgment byte* before the printer will send the subsequent byte. Furthermore, when a command requires a multi-byte response from the slave, the printer must sent the *acknowledgment byte* to the slave indicating it has received the previous byte before the slave can send the next byte in the response. This is often referred to as a "one byte down / one byte back" link communications protocol.
- Since this interface does not permit the slave to initiate communications with the master, the printer (master) must poll the slave to determine asynchronous data from the slave. This is done through the use of a general purpose status command (Query Device Status).
- The communications interface is a multi-drop serial interface that supports one master and multiple slaves. Each slave has a unique address allowing it to know specifically when the master is communicating with them via the common communications bus.
- For uniquely defined situations, the slave device can *alert* the printer that a major breakdown in communications has occurred.

Each slave device resides on a "trunk line". In general, a single trunk line can support multiple slave devices. Several trunk lines are predefined, and are identified in the *Link 3.0* section. For example, the "Bottom Attached Devices" trunk line consists of the stackable devices that are located under the printer. Each Paperport slave device is designed to exist on a given trunk line, and that particular trunk line is known at the time the device is designed and developed. So, the firmware within each slave device knows which predefined trunk line that slave will always reside on. The devices on a given trunk line are attached in a daisy-chain arrangement. Each slave device has the capability to connect or disconnect, from the serial bus, the slave device that is next in the daisy chain (farther away from the engine). A device does this by either enabling or disabling the engine's transmit signal to the next device. Devices do not gate the receive line.
An illustration of a possible printer / smart options system is given below:

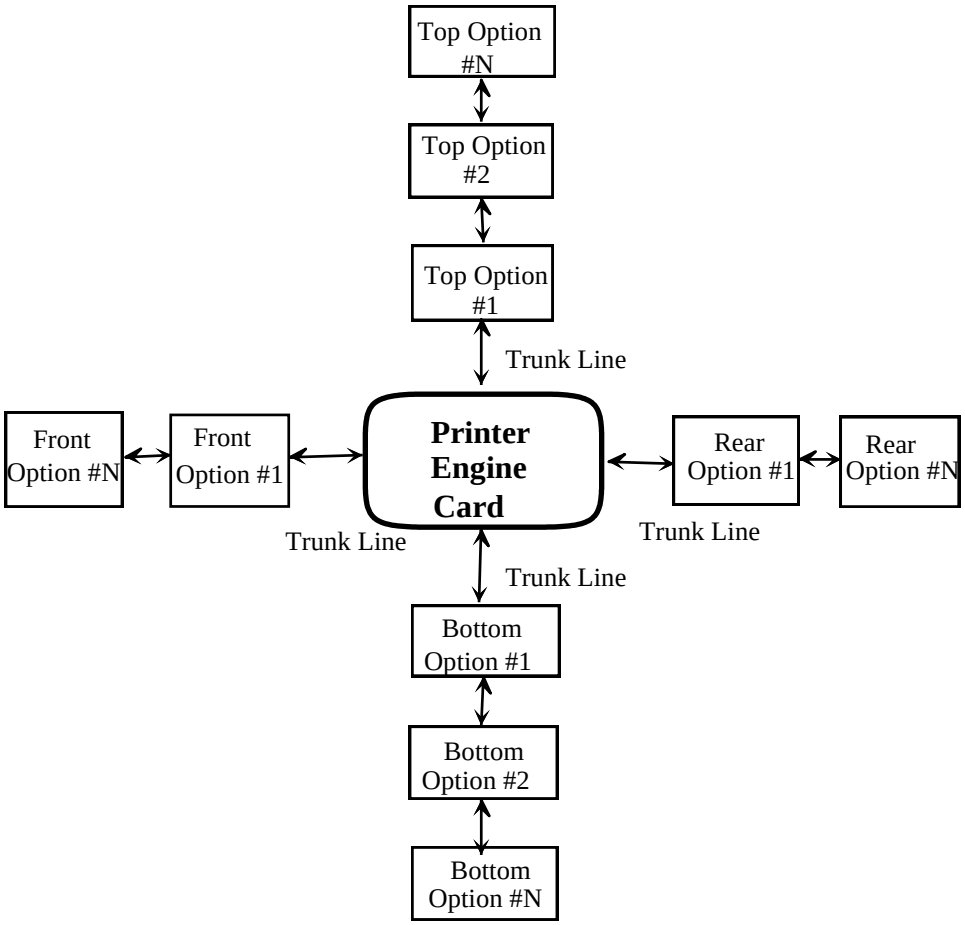


Figure 2.1 Master (Printer) / Slave (Devices) Illustration

1.5 FUTURE EXPANDABILITY

As indicated, the communications bus allows multiple slaves. This specification only addresses specific Smart Options (ie, devices or slaves). As new options are created, and subsequent printer systems released, there are ongoing compatibility issues as the paperport architecture evolves subsequent features. A compatibility level control has been established for specific commands to reflect the implementation capability of a device. All smart options released support level 0 by default. Options report a higher level in response to the QDSR (Section 1.15.4) query sequence from the printer. Subsequent printers which wish to use or require extended functions of the architecture may declare downlevel devices Unsupported, or adapt its operation to the restricted capability of the option. Refer to the QDSR response for the currently active compatibility levels, and the specific commands affected.

For improved robustness, if there are additional command bytes that are not applicable to a particular option, the engine should be able to send those bytes without the option failing. In that case, the option should simply ignore the additional information without flagging an error condition.

Where possible, abbreviated commands should also be accepted to help with bus bandwidth.

COMMUNICATIONS INTERFACE

1.6 LINK

This section describes the "data link" -- how individual bytes of data are transmitted between the master (printer engine) and the smart options (paper-handling devices).

1.6.1 Physical

The communications link is a serial interface with separate transmit (Txd) and receive (Rxd) signal lines. The engine and devices transmit information byte-by-byte via UARTs (Universal Asynchronous Receiver/Transmitters). Although this hardware configuration could support full duplex communications (data transmitted simultaneously on the Txd and Rxd lines), the Paperport communications protocol, described below, specifies that the Txd and Rxd lines are not simultaneously active. For a more detailed description of the physical hardware see the *Electrical Physical Interface* section. The engine and the ensemble of paper-handling devices all effectively share a single Txd line and a single Rxd line, in a multi-drop serial bus topology.

Each byte of data is transmitted serially in an 11-bit package: a start bit (value 0), 8 data bits (LSB first), a programmable 9th data bit (used as the address bit), and a stop bit (value 1).

The use of the 9th bit allows the communications bus to support a single master with multiple slaves. The print engine is the master, and the paper-handling devices are slaves. The following multi-processor communications protocol is best implemented with UART hardware that specifically supports this use of the 9th bit. When the master wants to transmit data to one of the several slaves, it first sends an address byte which identifies the target slave device. The address byte differs from a data byte in that the 9th bit is 1 for address bytes and 0 for data bytes. Each device must initially configure its serial interface to be interrupted only when an address byte is received. With its UART configured this way, the device will not be interrupted when a data byte is received. Thus, an address byte (one with the 9th bit set) will interrupt all devices simultaneously, giving them the chance to determine if they are the device being addressed. The addressed slave device then sets its serial interface to be interrupted for the data bytes (9th bit = 0) that will follow. The master then transmits data bytes, which only the addressed device receives. All the other slave devices do not receive these bytes, since they will only be interrupted by bytes with the 9th bit set. When the master is done sending data to the addressed device, the master addresses the next device by transmitting the next device's address, with the 9th bit set. Protocol details related to how a transmission to a particular device is terminated, and how the next device is addressed, are provided in the next subsection *Communications Protocol*.

1.6.2 Baud Rate

Baud rate will be determined by the printer on which the options are used. However, the standard baud rate used for most early options was 58,252. As of this spec update (Feb 2019), and for many years previous, baud rate for our options is the standard 57,600 baud.

Historically, some of the options baud rates were slightly different from the engine baud rate due to processor clock frequency differences. This was done in part to help with EMC emissions. Any differences in baud rates must be carefully analyzed to assure reliable communications under worst-case conditions.

Communications are 11-bit (1 start bit, 8 data bits, 1 address bit, 1 stop bit).

1.7 COMMUNICATIONS PROTOCOL

This section describes the "link protocol" aspects of the interface -- the relationship between engine and devices, how transmissions are combined to form commands and status responses, how transmission errors are detected and handled, and how devices are assigned addresses.

1.7.1 Engine / Device Relationship

The engine and devices communicate in a master/slave arrangement. The engine is the sole master, and each device is a slave. The Paperport communications protocol allows the engine to 1) communicate with one slave device at any given time, or 2) communicate a given command to all slave devices simultaneously. The latter method is termed "broadcasting". The engine assigns each slave device a unique address at printer POR time. When broadcasting to all devices, the engine uses a predefined broadcast address that is recognized by all slave devices. Communications can be initiated by the engine only. Therefore, the engine must query the slave devices to detect any change in a device's status.

When the engine communicates to a single device, for each single byte sent by the engine, the addressed slave responds with a single byte. Even when the engine transmits several bytes to a specific device, the one-byte-down, one-byte-back alternation is continued. Neither the engine, nor device send two or more bytes in succession without receiving an intervening byte from the other party. Since the assignment of addresses by the engine, at POR time, is somewhat elaborate, those details will be described in a separate subsection, *Power-On Reset (POR) Protocols*.

When the print engine wants to send a command to ALL slave devices simultaneously, it does that by *broadcasting* the command. To broadcast a command, the print engine first sends out the special *Broadcast Address*, which causes all devices to be addressed simultaneously. The 9th bit is set for the Broadcast Address, as it is for all device addresses. After receiving the Broadcast Address, all devices set their serial interface to "data mode" -- they are interrupted on every byte received. No device sends a response byte to the Broadcast Address. An attempt by the devices to respond would

essentially garble the bus due to near-simultaneous overlapping transmissions from all devices. The print engine then sends the necessary data bytes for the rest of the broadcasted command. No device will send a response byte to any of the bytes transmitted by the engine.

1.7.2 Response Time Requirements

Each slave device is required to respond to a byte sent from the engine within 1 msec, maximum, after receiving the byte. Historically, this has been an absolute requirement, especially on system where a paperport command controls a time-critical mechanical operation (diverter move, motor start-stop) where delays cannot be tolerated. The 1.0 msec time-period starts when the last bit transmitted from the engine has arrived at the device, and terminates at the time the first response bit from the slave begins transmitting from the slave's UART. Slave devices are highly encouraged to minimize the time between receiving a byte, and transmitting the response to the engine. It is in the best interest of all bus devices to make the most efficient use of the available bus capacity.

If the engine has more than one byte of data to send to the currently-addressed device, the engine will start the transmission of the next byte within 1.0 msec (nominal) after receiving the final bit of the previous byte from the slave, unless the following limitation on the engine's overall byte transmission rate applies. When the engine is communicating to a single device (not broadcasting), the elapsed time between any two successive bytes sent by the engine will be no less than 1.5 msec *on the average*. The absolute *minimum* time between any two successive bytes sent by the engine to a single addressed device is 1.0 msec. This minimum time can occur on a maximum of 20% of the engine transmissions. The engine imposes the 1.0 msec minimum duration between bytes sent to allow a slave device to more easily manage its processor bandwidth.

When broadcasting a command, the elapsed time between any two successive bytes sent by the engine will be no less than 1.5 msec *on the average*, with an absolute *minimum* time of 1.0 msec, and an absolute *maximum* time of 2.0 msec. The minimum time can occur on a maximum of 20% of the engine transmissions. The maximum time can occur on a maximum of 20% of the engine transmissions. Recall that when a command is broadcasted, all bytes originate from the engine; no bytes are sent from any device to the engine.

1.7.3 Engine to Single Device Communications

The engine and devices communicate via Paperport *commands* and *status responses*. This section describes how single-byte transmissions, as described in the previous sections, are grouped together to form the commands and status responses, which allow information flow between the engine and a single device. Section 3.2.4, describes how the engine broadcasts information to all devices simultaneously. The *Command Set Description* section and the *Devices Response Set Description* section contain a detailed definition for each Paperport command and status response, respectively.

1.7.3.1 Standard Communications Syntax

The engine communicates to a single device by issuing *commands* to that device. The device returns information to the engine via *status responses*. The standard form of communications, or *Standard Communications Syntax*, utilized when the engine communicates with a *single* device, is shown in Figure 3-1. A *command* may consist of one or more *Command Data* bytes. A *status response* may consist of zero or more *Response Data* bytes. Note that some commands have no associated status response, i.e. "zero" Response Data bytes. Specific commands that are defined to have no associated status response are a special case of the Standard Communications Syntax, and are discussed at the end of this section.

The link protocol described in Section 3.2.1., in which the engine and device alternate single-byte transmissions, is used to pace the multiple transmissions that comprise each command and status response. For each byte transmitted by the engine, the device responds with a corresponding byte. In addition to any information conveyed by the numeric value of the device's response byte, the receipt of a response byte from a device lets the engine know that the device received the corresponding byte transmitted by the engine.

Referring to Figure 3-1, the first byte sent by the engine is the Command Op Code. This is a single byte that identifies the specific command that is being sent to the addressed device. The device sends this very same byte back to the engine in the Command Op Code Echo. The echoing contained throughout the syntax is used to support link error detection and error handling. Depending on the definition of the specific command being sent, the engine sends a specified number of Command Data bytes to the device. These bytes are identified as Command Data #1 through Command Data #m.

Each Command Data byte received by the device is individually echoed back to the engine. The echoed bytes are identical to the corresponding Command Data bytes sent from the engine. Note that the number of Command Data bytes specified by a particular command may be zero. In this case, the command is fully specified by the Command Op Code.

Some commands require a status response from the addressed device, and some commands do not. When the engine has sent a command that requires a status response from the device, after all command bytes have been sent to that device, and echoed, the engine turns around the direction of data flow by sending the Start of Response byte. The Start of Response byte is a single, defined, constant value that conveys no specific information beyond indicating that a status response has begun. After receiving the Start of Response byte, the device begins its status response. The response information is contained in the Response Data bytes, identified as Response Data #1 through Response Data #n, with the number n depending on the particular status response being sent. The response data bytes are individually echoed from the engine back to the device. The echoed bytes are identical to the corresponding Response Data bytes sent from the device. To comply with the master/slave one-byte-down, one-byte-back protocol, the device terminates the status response by sending the End of Response byte to the engine. The End of Response byte is a single, defined, constant value. It does not contain any specific information.

The Start of Response byte assigned value is hexadecimal **AA**. The End of Response byte assigned value is hexadecimal CC.

As noted above, the definition of a particular command may specify that there is *no* corresponding status response. The *Hard Reset* command is an example. The Standard Communications Syntax is modified slightly for commands that have no corresponding status response. For such commands, the *entire* status response is eliminated -- i.e., there is no Start of Response byte, no Response Data bytes, no Response Data Echo bytes, and no End of Response byte. This modification of the syntax is done to optimize the overall usage of the Paperport bus.

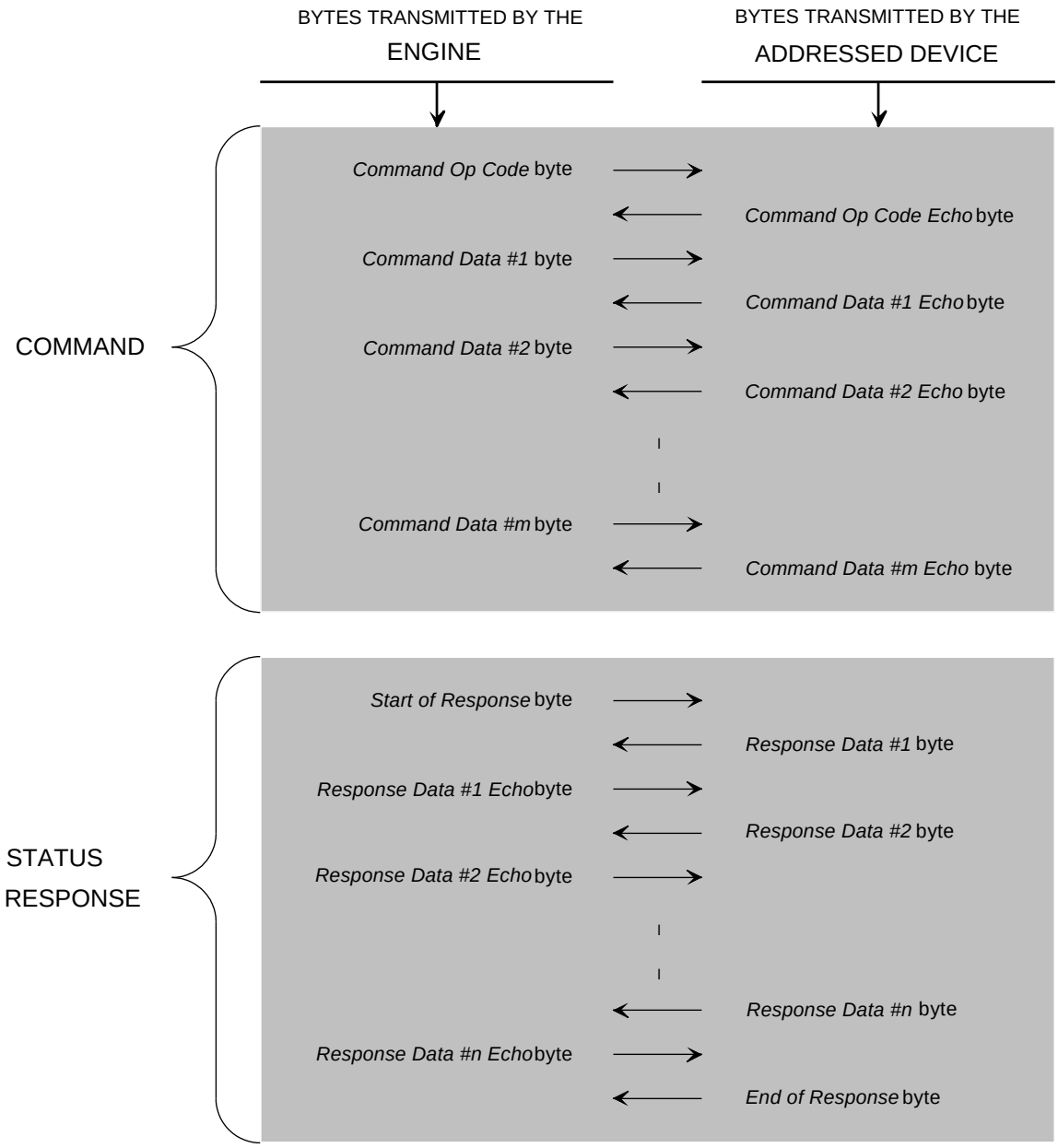


FIGURE 3-1 **Standard Communications Syntax**

1.7.3.2 Non-Echoing Communications Syntax

The echoing that is included in the definition of the Standard Communications Syntax is used to aid in the detection of transmission errors. However, there are instances when the Paperport bus bandwidth is used more efficiently if the redundancy of echoing is *not* present in the engine/device

communications. The ability to detect transmission errors is obtained in an alternate way. The syntax for these non-echoing commands is called the *Non-Echoing Communications Syntax*. We will attempt to minimize the number of commands that do not use the Standard Communications Syntax. Commands that are defined to use the Non-Echoing Communications Syntax are done so out of necessity, in order to avoid traffic bottlenecks on the Paperport bus. The Non-Echoing Communications Syntax is shown in Figure 3-2.

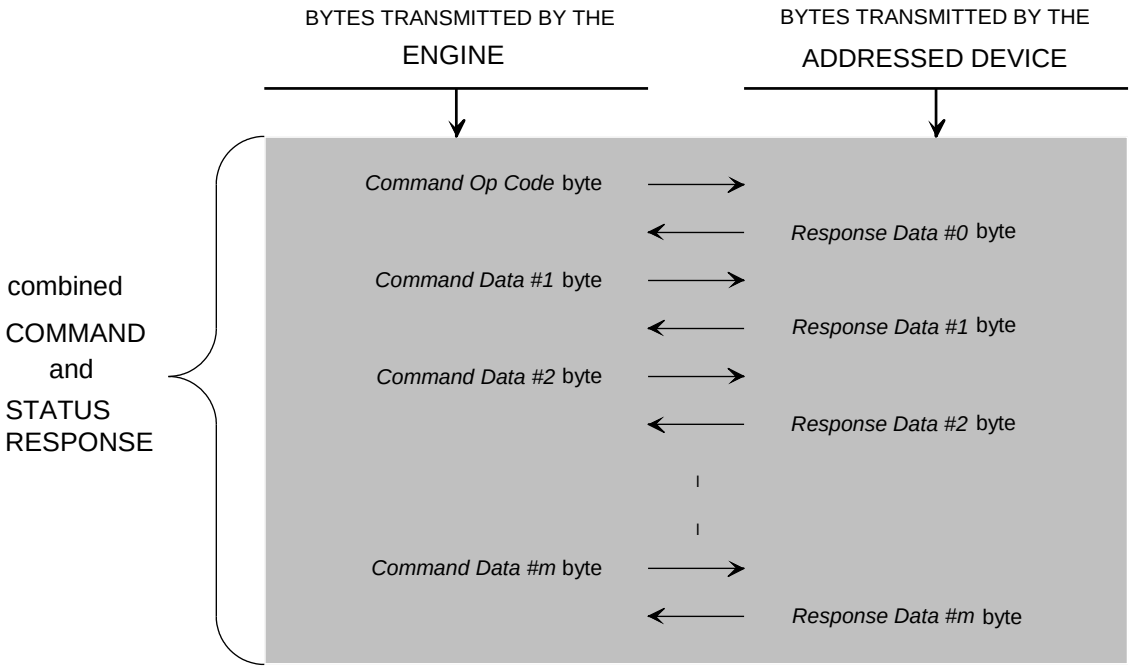


FIGURE 3-2 **Non-Echoing Communications Syntax**

The device does not echo the command data bytes sent from the engine, and the engine does not echo the response data bytes returned from the device. The information being sent from the engine is intermingled with the information being sent from the device to the engine.

If the engine desires to detect a transmission error when sending a non-echoing command, it must do so by sending the command multiple times, and comparing each successive response with the prior response. The engine will repeat the command until two successive responses are identical. If two successive responses are identical, the engine determines that the last received response is error-free.

An example of a command that utilizes the Non-Echoing Syntax is *Query Device Status 1*. (Note that this is currently the *only* example.) The specification for the Query Device Status 1 command calls for just the Command Op Code and single Response Data byte. An input device's response to the Query Device Status 1 command is bit-encoded and contains two bits that provide the current status (covered/uncovered) of the first two paper sensors in that input device. (If this particular device has less than two sensors, then only the response bits that correspond to actual sensors are valid, and are read by the engine.) In this example let's assume that the engine is trying to track the position of the leading edge of a sheet as the leading edge approaches sensor #1 in the input device. The engine will send Query Device Status 1 many times, at a relatively high frequency, until the response bit corresponding to sensor #1 finally changes status from "uncovered" to "covered". The engine will then send Query Device Status 1 one more time, to verify that the previous response byte was valid. Note that, if the engine is starting a critical timing when the status changes to "covered", it may opt to start the timing on the initial change in status, and not wait until the verification response has been received. In the somewhat rare event that the status bit change was due solely to a transmission error, then the engine can abort the timing just started, and go back to waiting for the status change. So, in summary, the validity of the response to a non-echoing command is determined and guaranteed by logically filtering the status response, as opposed to relying on echoes embedded in the command and status response.

Now let's assume that the device being polled first reports the aforementioned change in status on the Nth query. The engine then sends the command one more time to the device to verify the status change. In situations where N is "large enough", sending the command one more time for filtering consumes less overall bus bandwidth than sending the extra bytes to support echoing on *each* of the N queries. It is anticipated that any commands that are designed to utilize the Non-Echoing syntax will fit this scenario in which N will typically, or often, by "large enough" to achieve a net reduction in the overall bus bandwidth used.

1.7.3.3 Single Device Addressing

It is important to note that Figures 3-1 and Figure 3-2 do not show the transmission of the device address. The device address is not considered to be a part of the Standard Communications Syntax or the Non-Echoing Communications Syntax because a command is not always immediately preceded by a device address. The *Communications Session* subsection explains why. Figure 3-3 shows how the device address is sent before a command to address a particular slave. The address returned by the responding slave is that slave's assigned address, as opposed to the address byte read from the UART buffer. This is to allow the engine to validate that it is communicating to the correct slave device.

In Figure 3-3 the command following the device address transaction is a command that specifies the Standard Communications Syntax. Device addressing is no different for commands that specify the Non-Echoing Communications Syntax. Instead of showing separate command and status response transactions, Figure 3-3 would show the combined command / status response (as in Figure 3-2) for a command that specifies the Non-Echoing Command Syntax.

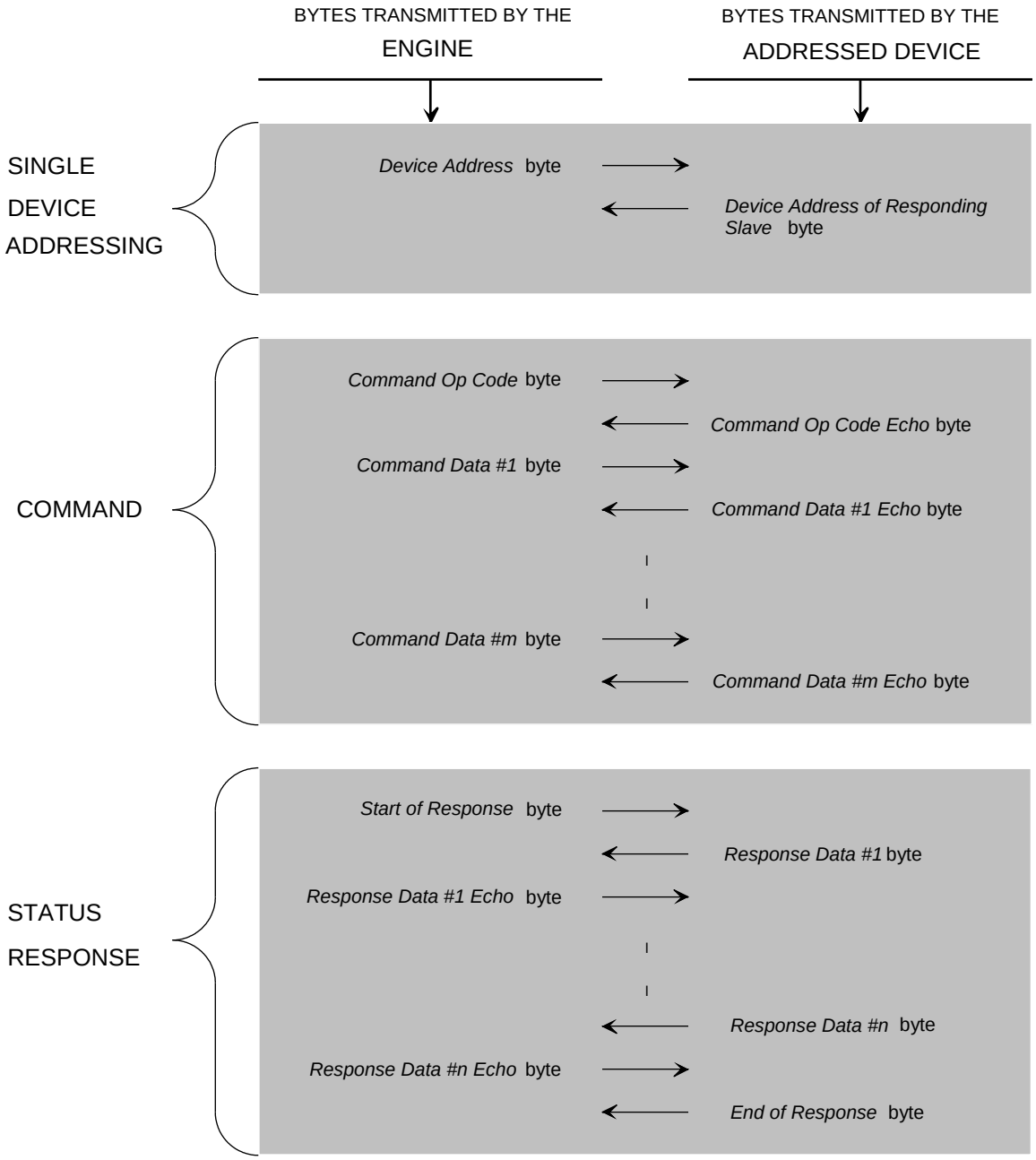


FIGURE 3-3 **Single Device Addressing**

1.7.4 Broadcast Communications

The engine sends a command to all devices simultaneously via broadcasting. Devices do not transmit any bytes to the engine, in the process of receiving a command that is broadcasted by the engine.

To allow the engine to preserve some form of data communications checking, the engine can select a single device to treat the broadcast communications the same as standard communications syntax; thus, that device will support the "receive a byte / send a byte back" communications. The engine can use this for communications checking. The engine selects a device to provide this service by sending it the **Broadcast Communications Mode Echo** command prior to sending broadcast communications. All the other devices do not return any communications, but operate as described above and in the following subsections (since they had not received the **Broadcast Communications Mode Echo** command). The following subsections further explanation of broadcast communications is without the special case of the **Broadcast Communications Mode Echo** command.

1.7.4.1 Broadcast Communications Syntax

Figure 3-4 shows the *Broadcast Communications Syntax* that is utilized by engine and devices when the engine broadcasts a command. Note that no devices provide any response bytes.

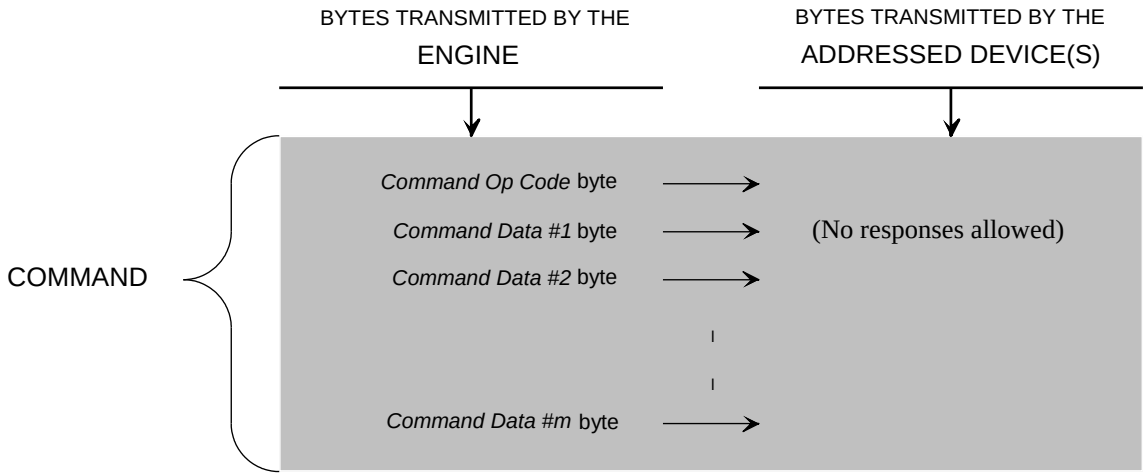


FIGURE 3-4 **Broadcast Communications Syntax**

1.7.4.2 Broadcast Addressing

Figure 3-5 shows how the broadcast address is sent before a broadcast command to address all slaves simultaneously. Note that no slave device echoes back the broadcast address - or its own address either.

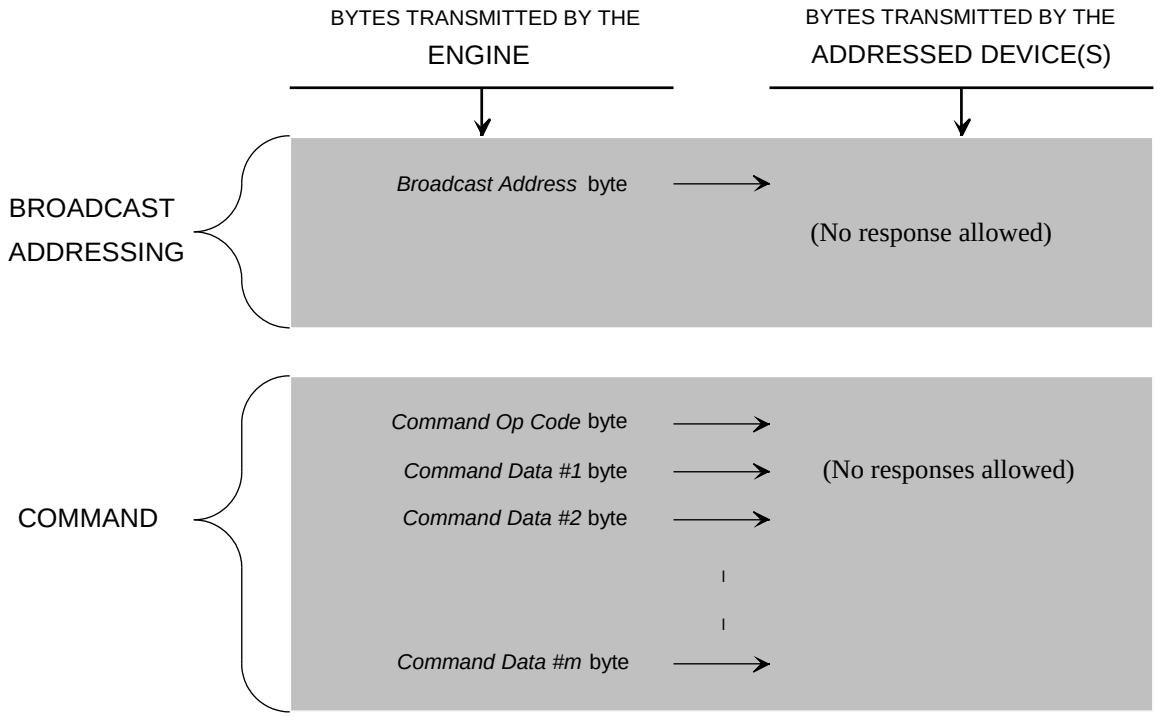


FIGURE 3-5 **Broadcast Addressing**

1.7.5 Communications Session

The *communications session* is a feature of the link design that pertains to situations in which the engine sends several commands in succession to one particular device address. In this scenario, the engine can avoid sending the device address before each successive command by creating a communications session. The general format for a communications session is shown in Figure 3-6. First, the device is selected via its address. Then, the communications session is opened with the Begin Communications Session command. Next, the engine sends the desired string of commands to the device. Finally, the communications session is terminated with the End Communications Session command. The engine does not send the device address between the Begin Communications Session command and the End Communications Session command. After the End Communications Session command is received, the device returns to the mode in which it is interrupted only if an address is received.

This version of the Paperport Interface specifies that a communication session cannot follow the Broadcast Address - i.e., a communication session containing broadcast commands is not allowed. Note also that, although Figure 3-6 depicts commands that use the Standard Communications Syntax, a session can also contain commands using the Non-Echoing communications syntax. Both syntaxes may be contained in the same session.

The advantage of sending several commands within one session is that the device's address does not need to be sent before each command. The overhead of addressing the device is spread over several commands. Bundling multiple commands together in a session is not a requirement. A succession of individual commands may be sent, with the device address sent before each command. The advantage of a session is that the total transmission time to a given device for a set of commands is less than would be obtained if the device address were sent before each command.

A disadvantage of combining several commands in a session is that the overall duration of the transmission to the addressed device is now increased, which may delay the engine's next command to a different device. If a session with any one device consumes too much time, an urgent command that is queued and waiting to be sent to some other device may have to wait too long. All of the above factors must be examined when considering the use of a communications session.

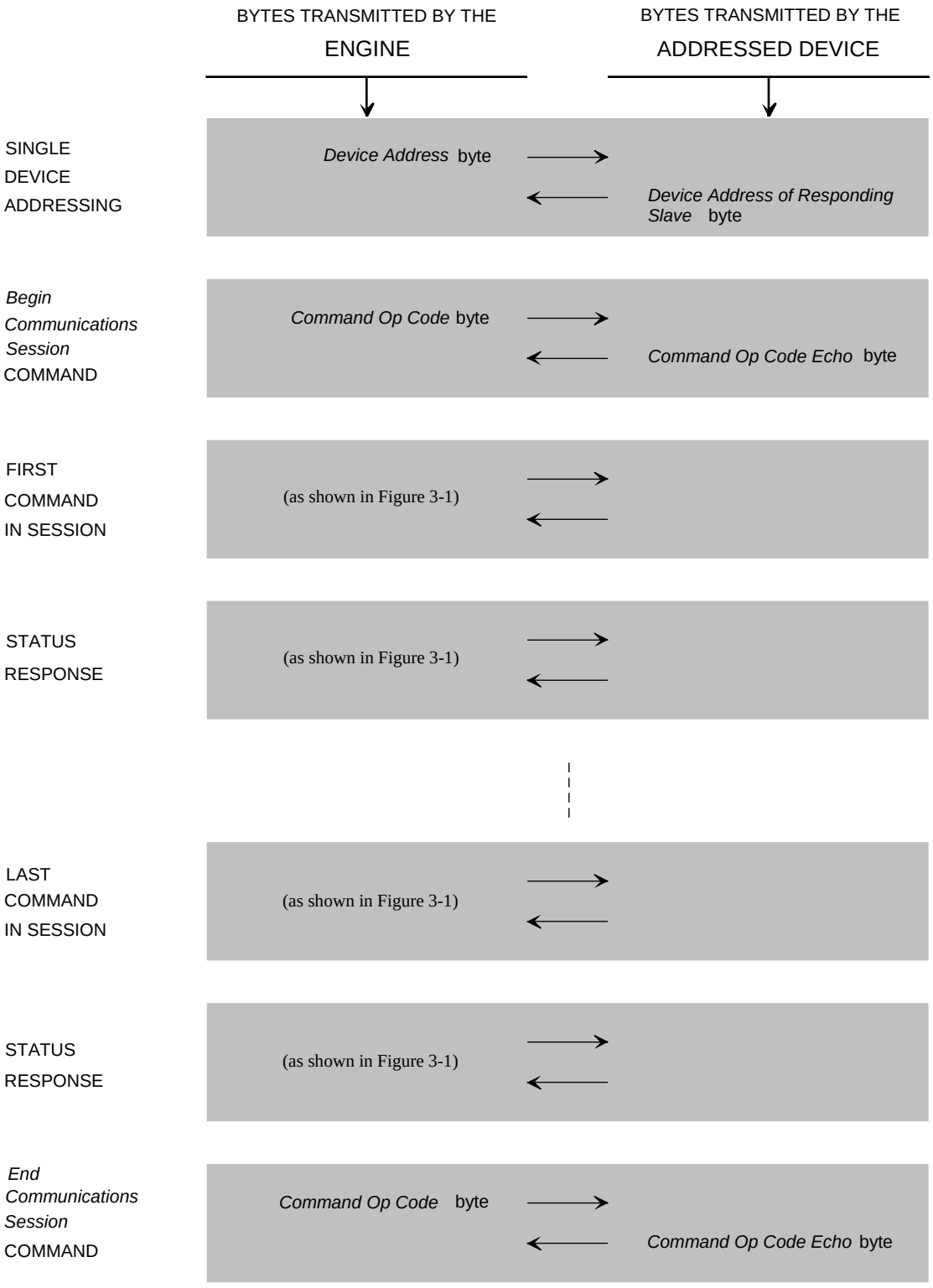


FIGURE 3-6 **Communications Session**

1.7.6 Link Error Detection and Error Handling

The echo bytes specified in the Standard Communications Syntax allow the engine and devices to detect certain transmission errors. Also, the specific UART hardware employed by the engine and/or devices may provide additional link error detection capabilities. The Paperport design does not specify a requirement for engine/device UART link error-detection capabilities, but it does allow these additional capabilities to be utilized if they are available to the engine or device.

The link error detection capabilities included in the Paperport design allow the detection of communication errors arising from: 1) intermittent signal connections due to poorly mating connectors or loose cables, and 2) electrostatic discharge. In the first scenario, intermittent connections, the engine may declare a fatal system error, based on the frequency and quantity of detected link errors. In the second scenario, occasional and infrequent link errors due to electrostatic discharge, the engine may attempt to recover without operator intervention, if possible. The link error detection is not aimed at making the Paperport design usable with a noisy communications channel. The Paperport design assumes that the serial bus is normally noise-free, as long as the printer and devices do not have any faulty hardware subassemblies. The ability to detect link errors prevents corrupted data from causing unpredictable machine behavior, software errors, and other hard-to-diagnose errors from occurring -- for both the engine and the slave devices.

1.7.6.1 Link Errors Detected by the Engine

If the engine's UART has the capability to detect certain link errors, such as framing errors, then the engine could potentially declare a transmission error for any byte that arrives at its UART. If the engine does detect a transmission error in a received byte, it will abort its internal processing of the current command, and will send the *Link Reset* special address value. All devices will receive the Link Reset, since the 9th bit is set. Upon receiving Link Reset, a device aborts any command that it may be processing, and returns its UART and link/command processing to the state in which the device is interrupted only on device address bytes (ninth bit set). Each given device still responds to the same uniquely assigned address that it received from the engine on POR, i.e., a Link Reset does *not* cause devices to return to their trunk line addresses. Note that this scheme forces an addressed device, when processing a command or status response, to examine the ninth bit of *all* bytes received from the engine. After the *Link Reset*, each device will then be ready to process subsequent commands that are addressed to that device. If the *link reset* occurred on a query command, the engine may "retry" that query command and the device is to return the correct query status for the device (ie; IR condition, cover open, tray pulled, etc). If the *link reset* occurred on a command other than a query command, the engine may "retry" that command, but the results are unpredictable by the device since they may affect the device's internal control or operation of the device's mechanism. Architecturally the engine may retry any command of which a *link reset* has occurred; however, it is recommended if it is not a query command that the engine simply post a user or service message on the engine's control panel.

The engine also has the capability to detect link errors based on the value of a byte that has been received, in certain cases. The engine's subsequent error-handling actions in these cases are the same as those described above for UART-detected errors. .

The engine will examine the Device Address Echo, and compare the echo to the value that was sent by the engine. If the two values differ, the engine determines that a transmission error has occurred. The engine will then abort its internal processing of the current command, and will send the Link Reset special address value. The assumed scenarios are 1) the Device Address was corrupted into another valid device address, and that device echoed its address, or 2) the Device Address was received by the intended device, but the echo was corrupted. In either case, one device is waiting for the Command Op Code to be sent next. When the Link Reset is received, that device will return to the state in which it is waiting for a Device Address. In the scenario where the Device Address sent by the engine is corrupted into an address value that is not currently assigned to any device in the system, the engine will never receive any echo. This situation is discussed below, in the section describing time-outs.

Likewise, if the engine determines that a transmission error has occurred for the Command Op Code or any of the Command Data bytes (ie, Length Field and/or Command Parameter bytes), the engine will abort the current command immediately after the invalid echo, and will send the Link Reset. In these scenarios, a smooth recovery with no side-effects may not be possible for all commands. By the time the engine detects the error, the device may have embarked on an incorrect operation, or may have stored away incorrect data. Depending on the command that was aborted the engine may 1) retry the failed command, 2) put the system in an Intervention Required state, and wait for an operator to initiate recovery, or 3) declare a fatal system error, which does not offer the option of attempting recovery. The assignments of particular commands to one of these three categories is TBD. Additionally, for each of the three categories, the engine will be limited to sending a restricted set of commands (which depends on the category) after the link error has occurred.

In the case of the Response Data bytes, the engine will *not* declare a link error if a received byte value appears to be invalid, and no UART-detected error occurred. The invalid value could be the result of a transmission error, or possibly a slave device software error. The engine will echo the invalid value in the corresponding Response Data Echo byte to give the device an opportunity to declare a link error. If no link error is declared by the device, the engine may then opt to declare a device software error. If the engine detects an error in the transmission of the End of Response byte, it will send the Link Reset.

Other potential transmission problems lead to the need for time-outs. For example, if the start bit of a byte transmission is noisy, the engine's UART hardware may not continue receiving the byte. Many UARTs will not report the failed start bit. In this situation, the engine will never see the byte that had been transmitted from the slave device. If the engine has not received a response from a device within 2.0 msec after sending a byte to the device (the timer starts when the first bit leaves the engine's UART), the engine will wait for the response for an additional 20.0 msec. If no response arrives, then the engine will assume that either 1) the byte sent from the engine was lost, or 2) the return byte sent from the device has been lost. The engine will then send the Link Reset to all devices. If a response arrives after the initial 2.0 msec time-out, but before the second 20.0 msec period has ended, the engine will post a Paperport protocol error, and identify the non-complying device. This device has violated the 1.0 msec response time requirement stated previously in the *Response Time Requirements* section.

For the most part, recovery procedures following the declaration of a transmission error will be determined by the engine code implementation, and will not be precisely specified in this document. For example, the engine code may retry a particular failed command transmission when the machine is quiesced, but may opt to post a machine error if the transmission error occurs during printing. This is just a hypothetical example. The engine will not retry a failed command transmission if the device itself or the overall system operation cannot recover from the original error. The functional specification for a Paperport slave device must identify commands for which that device cannot guarantee successful handling of a command retry. A likely example of a command that cannot be retried is the Pick Page command. Currently, we don't anticipate designing and specifying the protocol necessary for the engine and devices to appropriately recover from failed commands that might have caused a device to make an irrevocable action, like picking a sheet from a tray.

The following table gives a summary of link errors detected by the engine and the action taken by the engine.

Type Link Error	Engine's Action
Corrupted Address Byte	▪ Analyzes the byte echoed back from the device and determines it is invalid (ie, has been corrupted) ▪ Raises a Link Reset to devices
Corrupted Op Code Byte	▪ Analyzes the byte echoed back from the device and determines it is invalid (ie, has been corrupted) ▪ Raises a Link Reset to devices
Corrupted Length Field Byte	▪ Analyzes the byte echoed back from the device and determines it is invalid (ie, has been corrupted) ▪ Raises a Link Reset to devices
Corrupted Command Parameter Byte	▪ Analyzes the byte echoed back from the device and determines it is invalid (ie, has been corrupted) ▪ Raises a Link Reset to devices
Notes: [1]Engine detects link errors during Standard Communications Syntax and Broadcast Communications Syntax when a device has been selected to "repeat" transmissions Only. For Broadcast Communications Syntax when which a device has not been selected to "repeat" transmissions and Non-Echoing Communications Syntax, the engine is unable to detect link errors.	

Table 3.1 Summary: Link Errors Detected by Engine[1]

Link Errors Detected during Broadcast Communications Syntax -

The engine can identify a device to "repeat" the transmissions during Broadcast Communications Syntax via the **Broadcast Communications Mode Echo** command of which the device chosen detects and reports link errors the same as Standard Communications Syntax; thus, providing a means of link error detection, reporting and possible recovery for link transmission errors during Broadcast Communications mode. The specific link error recovery method, if any, is printer engine specific (see individual printer engine specification for details of a specific printer engine's link error recovery procedure).

1.7.6.2 Link Errors Detected by a Slave Device

If a slave device's UART has the capability to detect certain link errors, such as framing errors, then such a device could potentially declare a link error for any byte that arrives at its UART. The Paperport specification provides a means for devices to do so. If a device determines that a link error has occurred, for any byte that the device receives, the device will indicate this to the engine by setting the ninth bit in the next byte transmitted to the engine. The data contained in the byte with the ninth bit set is to be the same data as it would have been if no error were being declared - i.e., no special value needs to be sent by the device in this situation, the engine will react to the set ninth bit. After sending the byte with the ninth bit set; ie, raising the *Link Reset*, the device aborts any command that it may be processing, and returns its UART and link/command processing to the state in which the device is interrupted only on device address bytes (ninth bit set). Each given device still responds to the same uniquely assigned address that it received from the engine on POR, i.e., a Link Reset does *not* cause devices to return to their trunk line addresses. Note that this scheme forces an addressed device, when processing a command or status response, to examine the ninth bit of *all* bytes received from the engine. After the *Link Reset*, each device will then be ready to process subsequent commands that are addressed to that device. If the *link reset* occurred on a query command, the engine may "retry" that query command and the device is to return the correct query status for the device (ie; IR condition, cover open, tray pulled, etc). If the *link reset* occurred on a command other than a query command, the engine may "retry" that command, but the results are unpredictable by the device since they may affect the device's internal control or operation of the device's mechanism. Architecturally the engine may retry any command of which a *link reset* has occurred; however, it is recommended if it is not a query command that the engine simply post a user or service message on the engine's control panel. When the engine receives a byte from a slave device that has the ninth bit set, the engine will next send the Link Reset.

Slave devices also have the capability to detect link errors based on the value of a byte that has been received, in certain cases. The device's error-handling actions in these cases are the same as those described above for UART-detected errors. The device sets the ninth bit in the next byte transmitted to the engine, and will not continue to process the current command/status response. The device will return its UART and link/command processing to the state in which the device is interrupted only on device address bytes, and will be ready to process any command that is addressed to that device.

If the addressed slave device determines that the Command Op Code value received from the engine is not a valid op code value, for that particular device, the device will assume that a link error has occurred. For the case the command was transmitted under Standard Communications Syntax, the device will set the ninth bit in the next byte transmitted to the engine, which is the Command Op Code Echo. In this case, the value sent as the Command Op Code Echo is the illegal op code value received by the device. For the case the command was transmitted under Broadcast Communications Syntax, the device can not raise the 9th bit; but instead raises an *Invalid / Unsupported Paperport II Command* Equipment Check (see Query Equipment Check Status response section for specific value). A further discussion of link error detection during Broadcast Communications Syntax is given at the end of the section.

Slave devices are *not* permitted to set the ninth bit in a Command Data Echo transmission to indicate that a Command Data byte value is invalid *and* no UART-detected error has occurred.

If the addressed slave device determines that the Start of Response byte is invalid, the device will assume that a link error has occurred, and set the ninth bit in the transmission of the Response Data #1 byte. Likewise, if the device compares a received Response Data #n Echo byte to the corresponding Response Data #n that it sent, the device determines that a transmission error has occurred, and will set the ninth bit in the next byte transmitted to the engine. If an error is detected in the last Response Data Echo, the device will set the ninth bit in the End Of Response byte.

The following table gives a summary of communication errors (including link errors) detected by the slave device and the action taken by the device.

Condition Detected	**Device's Action**		
	Standard Communications Syntax	Broadcast Communications Syntax without Broadcast Echoing Device[1]	Broadcast Communications Syntax with Broadcast Echoing Device[2]
Unsupported Op Code byte	▪ Terminate command processing ▪ Returns to Address Mode ▪ Raises Link Error (9th bit) with echo of Op Code	▪ Terminates command processing ▪ Returns to Address Mode ▪ Raises *Invalid / Unsupported Paperport II Command* Equipment Check IR	▪ Terminate command processing ▪ Returns to Address Mode ▪ Raises Link Error (9th bit) with echo of Op Code
Corrupted Command Length Field byte	▪ Echoes back length field byte and engine detects corrupted length field	▪ N/A	▪ Echoes back length field byte and engine detects corrupted length field
Corrupted Command Parameter byte	▪ Echoes back parameter byte and engine detects corrupted parameter byte	▪ Terminates command processing ▪ Returns to Address Mode ▪ Raises *Invalid / Unsupported Paperport II Command* Equipment Check IR	▪ Echoes back parameter byte and engine detects corrupted parameter byte

Condition Detected	Device's Action		
	Standard Communications Syntax	Broadcast Communications Syntax without Broadcast Echoing Device[1]	Broadcast Communications Syntax with Broadcast Echoing Device[2]
Corrupted Start of Response byte	▪ Terminate command processing ▪ Returns to Address Mode ▪ Raises Link Error (9th bit) when next byte sent (response byte #1 or end of response byte)	▪ N/A	▪ N/A
Corrupted Response Length Field byte	▪ Device receives length field byte echoed from engine and detects it is corrupted ▪ Terminate command processing ▪ Returns to Address Mode ▪ Raises Link Error (9th bit) when next byte sent (response byte #1 or end of response byte)	▪ N/A	▪ N/A
Corrupted Response Echo byte	▪ Device receives length field byte echoed from engine and detects it is corrupted ▪ Terminate command processing ▪ Returns to Address Mode ▪ Raises Link Error (9th bit) when next byte sent (response byte #1 or end of response byte)	▪ N/A	▪ N/A
Note:	[1]For device not chosen to "repeat" transmissions during Broadcast Communications mode [2]For device chosen to "repeat" transmissions during Broadcast Communications mode		

Table 3.2 Summary : Communication Errors Detected by the Device

Link Errors Detected during Broadcast Communications Syntax -

The engine can identify a device to "repeat" the transmissions during Broadcast Communications Syntax via the **Broadcast Communications Mode Echo** command of which the device chosen detects and reports link errors the same as Standard Communications Syntax; thus, providing a means of link error detection, reporting and possible recovery for link transmission errors during Broadcast Communications mode. The specific link error recovery method, if any, is printer engine specific (see individual printer engine specification for details of a specific printer engine's link error recovery procedure). Devices not chosen to repeat the transmissions, process link errors during Broadcast Communications Syntax as described in the above paragraphs.

1.7.7 Device Address Assignment

At printer power-on reset, the engine assigns a unique device address to each Paperport device, as a part of the engine's normal initialization activities. The engine assigns an address for each slave device on each trunk line.

At device power-on reset, the initial hardware/software state of each slave device is such that the next device in the daisy-chain (on the same trunk-line) is disconnected from the Paperport bus. So, initially after power-on reset, the engine can communicate only to the device on each trunk line that is physically closest to the engine. On power-up, each slave device has not yet been assigned an address from the engine. Each slave device is designed to respond to its predefined *trunk line address* on power-up. The engine communicates first to the device that is on the lowest-valued trunk line address. Based on the current Paperport specification this would be the closest device on top of the printer, on the "Top Attached Devices" trunk line. The printer initializes that device and assigns that device its unique address. From that point on, the device will respond only to that address.

The engine will repeat this process of initialization and device address assignment for each successive slave device on the trunk line, until all of the devices on the trunk line are brought on-line. After this particular trunk line is initialized, then the engine does the same procedure for the next trunk link, assigning device addresses successively to each slave device. When all devices on all trunk lines have been initialized and have received device addresses from the engine, the initialization process between the engine and the devices has been completed. See the *Command / Response Protocol* section *(Power-On Reset Protocols* subsection*)* for an illustration of how slave devices initialize their serial communications interrupt system and related information.

The address space must accommodate all of the slave device addresses (assigned by the engine on system power-up), the trunk line addresses (predefined), the Broadcast Address, and the Link Reset address, in addition to reserved address values. All addresses are a single byte in size. The various address ranges and special addresses are assigned numerical values from the 8-bit address space as shown in Figure 3-7.

Address	Device
0x00	Top Attached Devices Trunk Line
0x01	Bottom Attached Devices Trunk Line
0x02	Front Attached Devices Trunk Line
0x03	Rear Attached Devices Trunk Line
0x04	Left Side Attached Devices Trunk Line
0x05	Right Side Attached Devices Trunk Line
0x06 - 0x7E	Reserved
0x7F	Broadcast Address
0x80 - 0xC0	Up to 64 Slave Device Addressed (assigned by engine on power-up)
0xC1 - 0xFD	Reserved
0xFE	Link Reset
0xFF	Reserved

FIGURE 3-7 **Address Space Definition**

COMMAND SET DESCRIPTION

1.8 COMMAND STRUCTURE

All Paperport II commands fall into one of two categories: *Single Byte Command* or *Multibyte Command*. Single byte commands are as their name implies; one byte long. Likewise, multibyte commands are as their name implies; several bytes long. The syntax for each command type is listed below:

Single Byte Command:	OpCode
MultiByte Command:	OpCode, Length Field, $Parm_1$, $Parm_2$, ... $Parm_n$ (Length Field = # bytes following the OpCode, including itself)

The Command OpCode Space is given below:

0x00 0x7F	Single Byte Commands
0x80 0xFE	MultiByte Commands
0xFF	Reserved

1.9 SYNCHRONIZATION COMMANDS

The following commands are "synchronization" commands. These commands are used by the master to establish synchronization for the entire system; ie, between itself and all the devices.

1.9.1 System Configuration Reset (SCR)

Op Code	Mnemonic	Command Name
0x00	SCR	System Configuration Reset

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

A **System Configuration Reset** is the same as a **Hard Reset** command (see section 1.9.2) with the following exception / addition: At the end of reset processing, the device disconnects the device "below" it and returns to only responding to its Trunk Line Address. The print engine uses this command to force all the devices back to their Trunk Line Address so it can re-initiate the processing of bringing the devices on-line (including address assignments). The print engine may need to do this if it detects a device has lost its address, or if the printer suspects a device has been added to the configuration.

Per protocol, the device does not perform a mechanical reset (or homing) as part of this reset processing; thus, the printer must send the device a **Mechanical Reset** command to allow the device to "home" itself mechanically before processing any print jobs.

Note: The device doesn't actually begin the "reset processing" until the transmission of the last byte of the link protocol for this command is completed.

For more information on reset processing, including timing requirements and a description of the "Limited Paperport Command Set" allowed to be sent to the device during this type reset; see section 0, POWER-ON RESET (POR) / RESET PROCESSING .

1.9.2 Hard Reset (HR)

Op Code	Mnemonic	Command Name
0x01	HR	Hard Reset

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

When the printer requires the device to return to its power-on reset (POR) state, the printer may issue the **Hard Reset** command. After receiving the Hard Reset command, the device will perform an electrical and software reset. The electrical reset consists of returning any functions controlled by hardware to their POR default state and executing any resident Basic Assurance Tests (BAT). The software reset consists of resetting the device's microcode to its POR default state. This means any previous commands setting the device, such as, Set Process Speed will have to be resent to the device by the print engine. Also, All the device's status bits will be reset to their POR default values. The print engine will Not have to re-establish the device's address; the device is to retain its address assignment across Hard Reset boundaries.

If during the Hard Reset processing the device detects an Intervention Required condition, then it will set the appropriate status bit indicating the IR condition and the type IR encountered.

Per protocol, the device does not perform a mechanical reset (or homing) as part of this reset processing; thus, the printer must send the device a **Mechanical Reset** command to allow the device to "home" itself mechanically before processing any print jobs.

Note: The device doesn't actually begin the "reset processing" until the transmission of the last byte of the link protocol for this command is completed.

For more information on reset processing, including timing requirements and a description of the "Limited Paperport Command Set" allowed to be sent to the device during this type reset; see section 0, POWER-ON RESET (POR) / RESET PROCESSING..

Note: At this time neither the printer, nor any device has identified a need for this command. Implementation is not required to meet current compatibility levels.

1.9.3 Soft Reset (SR)

Op Code	Mnemonic	Command Name
0x02	SR	Soft Reset

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

A **Soft Reset** is the same as a Hard Reset except the BAT Test Not performed and full power is not restored if the device is in the low power state.

Note: The device doesn't actually begin the "reset processing" until the transmission of the last byte of the link protocol for this command is completed.

For more information on reset processing, including timing requirements and a description of the "Limited Paperport Command Set" allowed to be sent to the device during this type reset; see section 0, POWER-ON RESET (POR) / RESET PROCESSING.

Note: At this time neither the printer, nor any device has identified a need for this command. Implementation is not required to meet current compatibility levels.

1.9.4 Mechanical Reset (MR)

Op Code	Mnemonic	Command Name
0x22	MR	Mechanical Reset

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

When the printer requires the device to initialize (or reset) itself mechanically in preparation for processing print jobs, the printer sends the device this command. Normally this command will be sent by the printer following completion by the device of POR processing, System Configuration Reset processing, Hard Reset processing, Soft Reset processing, or IR Clear Notification processing.

When the device receives this command it performs a mechanical initialization including mechanical homing if required to prepare itself for accepting and processing media. If during the mechanical reset processing the device detects an IR Condition (for example: cover open, paper in its paper path, etc), or an Equipment Check (for example: feed motor inoperable); the device is to raise an Intervention Required condition by asserting either the **Intervention Required Condition - Homing Allowed** or **Intervention Required Absolute Immediate Stop** signal (whichever is appropriate for the type error encountered) of the device's *General Status* response and the appropriate signals in the detailed query response bytes.

The printer is only to send a device the Mechanical Reset command when the printer and device are in the Idle state; ie, never when the device is processing media. If this command is received by the

device when the device is not either in the Idle state, the device will return an Invalid Paperport Protocol IR Condition response.

Notes:
1. The device doesn't actually begin the "reset processing" until transmission of the last byte of the link protocol for this command is complete.
2. The mechanical reset is different from other reset commands in that it does not terminate a communications session.

For more information on reset processing; including timing requirements for this type reset; see section 0, POWER-ON RESET (POR) / RESET PROCESSING.

1.9.5 Query Reset Complete (QRC)

Op Code	Mnemonic	Command Name
0x03	QRC	Query Reset Complete

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.1

Description:

After issuing either a **System Configuration Reset, Hard Reset**, **Soft Reset**, **Mechanical Reset** or **Intervention Required Clear** command, the printer utilizes this command to determine when the device has completed execution of the reset.

1.9.6 Disconnect Communications To Next Device (DCND)

Op Code	Mnemonic	Command Name
0x04	DCND	Disconnect Communications To Next Device

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

The printer sends this command to a device to have that device disconnect the communications (printer's transmit line) to the device that may be cascaded to it (for example: the device below it for those stacked underneath the printer). The printer may want to do this if it believes that device is corrupting the communications bus.

Note: At this time neither the printer, nor any device has identified a need for this command. Implementation is not required to meet current compatibility levels.

1.9.7 Connect Communications To Next Device (CCND)

Op Code	Mnemonic	Command Name
0x05	CCND	Connect Communications To Next Device

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

The printer sends this command to a device to have that device reconnect the communications (printer's transmit line) to the device that may be cascaded to it (for example: the device below it for those stacked underneath the printer).

1.9.8 Assign Device Address (ADA)

Op Code	Mnemonic	Command Name
0x80	ADA	Assign Device Address

Command Structure: OpCode, Length Field, N1
0x80, 02, N1

Pa*rameters:* N1 Device's Unique Address
See *Link Section* 0 for valid device addresses.

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

The printer sends this command to assign an address to a specific device. The device begins utilizing this 1 byte address, N1, as its unique device address at the conclusion of processing this command (ie, upon the transmission of the "start of response" byte of this command). This address assignment remains the device's address until it receives another **Assign Device Address** command at which time it disregards the old address and begins using the new address, or until power-off or receiving the **System Configuration Reset** command at which time the device reverts back to only responding to its Trunk Address.

Whenever the printer wishes to establish an engine to device communications session with the device in order to send commands to the device, the printer must first "address" the device with this unique 1 byte device address, N1. The printer is responsible for assigning addresses to all the devices and ensuring each device has a unique address.

1.9.9 Broadcast Communications Mode Echo (BCME)

Op Code	Mnemonic	Command Name
0x8F	BCME	Broadcast Communications Mode Echo

Command Structure: OpCode, Length Field, N1
0x8F, 02, N1

Pa*rameters:* N1 Echo Control
0x00 Echo Off
0x01 Echo On

0x02 - 0xFF reserved

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used by the engine to identify a specific device to utilize the Standard Communications Syntax, including link error detection and processing, during Broadcast Communications mode. This allows for link error detection and processing during Broadcast Communications mode.

N1 = 0x00 Echo Off

If the device has been set to *Echo On* for Broadcast Communication mode from a previously received **Broadcast Communications Mode Echo** command and receives this command with N1 = *Echo Off*, the device returns to processing Broadcast Communications mode normally; ie, without any echo responses. If the device had not previously been set to *Echo On* for Broadcast Communication mode and receives this command with N1 = *Echo Off*, the device performs no operation (ie, continues to process Broadcast Communications mode normally.

N1 = 0x01 Echo On

When the device receives this command with the *Echo On* parameter, it is to process Broadcast Communications mode utilizing Standard Communications Syntax; ie, echo back the broadcast address, Op Code, length, parameters, etc and raise Link Errors as defined for Standard Communications Syntax (see Link section 0 for syntax details). If the engine desires echoing of Broadcast communications mode commands, it must set only one device as *Echo On* for Broadcast Communications mode (otherwise unpredictable results will occur from more than one device attempting to communicate on the serial communications bus at the same time).

At power-on, all devices default to *"Echo Off"*; ie, no device provides the echo for Broadcast Communications mode until the engine explicitly selects it to do so via this command.

If the device that has been selected for *Echo On* during Broadcast Communications mode receives a **System Configuration Reset**, **Hard Reset**, or **Soft Reset** command, it will terminate *Echo On* and return to processing Broadcast Communications mode normally. If one of these commands were broadcast, the device set with *Echo On* utilizes Standard Communications Syntax for this command and then upon completion of its processing reverts back to processing Broadcast Communications mode normally; ie, terminates *Echo On.*

Note: There is no affect; ie, change in *Echo On* or *Echo Off* condition of the device when a **Mechanical Reset** command, **Intervention Required Clear** command, or Link Reset is received.

Link Errors detected during Broadcast Communications mode when a device is selected with *Echo On* is handled the same as link errors during Standard Communications Syntax (see the Link section 0 for more details).

1.10 CONFIGURATION COMMANDS

The master uses these commands to establish the *characteristics* of the devices attached; ie, the type devices attached and their specific characteristics.

1.10.1 Query Device Type (QDT)

Op Code	Mnemonic	Command Name
0x06	QDT	Query Device Type

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.2

Description:

The printer issues this command to a device to identify the type of device it is (ie, for determining what it is). The physical attributes (or model) assumed by the engine for a given *device type* is shown in *Appendix D - Device Type(s) Physical Model*.

1.10.2 Query Device Characteristics (QDC)

Op Code	Mnemonic	Command Name
0x8A	QDC	Query Device Characteristics

Command Structure: OpCode, Length Field, N1
0x8A, 02, N1
0x8A, 03, N1, N2

Pa*rameters:* N1 Device's Tray # of Which Characteristics Are to be Reported
Values:

0x00	For Device's Without a Tray
0x01 - 0x80	For Devices with Trays #1 thru 128
0x81	For Device's Stapler
0x82	For Device's Accumulator
0x83	For Device's Jogger
0x84	For Device's Aligner
0x85	For Device's Hole Puncher
0x86	For Device's 2nd Stapler
0x87	For Devuce's Fold Finisher
0x88 - 0x8F	Reserved for future finishing related systems
0x90	For Device's Lighting System (Compatibility Level >= 18 for Input Trays)
0x91	For Device's Diverter Mechanism
0x92	For Device's In-System Programming
0x93	For Device's Media Length Zone Sensor
0x94 - 0xFF	Reserved

N2 Sub-Command – see device specific response.

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.16

Description:

The printer issues this command after identifying the type device via the **Query Device Type** to determine the specific characteristics of that device. The printer specifies the particular Tray of the devices it wishes to learn its characteristics. The tray # equally applies to both input and output devices that have trays (bins). For devices that do not have a tray, the printer issues N1 = 0x00 to indicate it is querying a device without a tray. If the printer selects a tray the device does not support (example: tray #4 of a device that only has 2 trays), the device should report back an "unsupported" tray response in its characteristics response.

The specific response to the query is device dependent; see the "Device Dependent Responses" (see section 1.16 for details).

Note:

1. Query for Device's 2nd Stapler is only sent when if the device has a 2nd stapler installed, as indicated in the QDC Response for Device's With Output Trays (N1 = 0x01 thru 0x80) under Section 5.3.1.5.2

1.10.3 Query Paper Path Characteristics (QPPC)

Op Code	Mnemonic	Command Name
0x8D	QPPC	Query Paper Path Characteristics

Command Structure: OpCode, Length, N1
0x8D, 0x02, N1

Parameters: N1 Paper Path Characteristics Requested

0x00 Device's Height
Units: 1 mm

0x01 Distance from Device's 1st PassThru Paperpath's Sensor to device's top
Units: 1 mm

0x02 Distance from Device's PassThru Paperpath's 2nd Sensor to 1st Sensor
Units: 1 mm

0x03 Distance from Device's PassThru Paperpath's 3rd Sensor to 2nd Sensor
Units: 1 mm

: :
: :

0x3F Distance from Device's PassThru Paperpath's 63rd Sensor to 62nd Sensor
Units: 1 mm

0x40 Sensors' Distribution.

0x41 Distance from Device's 1st Feed Paperpath's Sensor to Device's edge closer to the printer.

0x42 Distance from Device's 2nd Feed Paperpath's Sensor to 1st Feed Paperpth's Sensor.

: :
: :

0x7F Distance from Device's 63rd Feed Paperpath's Sensor to 62nd Feed Paperpth's Sensor.

0x80 to 0xFC Reserved

0xFD Distance from Device's Bottom to 1st PassThru Nip - Feed Paperpath -
Units: 1/16 mm

0xFE Speed Ratio of Device's 1st PassThru Nip
(Nip Closest to the Bottom) Compatibility Level >= 14
Units: 1/256 mm/Encoder (Used in conjunction with TS)

0xFF Distance from Device's Bottom to 1st PassThru Nip
(Nip Closest to the Bottom) – PassThru paperpath-
Compatibility Level >= 14
Units: 1/16 mm

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.3

Description:

Note : Need to Add Distance from Duplexer Input to First Sensor ?

The printer issues this command to a device to obtain information (ie, the characteristics) about the device's paperpath. Such information includes: the device's height (or "PassThru" paperpath length; ie, the distance paper from the device "below" must travel thru this device), distance between "PassThru" sensors, etc.

Notes:

1. Compatibility Level 14 and Less: The printer determines the number of sensors a device has in its PassThru Paperpath by the device's responses to the Query Device Type and Query Number of Trays commands.
2. Compatibility Level 15 and More: The printer determines whether a device has a feed Paperpath that is distinct than its PassThru Paperpath by the device's response to QDT. See Appendix C for more details.

3. Compatibility Level 15 and More: The printer determines the number of sensor a device has in its Passthru Paperpath and number of sensors that device has in its Feed PaperPath by the device's response to QPPC: Sensor's Distribution.

1.10.4 Query Device Inventory (QDI)

Op Code	Mnemonic	Command Name
0x8B	QDI	Query Device Inventory

Command Structure: OpCode, Length Field, N1
0x8B, 02, N1

Pa*rameters:* N1 Device's Tray # of Which Inventory is to be Reported
Values:

0x00	For Device's Without a Tray
0x01 - 0x80	For Devices with Trays #1 thru 128
0x81	For Device's Stapler
0x82	For Device's Accumulator
0x83	Reserved
0x84	Reserved
0x85	For Device's Hole Puncher
0x86	For Device's 2nd Stapler
0x87	For Devuce's Fold Finisher
0x87 – 0x92	Reserved
0x93	For Device's Media Length Zone Sensor
0x94 - 0xFF	Reserved

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.16

Description:

The printer issues this command to a device whenever it thinks the device's inventory may have changed of which the device responds with its current inventory status. For example: An operator may have adjusted a tray from letter size paper to legal size paper, or an output stacker from 10 output bins to 3 output bins. The printer specifies the specific tray via parameter N1 for the tray its inventory is to be reported. The tray # equally applies to both input and output devices that have trays (bins). For devices that do not have a tray, the printer issues N1 = 0x00 to indicate it is querying a device without a tray. If the printer selects a tray the device does not support (example: tray #4 of a device that only has 2 trays), the device should report back an "unsupported" tray response in its inventory response.

The specific response to the query is device dependent; see the "Device Dependent Responses" (see section 1.16 for details).

Note: Query for Device's 2nd Stapler is only sent when if the device has a 2nd stapler installed, as indicated in the QDC Response for Device's With Output Trays (N1 = 0x01 thru 0x80) under Section 5.3.1.5.2

1.10.5 Query Device Software Revision (QDSR)

Op Code	Mnemonic	Command Name
0x0A	QDSR	Query Device Software Revision

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.4

Description:

The printer issues this command to a device to identify the device's microcode version for printing on the printer's test and configuration pages, and for enabling / disabling the correct device driver for the device.

1.10.6 Query Device Boot Software Revision (QDBSR)

Op Code	Mnemonic	Command Name
0x2D	QDBSR	Query Device Boot Software Revision

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.5

Description:

The printer issues this command to a device to identify the device's microcode version for printing on the printer's test and configuration pages, and for enabling / disabling the correct device driver for the device. Unlike the Query Device Software Revision which identifies device’s application code version ,Query Device Boot Software Revision identifies the device’s boot code version.

1.10.7 Printer Characteristics Notification (PCN)

Op Code	Mnemonic	Command Name
0x81	PCN	Printer Characteristics Notification

Command Structure: OpCode, Length Field, N1, N2, ... Nm

Parameters: N1 Printer Characteristics (bit encoded)

Bit 0 – Inactive state of printer's first paper path sensor
Values: 0 = Active high; 1 = Active Low
Bit 1 – Reserved
Bit 2 – Reserved
Bit 3 – Reserved
Bit 4 – Reserved
Bit 5 – Reserved
Bit 6 – Reserved
Bit 7 – Reserved

N2 Printer's Exit Paperpath Length

The distance in mm along the paperpath from the printer's exit sensor to the attached device's paperpath entry point.

Values: 0x00 - 0xFF
where 0x00 = 0mm
0x01 = 2mm
0xFF = 510mm

N3 PEXSN Offset

The distance in mm along the paperpath of which the incoming sheet's leading edge is located from the device's entry point when the engine sends the PEXSN notification to the device.

Values: 0x00 - 0xFF
where 0x00 = 0mm
0x01 = 2mm
0xFF = 510mm

Note: The PEXSN Offset (N3), depending upon the engine's application, may be process speed dependent. If so, the engine must send the PCN with the PEXSN Offset before each SPS command representing a process speed change to allow the device to calculate its parameters accordingly.

N4 DMID Offset

The distance in mm along the paperpath of which the incoming sheet's leading edge is located from the device's entry point when the engine sends the DMID notification to the device.

Values: 0x00 - 0xFF
where 0x00 = 0mm
0x01 = 2mm
0xFF = 510mm

Note: The DMID Offset (N4), depending upon the engine's application, may be process speed dependent. If so, the engine must send the PCN with the DMID Offset before each SPS command representing a process speed change, to allow the device to calculate its parameter accordingly

N5,N6 Distance from Printer's Bottom To Printer's First Nip

The distance in 1/256 mm along the paperpath from the bottom of the printer to printer's first nip or first drive roller.

N5 = Most significant byte of Distance from Printer's Bottom To Printer's First Nip
N6 = Least significant byte of Distance from Printer's Bottom To Printer's First Nip

Valid Range - Two byte unsigned integer
Default Units: 1/256 mm

Note: Some Color printers use 1/16 mm

N7,N8 Distance from Printer's Bottom To Printer's First Paper Path Sensor

The distance in 1/256 mm along the paperpath from the bottom of the printer up to when the printer's first sensor makes or trips.

N7 = Most significant byte of Distance from Printer's Bottom To Printer's First Paper Path sensor.
N8 = Least significant byte of Distance from Printer's Bottom To Printer's First Paper Path Sensor.

Valid Range - Two byte unsigned integer
Default Units: 1/256 mm

Note: Parameters N3 and N4 may be recomputed by the engine based on process speed. If so, the engine must send an updated PCN command before each SPS command representing a process speed change, to allow the device to calculate its parameter accordingly.

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:
This command allows the printer to notify the device of any specific printer (master) "characteristics" of which the device may need to adjust its configuration, timings, protocols, etc to the uniqueness of the specific printer attached.

1.10.8 Track Speed (TS)

Op Code	Mnemonic	Command Name
0x97 0xA5	TS	Track Speed

Note : This command used to apply to selected input options only. The original op code is shared with the SLPQ command (section 1.12.17) for some output options. Therefore, this command should not be broadcast to all options. Beginning with Compatibility level 16, the command is being extended to be usable on all options. The new Op code of 0xA5 should be used from here on. Also this command may be broadcast with the new op code. If the command is broadcast, N5 should be used to determine if the command applies to the device receiving it or not. The engine may chose to send the command with length byte less than or equal to 0x06. If the command is sent as a Standard Communications Syntax or Broadcast Syntax with length byte less than 0x06, the receiving option should process it by default.

Command Structure: 0x97, 0x03, N1, N2.
0x97, 0x05, N1, N2, N3, N4
0xA5, 0x06, N1, N2, N3, N4, N5 Compatibility level >= 16

Parameters: For length = 0x03:
N1, N2 Distance after sensor trip to start tracking
N1 = Most significant byte
N2 = Least significant byte
Default Units: 1/16 mm

Valid Range - Two byte unsigned integer.
N1,N2 = 0x0000 Always Track Speed.
N1,N2 = 0xFFFF Never/Stop Track Speed.

For length = 0x05:

N1, N2 Speed Scale Factor
N1 = Most significant byte
N2 = Least significant byte
Default Units: 1/256 mm / encoder

N3, N4 Distance after sensor trip to start tracking
N3 = Most significant byte
N4 = Least significant byte
Default Units: 1/16 mm

Valid Range - Two byte unsigned integer.
N3,N4 = 0x0000 Always Track Speed.
N3,N4 = 0xFFFF Never/Stop Track Speed

.

N5 = FD Apply to all devices
= FE Apply to Output Devices (such as stackers)
= FF Apply to Input Devices (such as trays)

Link Pacing: - Engine to Device: Standard Communications Syntax
Broadcast Syntax

Response: None

Description:

This command is used by the printer to communicate to the device the distance to feed the media at process speed before matching the speed of a specified motor in the printer. The printer provides an encoder pulse-train to enable this function. The N1 and N2 parameters when the length byte is 0x05/0x06 determine the scaling factors which the option should use after measuring the edge-edge timing of the encoder pulse train. At this time, the falling edge is the most accurate to use.

The parameter N1,N2 (or N3, N4 when length byte is 0x05/0x06) bytes specify the distance to feed at process speed. The distance is measured from the point at which the leading edge of the media reaches the device's first pass-thru sensor. N1,N2 (N3, N4) = 0x0000 means to immediately track speed once the media has reached the pass-thru sensor for input options and once the motor is started and reached steady state process speed for output options. N1,N2 = 0xFFFF means to always feed paper at process speed. Any other value should be interpreted as a distance in 1/16 millimeters with the high byte sent as N1(N3).

For the HCIT and other feed devices that open the feed rollers instead of tracking the specified motor speed, only the 2 parameters corresponding to the distance are used to determine the point at which to open the rollers.

The distance to be sent to the device will be determined by the printer based on the distance of the device's pass-thru sensor from printer's motor to tracks nip. At this distance the media should be in

the printer's motor nip. This distance should be calculated to allow the device sufficient time to start matching the speed before the motor starts making adjustments to correct for error.

After a power-on reset or receiving a **System Configuration Reset**, **Hard Reset** or **Soft Reset** command, if the device is instructed by the printer to feed media and has not received the **Track Speed** command, it should always feed at process speed by default.

1.10.9 Distance to End Track Speed End (DETS)

Op Code	Mnemonic	Command Name
0xB1	DETS	Distance to End Track Speed

Command Structure: *0xB1*, 0x03, N1, N2

Parameters: N1 - High Byte of distance at which to end track speed
N2 – Low Byte of distance at which to end track speed
Units: 1/16 mm

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command sends a distance in 1/16 mm precision to the tray option. When the trailing edge of a page reaches this distance beyond the trays pass thru sensor the tray option should stop tracking the speed signal sent by the engine. This command in combination with the TS, Track Speed, command define the distances at which the tray option should start and stop tracking the speed of the Engine's motor. The distance set by this command should ignore sensor hysteresis. The device code will account for the sensor hysteresis.

If this command is not sent, the tray option will discontinue its printer speed tracking when the trailing edge of the sheet has cleared its pass-thru sensor.

1.10.10 Media Sensor Analog to Digital Threshold Offset (MSADTO)

Op Code	Mnemonic	Command Name
0xB2	MSADTO	Media Sensor Analog to Digital Threshold Offset

Command Structure: *0xB2*, 0x02, N1

Parameters: N1 Offset in 0.0196V resolution
-102d <= N1 <= 127d
i.e. –2.00V <= N1 <= 2.49V
i.e. 0x9A <= N1 <= 0x7F

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used to set an offset for the transparency sensor reading. The device will add this value to its default comparison value. Every time the device reads its analog to digital converter it will compare the result of that reading to the new comparison value. (The new comparison value is the default comparison value plus the offset value.) Any reading above the new comparison value will be reported as transparent media and anything below the new comparison value will be reported as non-transparent media. The option will NOT store the offset value in NVRAM, therefore the printer must send this value after every POR. The device will retain this offset value even if a SCR, IRC, HR, or SR is received. If multiple MSADTO commands are sent, the device will retain the offset sent in the most recent command.

1.10.11Sequence Configuration Command (SCC)

Op Code	Mnemonic	Command Name
0xA7	SCC	Sequence Configuration Command

Command Structure: *0xA7*, Length Field, N1, N2, N3…..Nm

Parameters: N1 – Nm: Each parameter is printer-specific.

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description: This command is sent by the engine to an option to configure a particular event sequence or to send project-specific data. Please see the associated project-specific document for this command's specific usage in a particular printer.

1.11 STATUS COMMANDS

The master uses these commands to poll the devices to determine their respective status.

1.11.1 Query Device Status 1 (QDS1)

Op Code	Mnemonic	Command Name
0x0B	QDS1	Query Device Status 1

Command Structure: OpCode

Link Pacing: - Engine to Device: Non-Echoing Communications Syntax

Response: see Section 1.15.6

Description:

This is the printer's ***general purpose*** status request and the only way a device alerts the printer of its particular state. Thus, the printer must poll the device by issuing this command to determine the device's current general status. The *general status* consists of two parts: the status of the device's "PassThru" sensors (ie, paper present or not present at the sensor), and the basic status of the device (ie, is it capable of processing print jobs or is it currently in an Intervention Required condition).

1.11.2 Query Non-Critical Status Change (QNCSC)

Op Code	Mnemonic	Command Name
0x0C	QNCSC	Query Non-Critical Status Change

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.7

Description:

A device indicates in its response to the *general purpose* status command (**Query Device Status 1**) that a "Non-Critical" status change has occurred. Upon receiving that *general purpose* status indication, the printer issues this command to query the specific "non-critical" status change reported by the device. The device reports the status change on a tray basis; ie for the specific tray (or bin) within the device that had the status change.

1.11.3 Query IR Condition - Homing Allowed (QIRCHA)

Op Code	Mnemonic	Command Name
0x0D	QIRCHA	Query IR Condition Homing Allowed

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.8

Description:

A device indicates in its response to the *general purpose* status command (**Query Device Status 1**) that an IR condition has arisen, but the printer can perform a "homing" operation if needed; ie, the IR condition is not such that the device requires the printer system to immediately shut down. The printer issues this command to query the specific details of the *IR - Homing Allowed* condition. Examples of IR conditions where homing may be allowed by a device is an output device that has detected that a sheet has failed to reach its input sensor; the paper jam needs to be raised, but the device can allow the print engine to "home" before raising the paper jam.

1.11.4 Query IR Absolute Immediate Stop (QIRAIS)

Op Code	Mnemonic	Command Name
0x0E	QIRAIS	Query IR Absolute Immediate Stop

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.9

Parameters: None

Description:

A device indicates in its response to the *general purpose* status command (**Query Device Status 1**) that an IR condition has arisen of which the device requires the printer system to immediately shut down and stop processing print jobs. The printer issues this command to query the specific details of the *IR Absolute Immediate Stop* condition. *IR Absolute Immediate Stop* conditions are ones of which printing can not continue until this condition is corrected by the operator. Examples are: a certain paper jams detected by the device, and a cover open detected by the device.

1.11.5 Query Equipment Check Status (QEQCKS)

Op Code	Mnemonic	Command Name
0x10	QEQCKS	Query Equipment Check Status

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.10

Description:

A device indicates in its response to the **Query IR Absolute Immediate Stop** command if the condition is an "equipment check" (ie, equipment failure) condition. The printer issues this command to query the specific details: equipment check identification number.

1.11.6 Query Number Trays (QNT)

Op Code	Mnemonic	Command Name
0x11	QNT	Query Number Trays

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.11

Description:

The printer issues this command to a device to query the device to determine the number of trays the device contains. This command is issued to both input and output devices; to input devices to learn how many input sources the device has, and to output devices to learn how many output destinations the device has. For devices that do not have input sources or output destinations, such as the duplex option, "no trays" is returned by the device.

1.11.7 Query Paper Position Status (QPPS)

Op Code	Mnemonic	Command Name
0x8E	QPPS	Query Paper Position Status

Command Structure: OpCode, Length, N1
0x8E, 0x02, N1

Parameters: N1 Paper Status Requested
0x00 Number of Sheets in Device's Paperpath
0x01 Position of Sheet #1 in Device's Paperpath
(Sheet 1 should be the offending sheet for the jam)
0x02 Position of Sheet #2 in Device's Paperpath
: :
: :
0xFF Position of Sheet #255 in Device's Paperpath

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.12

Description:

During an IR condition, the printer sends a device this query to determine the # of sheets that reside within the device and their respective locations along the device's paperpath that must be removed by the operator to clear the IR condition.

1.11.8 Query Cover Open Location (QCOL)

Op Code	Mnemonic	Command Name
0x13	QCOL	Query Cover Open Location

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.13

Description:

A device indicates in its response to the **Query IR Absolute Immediate Stop** command if the condition is an "cover open" (ie, one of the device's covers has been opened) condition. The printer issues this command to query the specific details: which cover(s) is open.

The printer also issues this command once a "cover open" condition has been identified to learn what cover has been subsequently closed.

1.11.9 Query Tray Pulled Location (QTPL)

Op Code	Mnemonic	Command Name
0x27	QTPL	Query Tray Pulled Location

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.14

Description:

The input or output device indicates in its response to the **Query IR Condition - Homing Allowed** command if it has the condition of one of its "trays pulled" (ie, one of the device's trays (or bins) has been pulled out and can no longer be used to feed media or receive media). The printer issues this command to the device to query the specific details: which tray(s) or bin(s) is pulled out.

The printer also issues this command once a "tray pushed in" condition has been identified to learn what specific tray has been subsequently pushed in.

1.11.10Query Device Pick Time (QDPT)

Op Code	Mnemonic	Command Name
0x82	QDPT	Query Device Pick Time

Command Structure: OpCode, Length Field, N1
0x82,02,N1

Parameters:

N1	Bin #	
	00	For Input Devices without a Tray
	01 - FF	Input Tray (Bin) Number

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.15

Description:

The printer sends a device this query to determine the device's pick time; ie, the time it takes the device to pick a sheet from Bin#, N1, and have it reach the first sensor in the "Pass Thru" paper path to the printer.

The device determines the pick time off the printer's process speed indicated in the last received **Select Process Speed** command prior to this command. If the device has not received a **Select Process Speed** command, then it uses it's power-on reset default process speed to determine the pick time.

Upon receiving this command the device's response is dependent upon the state of the printer system. If this command is received immediately after a POR, **System Configuration Reset**, **Hard Reset**, or **Soft Reset** command, the device returns its "Minimum" pick time; ie, the minimum amount of time it could take for the device to pick a sheet and feed it to the first "Pass Thru" sensor for the previously received **Select Process Speed**. During this time an operator could have altered the supplies capacity in the tray thus changing the pick time required. After an **IR Clear** command, the device returns its "Minimum" pick time if it can sense the operator may have altered the supplies capacity. If the device can sense the operator did not alter the supplies capacity prior to the **IR Clear** command, the device returns its current required pick time. If the device is incapable of sensing if the operator has altered the supplies capacity prior to the **IR Clear** command, it should always return its "Minimum" pick time. For any time other than those listed above, when this query is received the device is to return its currently required pick time.

C750 : For 500-sheet trays used on the C750, this command will actually return the measured value for the last measured pick. The tray will then do auto-correction of leading edge page position to match the reported pick time. At POR, a best guess of the pick time will be used based on media level sensor in the tray.
The duplexer also supports QDPT. This command would return a duplexer pick time value that would depend on the Process Speed the engine is running.

Notes:
1. For Input devices that do not have an input tray such as the duplex option, the printer would send: 0x82, 02, 00.

2. For Input devices that only have one input source such as "single input tray" Smart Input Tray or Feeder, the printer would send: 0x82, 02, 01.

1.11.11Query Device Minimum Interpage Gap (QDMG)

Op Code	Mnemonic	Command Name
0x83	QDMG	Query Device Minimum Interpage Gap

Command Structure: OpCode, Length Field, N1, N2
0x83, 03, N1, N2

Parameters:
N1 Sheet's Output Bin Destination
00 - FF Bin Number
N2 Operation Parameter (of sheet)
0x00 Do Not Accumulate Sheet
0x01 Accumulate Sheet
0x02 - 0xFE Reserved
0xFF Operation Parameter Found in Next Byte

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.16

Operation Parameter (of sheet):

Do Not Accumulate Sheet -
When the engine desires the sheet to be treated as a single, individual sheet and go directly to the output bin destination without any "finishing" operations (such as stapling, hole punching, binding, etc), the engine selects this operation for the sheet.

Accumulate Sheet -
When the engine desires the sheet to be "accumulated" with other sheets for some "finishing" operation (such as stapling, hole punching, binding, etc) the engine selects this operation for the sheet.

Description:

Input Devices:
For Input devices such as trays, feeders, etc; this command does not apply.

Output Devices:
The printer sends an output device this query to determine the Minimum Interpage Gap required by the output device to support accepting this sheet into the output device to perform the operations identified.

Output Devices goal should be to support the printer's minimum interpage gap for output directed to any output bin and for any "finishing" operation. However, this command allows the printer to determine the needed gap for those devices that must force a larger gap due to physical requirements of that device (for example: the output device can't move its deflector to route sheet#1 into bin #n and then move it up into position for sheet#2 into bin#1).

The device upon returning a valid inter-page minimum gap (ie, value other than 0xFFFF) to the engine, logically accepts this sheet into its "page queue" to physically accept the sheet later upon receiving the **Divert Media Into Device** command.

Protocol Note:
The engine must send the **Select Output Device** command to provide the Pg Id and Media Type for the sheet to the device just prior to sending this command to allow the device to properly "calculate" the needed inter-page gap knowing the full characteristics of the sheet and the operations intended for the sheet..

Alternately, the engine can send the new SLPQ (**Submit Page to Logical Queue**) command prior to issuing this QDMG command for options such as the HCOF that must support the SLPQ command. In this case, such as for the HCOF device, the parameters N1 and N2 of the QDMG command are redundant and should be ignored by the option.

Input / Output Devices:
For input and output devices this command may or may not apply.

For the duplex option, an input & output device, this command is supported and returns the Minimum Interpage Gap required between side 2 and side 4 of a 2-4-1 sequence print job (ie, the 2 consecutive sheets feed into the duplex option). The Minimum Interpage Gap returned is printer model dependent provided by the **Printer Characteristics Notification** command and the process speed provided by the **Select Process Speed**. If these commands are not sent prior to the Query Device Minimum Interpage Gap command, the last received value for these commands will be used. If these commands have not been received since a POR or reset then the device will use its POR defaults in determining the Minimum Interpage Gap value to return.

1.11.12Query Page Complete ID (QPCID)

Op Code	Mnemonic	Command Name
0x14	QPCID	Query Page Complete ID

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.17

Description:

When an output device signals to the printer that a sheet has successfully reached its output tray (bin) via the *Page In Output* bit in the **Query Device Status** command response, the printer sends the output device this query command to determine which sheet has successfully reached its output destination (commonly called "print complete").

Note: For Input devices such as trays, and for Input & Output devices such as the duplex option; this command does not apply.

If the device has raised the *Page In Output* signal (via the Query Device Status 1 response) to indicate to the engine that the sheet has made it into the output device and an intervention required condition has occurred; the engine must first send the device the **Query Page Complete ID** command to obtain the ID of the sheet that just got deposited into the output device before issuing the **Printer Intervention Required Notification** command (otherwise the ID of the sheet successfully outputted will be lost).

1.11.13Query Device Quiesce (QDQ)

Op Code	Mnemonic	Command Name
0x26	QDQ	Query Device Quiesce

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.18

Description:

The printer issues this command to the device to determine if the device's mechanism is moving or is stopped (ie, quiesced).

1.11.14Printer Intervention Required Notification (PIRN)

Op Code	Mnemonic	Command Name
0x84	PIRN	Printer IR Notification

Command Structure: OpCode, Length Field, N1

0x84, 02, N1

Parameters: N1 Printer Raised Intervention Required Type

0x00	IR Condition Requires Mechanical Homing
0x01	Print Engine Halted
0x02	Print Engine Halted No Jam Release (Compatibility Level >= 13)
0x03	Option Tray Feed Error (Compatibility Level >= 19)

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

When the printer detects an Intervention Required condition (ie, paper jam, cover open, etc) that requires the printer to cease printing, the printer issues this command to alert the device that the printer has entered the Intervention Required state and the device is to do the same.

For the Intervention Required state, either detected by the printer or the device itself, the device is to process the current sheet according to the type intervention required condition raised. When the printer raises an *IR Condition Requires Mechanical Homing* type IR condition, the device should enter the IR (or error) state but still process commands that allow the current sheet being processed to transport through the printer system. When the "homing" has been completed, the printer will issue the *Print Engine Halted* type IR condition at which time the device should stop all mechanical movement and process any needed internal processing for the IR state, and remain in the IR state awaiting the printer to issue the **Intervention Required Clear Notification** command indicating to the device that the IR condition has been resolved by the operator. An example of an *IR Condition Requires Mechanical Homing* type IR would be a paper jam for a printer with a pick technology requiring it to home before shutting down (for example: D-roll pick technology used on the Lexmark Optra LaserPrinters).

When the printer detects an IR condition that does not require any mechanical homing (for example: cover open or equipment check hardware failure) the printer will issue the *Print Engine Halted* type IR condition. The device should stop all mechanical movement and process any needed internal processing for the IR state, and remain in the IR state awaiting the printer to issue the **Intervention Required Clear Notification** command indicating to the device that the IR condition has been resolved by the operator.

Print Engine Halted No Jam Release is the same as *Print Engine Halted*. It was added to maintain compatibility between newer options and older engines. Older engines would send always *Print Engine Halted* and expect the option to always automatically perform jam release upon receiving PIRN. The newer engines cause the option to perform jam release by sending the PTC

command.and therefore does not want the option to perform jam release when PIRN is received but the option still needs to support performing jam release upon receiving PIRN when attached to the older engine.

Option Tray Feed Error is sent to a specific option tray when the printer encounters a feed error condition when picking from that option tray. The option tray should immediately stop the pick motor when Option Tray Feed Error IR is received. The tray may keep other motors at their current state. The engine will follow Option Tray Feed Error IR with Mechanical Reset command if the engine is to retry picking the sheet otherwise the engine is follow with IR Condition Requires Mechanical Homing or Print Engine Halted. The engine will not send Intervention Required Clear Notification after Option Tray Feed Error IR. This PIRN subcommand is added to elevate the tray lift plate before retrying the pick after a feed failure since the feed failure could be due to under/not elevated lift plate.

Note: Except for Option Tray Feed Error IR, The device remains in the Intervention Required state until the printer issues the **Intervention Required Clear Notification** command.

1.11.15 Intervention Required Clear Notification (IRC)

Op Code	Mnemonic	Command Name
0x15	IRC	IR Clear Notification

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

The printer issues this notification to inform the device that printing is resuming; ie, the printer system is no longer to be in the Intervention Required state and should prepare itself for printing. Both printer and device detected Intervention Required conditions require the printer to issue this notification to indicate the Intervention Required condition has been serviced and the printer system is now able to resume printing.

The device upon receiving this notification will set All of its *device status* signals to their power-on reset default value. This includes the response for all the Intervention Required related commands:

1. **Query Device Status 1**
2. **Query Non-Critical Status Change**
3. **Query IR Condition - Homing Allowed**
4. **Query IR Absolute Immediate Stop**

5. **Query Equipment Check Status**
6. **Query Paper Position Status**
7. **Query Cover Open Location**
8. **Query Tray Pulled Location**

The device will also reset its software control signals to their power-on reset default values; thus, the printer can not depend upon previously sent commands, such as, Set Process Speed to be kept (they will be reset to their power-on reset values).

If this notification is received by the device when the device is not in the Intervention Required (IR) State and the device can not handle (ie, process without negative side affects) this command under this condition, then the device returns an *Invalid Paperport IR* Condition error response. However, if the device can handle (ie, process) the **Intervention Required Clear** command the same regardless of its IR State (in one or not in one), then the device processes the **Intervention Required Clear** command without raising an *Invalid Paperport Protocol IR Condition* error response.

The device should complete its IR Clear processing within 500 msec.

Per protocol, the device does not perform a mechanical reset (or homing) as part of this reset processing; thus, the printer must send the device a **Mechanical Reset** command to allow the device to "home" itself mechanically before processing any print jobs. If the printer sends the device a command that selects it or requires movement of the device's mechanism prior to the **Mechanical Reset** to complete the IR Clear protocol, the device will raise an *Invalid Paperport Protocol* error.

Notes: The device doesn't actually begin the "reset processing" until the transmission of the last byte of the link protocol for this command is completed.

For more information on reset processing, including timing requirements for this type reset; see section 0, POWER-ON RESET (POR) / RESET PROCESSING.

1.11.16Query Debug Message (QDM)

Op Code	Mnemonic	Command Name
0x16	QDM	Query Debug Message

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.19

Description:

This command provides a means for the device to report specific device state & status information (ie, state of sensors, value of state control variable, detailed error code, etc) to the printer when an Intervention Required or Equipment Check condition occurs. This information can be displayed on the printer's control panel for debug purposes.

The printer will send this query to the device that has raised the Intervention Required or Equipment Check condition to learn more "engineering" details of the device's condition. The printer may also send this command to devices that did not raise the IR or equipment check condition, to learn more "engineering" details of the device.

The printer will allow the print engine the ability to "scroll" thru all the devices; thus, displaying each of the devices' Debug Messages whenever an intervention required or equipment check condition occurs; even if they aren't the device identifying the error condition.

1.11.17Query Aux Power Status (QAPS)

Op Code	Mnemonic	Command Name
0x08	QAPS	Query Aux Power Status

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.22

Description:

The definition of an auxiliary power supply is a power supply source that a device may have other than the printer's power supply or its own that is used to power its system card that communicates to the Engine. For example: a device's separate auxiliary power supply to provide power to the device's motors.

This command provides a means for the device to report the status of its auxiliary power supply; ie, on or off to the Engine.

Devices with their own separate power supply that also provides power to its system card that communicates with the Engine, must comply with section 8.0, *Power Supply - Device*.

1.11.18Query Operation Complete (QOC)

Op Code	Mnemonic	Command Name
0x09	QOC	Query Operation Complete

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.23

Description:

Some devices must perform a "finishing" operation after the device has reported *Page In Output* via a response to the QDS1 command and *Print Complete* via a response to the QPCID command. The Engine uses this command to query the device for completion of its "finishing" operation. Only after the "finishing" operation is completed by the device and after all non-critical status changes associated with the "finishing" operation has been detected, reported to the Engine, and the Engine received them; can the device report back to this query that the *Operation Complete* (ie, the "finishing" operation is complete).

For Example:
A mailbox option with a stapler feature may accumulate the media to staple it, in which the device reports the "print complete" protocol based on the individual sheet reaching the accumulator / stapling staging area. But the printer system needs to know when the media is "finally" stapled and deposited into its final destination (mailbox bin), and if any non-critical status changes for the "finishing" operation has occurred before issuing other print jobs to this "finishing" operation. Thus, only after the "finishing" operation is completed and after all non-critical status changes associated with the "finishing" operation has been checked, and any changes in status reported back to the printer system, can the device respond *Operation Complete* to this query. Using this accumulating to staple example: a possible non-critical status change may be detected via the "bin full" sensor once the stapled media is deposited into the output bin. If the status of "bin full" has not changed then *Operation Complete* is reported back at that point. However, if the device has detected the "bin full" status has changed; ie, from not full to full, the device must report this non-critical status change to the Engine and the engine receive the status change before responding *Operation Complete* to this query command.

The printer also uses this query to determine if the device was able to execute the "finishing operation" successfully or not. And if not, what was the reason the "finishing operation" was not performed successfully. For example: a mailbox option with a stapler - the printer sends this command not just to learn with the device has finished its operation, but also if the stapling was performed successfully and if not, why the print job was not stapled.

1.11.19Query Nvram Read / Write Complete (QNRWC)

Op Code	Mnemonic	Command Name
0x0F	QNRWC	Query Nvram Read / Write Complete

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.24

Description:

After the engine has requested the device to read / write its nonvolatile memory , it must query the device to determine when the requested read / write operation has been completed by the device. This is accomplished by this command. For more information with respect to the protocol for reading / writing a device's nonvolatile memory, refer to the **Read Device's Nvram** command and **Write Device's Nvram** command sections respectively.

If this command is received by a device that does not contain nonvolatile memory available to the printer system should respond with a *Paperport II Unsupported Command Error*.

1.11.20Query HW IR (QHWIR)

Op Code	Mnemonic	Command Name
0x29	QHWIR	Query HW IR

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.16.6.3

Description:

The engine sends the device this command whenever a HW IR condition has been raised via the Query IR Absolute Immediate Stop response to learn of the specific hardware IR condition being raised by the device. The device's response is device dependent & specific.

Note: Whenever this command is sent to a device to determine a specific HW IR condition, the engine should also send the device the **Query Paper Position Status** command to learn of any sheets that may be in the device's paperpath, which may need to be cleared, as a result of the particular HW IR condition raised.

1.11.21Query Media Sensor Calibration Status (QMSCS)

Op Code	Mnemonic	Command Name
0x40	QMSCS	Query Media Sensor Calibration Status

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response Syntax: Single Byte NOT Bit Encoded

Response Values:
0x00 – Calibration Needed
0x01 – Calibration in Progress
0x02 – Calibration Done
0x03 – Calibration Failed
0x04 – 0xFF Reserved

Description:

The print engine will send this command following a **ICP**, **Initiate Calibration Process**, command. Immediately after receiving the **ICP** command, the device should set its calibration status to "Calibration in Progress". This should remain the calibration status until the calibration process is complete. Once the calibration process is complete, the device should set its calibration status to either "Calibration Done" or "Calibration Failed" to indicate that the calibration process was successful or unsuccessful, respectively. "Calibration Needed" should be used as the POR default calibration status. The device should maintain the "Calibration Needed" status until either the device determines that it has been calibrated or the print engine sends the **ICP** command. If the device receives a **SCR**, **System Configuration Reset**, command it should reset its calibration status to "Calibration Needed." The device should maintain the "Calibration Needed" status until either the device determines that it has been calibrated or the print engine sends the **ICP** command. The device need NOT change its calibration status for **SR**, **HR**, or **IRC** commands.

1.11.22Query Dot XX Error Code (QDXX)

Op Code	Mnemonic	Command Name
0x99	QDXX	Query Dot XX Error Code

Command Structure: OpCode
0x99

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.27

Description:

During an IR condition, the printer will send this command to determine what "dot xx" code to append to the main error code that will be displayed on the op-panel. This is of the form "YYY.XX". For mono printers the engine code should only send this command if the option's paperport compatibility level is 12 or greater (Compatibility Level >= 12).

STEADY-STATE OPERATION COMMANDS

These are the commands used by the master (ie, print engine or printer) to instruct the devices how to control the picking, feeding, and output of paper during normal operation; ie, "steady state".

1.11.23Select Input Device (SID)

Op Code	Mnemonic	Command Name
0x85	SID	Select Input Device

Command Structure: OpCode, Length Field, N1
0x85, 02, N1
- or -
0x85, 01

Parameters:

N1	Input Tray #	
	00	For devices without input trays (ie, duplex option)
	01 - FF	Specific Bin # to Feed Media From

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used by the printer to select a device for inputting the next media sheet. The parameter byte, N1, identifies the specific bin or tray the media is to be picked from. If the selected input device does not have the tray # specified, the device should raise an *Invalid Paperport Protocol - IR* condition.

This command must be issued before the **Pick Page** or **Pick Page At Set Gap** command.

The printer must select input and output sources for a sheet's physical side. Thus, for Input/Output devices, such as a duplex option, the printer would first select the input source as a tray and the output source as the duplexer for side #1 and then select the input source as the duplexer and the output source as the output stacker for side #2. After issuing the input source = duplexer for side #2, the printer would poll the duplexer for **Input Source Ready** to know that the sheet had reached the duplexer's stage 2 point and is ready to be feed into the Paper Feed "Pass Thru" path; ie, now the printer can issue the **Pick Page** command to the duplexer for side #2.

For Input devices with only one input tray and for Input & Output devices such as the duplex option, the printer does not have to specify an input tray, N1; ie, OpCode, 01 is acceptable (and preferred to minimize serial communications bus traffic). However, these devices should also support OpCode, 02, 00 as an acceptable standard selection.

1.11.24Select Output Device (SOD)

Op Code	Mnemonic	Command Name
0x86	SOD	Select Output Device

Command Structure: OpCode, Length Field, N1, N2, N3
0x86, 04, N1, N2, N3

Parameters:

N1 Output Bin #
00 - FF Specific Output Bin Output is to be Feed Into

N2 Page Identification Tag
00 Nop, Reserved
01 - FF Page ID Associated with the Media to be stored into Output Bin # (N1 above)

N3 Media Type
Bit Encoded (Bit 7 MSB)
Bit 7-4 Media Type
0000 Paper
0001 Labels
0010 Envelope
0011 Card Stock
0100 Transparencies
0101 - 0111 Unused
1000 - 1111 Reserved
Bit 3-2 Media Weight
00 Light
01 Normal
10 Heavy
11 Unused
Bit 1-0 Media Texture
00 Smooth
01 Normal
10 Rough
11 Unused

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command selects the output device for the next sheet. The parameter byte N1 specifies the output destination (ie, bin) within the selected output device. The parameter byte N2 specifies the page identification tag to be associated with this page (see **Query Page Complete ID** command and response for details). The parameter byte N3 specifies the "media type" of the sheet the output device is to receive.

The printer must send this command prior to the **Divert Media Into Device** and **Device Controls Media Notification** commands. Normally, the printer should try and send this command prior to the **Page At Printer's Input Sensor** notification to allow the output device as much time as possible to prepare for the incoming media.

The printer must select input and output sources for a sheet's physical side. Thus, for Input/Output devices, such as a duplex option, the printer would first select the input source as a tray and the output source as the duplexer for side #1 and then select the input source as the duplexer and the output source as the output stacker for side #2.

For devices that have their own paper path input deflector for an "output pass thru paper path" (such as a modular output stacker), the printer will ensure this command is sent before the sheet's leading edge reaches the device's diverter and "output paper path's pass thru sensor" when they are the device of which the sheet is the final output destination. Note: the **Select Output Device** command will Not be sent to an output device that is not the final output destination for the media; ie, output devices that may need to be instructed by the printer to position their paper path input deflector to divert the media away or into the "output pass thru paper path" will Not receive the **Select Output Device** (since they are not the final output destination for that sheet).

1.11.25Jog Output Tray (JOT)

Op Code	Mnemonic	Command Name
0x17	JOT	Jog Output Tray

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is sent to the output device to have it "jog" its output stack prior to receiving subsequent output.

The printer must send this command after the device has been selected as the destination for output; ie, after the **Select Output Device** command; but prior to the **Divert Media Into Device**. The

device will not jog its output stack until processing the sheet associated with this **Select Output Device** command.

Output devices that do not support tray jogging will process this command even though they can not jog their output stack.

Note: At this time neither the printer, nor any device has identified a need for this command. Implementation is not required to meet current compatibility levels.

TEMPORARY : For test purposes, this command can be used to alternate between output tray offset positions of the HCOF.

1.11.26Device Operation Command (DOC)

Op Code	Mnemonic	Command Name
0x92	DOC	Device Operation Command

Command Structure:

OpCode, Length Field, N1	(Compatibility Level < 24)
0x92, 02, N1	(Compatibility Level < 24)
OpCode, Length Field, N1, N2	(Compatibility Level >= 24)
0x92, 03, N1, N2	(Compatibility Level >= 24)

Parameters:

N1 Operation Parameter

- 0x00 No Operation - Reserved
- 0x01 Flush Physical Paperpath
- 0x02 Staple (See N2 if multiple patterns supported)
- 0x03 Prime Stapler
- 0x04 Home Stapler
- 0x05 Cut Media
- 0x06 Start Loading Media
- 0x07 End Loading Media
- 0x08 Alternate Cut Media
- 0x09 - 0xFF Reserved

N2 Staple Pattern (Compatibility Level >= 24)

- 0x00 No Staple
- 0x01 Top Left (aka Portrait Single)
- 0x02 Top Dual (aka Portrait Dual Top)
- 0x03 Left Dual (aka Portrait Dual Left)
- 0x11 Top Right (aka Landscape Single)
- 0x12 (previously Landscape Dual Top)
- 0x13 (previously Landscape Dual Left)
- 0x14 Top Left Diagonal
- 0x15 Top Right Diagonal

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used to force various finishing operations on the sheets accumulated in the stacker / finisher's accumulator in the case of output devices and various paper handling operations in input devices.

Rules():

- *When the "Flush Physical Paperpath" parameter is received, the device will flush the sheets in the accumulator to the output tray positioned at the offset position sent with the Submit Page Command.*
- *When the "Flush Physical Paperpath" parameter is received, the device will ignore the command if no media is in the accumulator.*
- *When the "Flush Physical Paperpath" parameter is received and no Submit Pages have been received prior, the device will flush to its default position. EX: POR and media in the accumulator. If the RIP wishes to flush, the device will flush to its default position.*
- *When the "Staple" parameter is received, the device will staple the sheets in the accumulator to the staple position sent with the Submit Page Command.*
- *When the "Staple" parameter is received, the device will ignore the command if no media is in the stapler throat.*
- *When the "Staple" parameter is received and no Submit Pages have been sent prior, the device will staple in the position specified in the DOC command.*
- *When the "Home Stapler" is received, the device will perform any homing necessary for the Stapler. NOTE: This device will home with full power. See "Stapler Power Levels" section for more details.*
- *When the DOC command can be sent:*
  1. *If the device is processing pages, the DOC command is sent after the device raises "Page Compete" for the sheet that is to have the finishing operation performed on it.*
  2. *The DOC command can be sent when the device is Idle. I.E. No pages have been submitted, device is not performing any tasks; such as, Reset Processing, Diagnostic Tests, etc. The Query Device Quiesce (QDQ) command can be sent to guarantee the device is Idle.*

- *After the DOC command is sent, the engine must query for operation complete. See "Query Operation Complete" section for more details.*

1.11.27Select Process Speed (SPS)

Op Code	Mnemonic	Command Name
0x87	SPS	Select Process Speed

Command Structure: OpCode, Length Field, N1, N2, ... Nm

Parameters: N1 Process Speed Selection

FE	Output Devices (such as stackers)
FF	Input Devices (such as trays)

N2,N3 to define a specific machine transport speed
N2 = Most significant byte
N3 = Least significant byte
Default Units: LSB=0.0625 (1/16) mm/sec, approx 0.0025 "/sec

Valid Range - Two byte unsigned integer
N2,N3 = 0x0000 Reserved
N2,N3 = 0x0001 to 0xFFFF Range

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

This command is used by the printer to communicate to the device the printer's process speed (ie, the speed at which the media moves through the printer's paper path and the Paper Feed "Pass Thru" path). Whenever the device is feeding the media that is also under the printer's control; ie, in both the device's and the printer's feed rollers, the device must feed the media at this process speed to avoid damage to the media

Note some printers support multiple process speeds (for example: one speed for 300/600 dpi printing and another for 1200 dpi printing). The device must be capable of supporting each of the printer's process speeds. Also, some devices are to be common to a printer product family of which throughout the product family several process speeds are used. In that case, the device must support the complete set of process speeds for the product family.

After a power-on reset or receiving a **System Configuration Reset**, **Hard Reset** or **Soft Reset** command, if the device is instructed by the printer to feed media and has not received the **Select Process Speed** command, it should default to its power-on reset default.

Notes:

1. Before the printer can change the process speed a device is to use, the printer must ensure the printer system is idle; ie, no devices are currently or about to process media. The printer can determine this by sending the device the **Query Device Quiesce** command. If the device receives a select process speed while processing media, unpredictable results may occur.
2. In order for the printer to *broadcast* this command to all the devices at the same time, it must ensure the printer system is idle; ie, no devices are currently or about to process media. The printer can determine this by sending each device the **Query Device Quiesce** command. Otherwise unpredictable results may occur to any media in process.
3. The Engine has the responsibility to ensure that a particular device supports the speed value requested (Reference command ADSMIN and QDSMAX). To ensure completion of option actuation of the new speed value, any speed dependent queries or operations should not be processed until handshake completion indicated by Device Quiesce (reference QDQ).

1.11.28Pick Page (PP)

Op Code	Mnemonic	Command Name
0x18	PP	Pick Page

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

The printer issues this command to a device to instruct that device to pick the media and deliver it into the Paper Feed "Pass Thru" path and to the first set of feed rollers. The device should feed the media at the same process speed the printer is operating the Paper Feed "Pass Thru" path when it enters the nip of these rollers. The device is to feed the media's leading edge beyond the nip of the Paper Feed "Pass Thru" feed rollers a distance to ensure the printer can successfully feed the media on through the printer system.

For devices with multiple sources ONLY, the printer is required to select this device as the input source via the **Select Input Device** command before issuing a **Pick Page** command. If not, the device will return an Intervention Required Immediate Stop general status as an Equipment Check = Invalid Command Protocol.

The printer is required to select the process speed the device is to match via the **Select Process Speed** before issuing a **Pick Page** command. If the device does not receive a **Select Process Speed** command prior to the **Pick Page** command, it will use the process speed specified by the most recently received **Select Process Speed** command. If the device had never received a **Select Process Speed** command, then it will use its power-on reset default process speed.

1.11.29Divert Media Into Device (DMID)

Op Code	Mnemonic	Command Name
0x19	DMID	Divert Media Into Device

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

The printer issues this command to a device to notify it that it can now position the printer's paper path's exit deflector or its own paper path's input deflector to route media into the device or the device's "output pass thru paperpath". This deflector movement must occur during the inter-page gap (ie, the time & space between 2 sheets being processed by the printer). The printer will issue this command at the beginning of the inter-page gap.

For devices of which the media is diverted into an "output pass thru paperpath" (such as a modular output stacker) and not directly into the device itself, the printer will explicitly notify the device when to restore the deflector via the **Divert Media Not Into Device** command.

For devices of which the media is diverted directly into the output device (such as the duplex option), there is no specific command instructing the device when to return the deflector back to its original position (ie, the position prior to receiving this command). The device is required to return the deflector implicitly at the trailing edge of receiving the sheet into itself.

1.11.30Device Controls Media Notification (DCMN)

Op Code	Mnemonic	Command Name
0x1A	DCMN	Device Controls Media Notification

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This notification is sent by the printer to a device when the trailing edge of a sheet is no longer in any of the printer's feed rollers; ie, the sheet is at a point it is under total control of the device. This allows the device to alter the sheet's transport speed, etc as needed for that particular device. For example: the duplex option accelerates the media to a speed greater than the printer's process speed once the media is internal to the duplex option in an attempt to get the media to its pick point as quickly as possible to enhance duplex printing performance. Likewise, an output device with a long paper path may need to accelerate the media to a faster speed to improve performance; ie, shorten the time in which it can report back to the printer that the page is safely & successfully in the designated output bin (ie, "print complete").

1.11.31Page At Printer's Input Sensor Notification (PISN)

Op Code	Mnemonic	Command Name
0x1B	PISN	Page At Printer's Input Sensor Notification

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

This notification is sent by the printer to a device when the media previously picked arrives at the printer's input sensor (ie, the leading edge of the sheet is at the printer's input sensor).

At this time, this command has no particular control purpose, but has been added to allow monitoring the communications bus and track a sheet's movement through the printer system's paperpath.

1.11.32Begin Communications Session (BCS)

Op Code	Mnemonic	Command Name
0x1C	BCS	Begin Communications Session

DELETED – Reserving Option to Restore this Command in the Future

1.11.33End Communications Session (ECS)

Op Code	Mnemonic	Command Name
0x1D	ECS	End Communications Session

DELETED – Reserving Option to Restore this Command in the Future

1.11.34Set Interpage Gap (SIG)

Op Code	Mnemonic	Command Name
0x24	SIG	Set Interpage Gap

DELETED – Reserving Option to Restore this Command in the Future. Perhaps this command could be useful when considering smaller gaps at higher pick rates where paperport latency becomes a factor in pick timing.

1.11.35Pick Page At Set Gap (PPASG)

Op Code	Mnemonic	Command Name
0x23	PPASG	Pick Page At Set Gap

DELETED – Reserving Option to Restore this Command in the Future. Perhaps this command could be useful when considering smaller gaps at higher pick rates where paperport latency becomes a factor in pick timing.

1.11.36Pick Retry (PR)

Op Code	Mnemonic	Command Name
0x28	PR	Pick Retry

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used by the engine to identify to an input device that has failed to pick the media and/or feed it to a particular point in the printer's paperpath to request the device to try and pick the sheet again.

The engine will only issue the **Pick Retry** command for the Input Source selected and after a failed pick from a **Pick Page** or **Pick Page At Set Gap** command.

The engine will wait until the device issues *Input Source Ready* via the **Query Device Status 1** response for the initial pick page before issuing a **Pick Retry** command. The engine is not to reselect the input source via the **Select Input Source** command before issuing the **Pick Retry** command.

Any picks of additional sheets following a successful feed from a **Pick Retry** command must use the **Pick Page** command.

Multiple "pick retries" are allowed by the engine, but subsequent **Pick Retry** commands must only be after a previous **Pick Retry** command utilizing the same protocol described above.

Input devices that receives this command that can not support a pick retry due to mechanical, technology, etc limitations; should accept this command and perform no operation for the command. Any device that is not an input device that receives this command should raise a Paperport Invalid Protocol error.

1.11.37Query Device Measured Inter-Page Gap (QDMIG)

Op Code	Mnemonic	Command Name
0x2E	QDMIQ	Query Device Measured Interpage Gap

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Description:

The printer issues this command to a device to query the last measured logical interpage gap at the device’s first paper-path sensor the sheet makes when it’s is moving downstream.

Response:

Syntax: Single Byte

Value: 0x00 – 255
Unit: milliseconds

1.11.38Cancel / Flush (Logical) Page Queue (CFPQ)

Op Code	Mnemonic	Command Name
0x07	CFPQ	Cancel / Flush (Logical) Page Queue

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

Some devices due to their paperpath need to queue up the sheets intended for that device prior to them actually being picked (for example: Mailbox option). These sheets are referred to as "logical" sheets. For these devices, at times the engine needs to "reset" the logical paperpath to being empty. Thus the engine sends this command to the device to instruct the device to cancel / flush (ie, remove) all the logical sheets from its page queue. For devices that do not queue up sheets, it should ignore this command.

Protocol:

While the device is canceling / flushing its logical page queue, it "holds off" receiving any additional logical sheets by reporting back an inter-page gap of 0xFFFF for any **Query Device Minimum Interpage Gap** command. Once the device has completed the cancel / flush of its page queue, it then calculates the gap needed to accept the sheet of any requested **Query Device Minimum Interpage Gap** command.

1.11.39Query Device Minimum Speed (QDSMIN)

Op Code	Mnemonic	Command Name
0x2A	QDSMIN	Query Device Minimum Speed

Command Structure: OpCode, Length Field, N1 ...

Link Pacing: Engine to Device: Standard Communications Syntax

Response: Length, R1, R2
Length = 3
R1 = High order byte of Minimum Speed
R2 = Low order byte of Minimum Speed

Description: Command to query device as to its minimum supported process speed.
Default Units: LSB=0.0625(1/16)mm/sec, approx 0.0025 "/sec

1.11.40Query Device Maximum Speed (QDSMAX)

Op Code	Mnemonic	Command Name
0x2B	QDSMAX	Query Device Maximum Speed

Command Structure: OpCode, Length Field, N1 ...

Link Pacing: Engine to Device: Standard Communications Syntax

Response: Length, R1, R2
Length = 3
R1 = High order byte of Maximum Speed
R2 = Low order byte of Maximum Speed

Description: Command to query device as to its maximum supported process speed.
Default Units: LSB=0.0625(1/16)mm/sec, approx 0.0025 "/sec

1.11.41Page At Printer's Exit Sensor Notification (PEXSN)

Op Code	Mnemonic	Command Name
0x2C	PEXSN	Page At Printer's Exit Sensor Notification

Command Structure: OpCode

Link Pacing: Engine to Device: Standard Communications Syntax

Response: *None*

Description: This command is sent by the engine to identify to a device where along the printer system's paperpath an incoming sheet to that device is located. The actual position of the incoming sheet's leading dege is given by the PEXSN Offset parameter provided with the PCN command (see the PCN command for more details).

1.11.42Query Pick Time Error (QPTE)

Op Code	Mnemonic	Command Name
0x2F	QPTE	Query Pick Time Error

Compatibility Levels: 22 and above

Command Structure: OpCode

Link Pacing: Engine to Device: Standard Communications Syntax

Response: Length, R1, R2
Length = 3
R1 = High order byte of Pick Time Error
R2 = Low order byte of Pick Time Error
Units: milliseconds
Range: 0x8001 to 0x7FFF (-32,767 ms to 32,767 ms)

Description:

This command queries a time difference in milliseconds from a source tray option. This command queries the difference between the pick time the source tray actually measured and the pick time reported to the engine in the previous QDPT. This command sends a signed quantity. A negative number means the page is ahead of where it should be. This command is used when source tray does not have the capability to correct for the pick time error itself but must rely on the tray above to do so. If the pick time error is unavailable at the time it is queried by the engine the option may respond with 0x8000 indicating to the engine that it may try again later.

1.11.43Select Input Device eXtended (SIDX)

Op Code	Mnemonic	Command Name
0xB3	SIDX	Select Input Device eXtended to include media type

Command Structure: 0xB3, 0x03, N1, N2 (Compatibility Level 1)
0xB3, 0x02, N1

Parameters: N1 Media Type (Lower Nibble)

0x0	Plain Paper	
0x1	Transparency	(Compatibility Level 0)
0x1	Labels	(Compatibility Level 1)
0x2	Envelope	(Compatibility Level 1)
0x3	Card Stock	(Compatibility Level 1)
0x4	Transparency	(Compatibility Level 1)0x5
Glossy Paper	(Compatibility Level 1)	
0x5	Vinyl Labels	
0x6	Cotton	
0x7	Glossy	(Cerberus Engines)
0x8 - 0xE	Reserved for Media Type Identifiers	
0xF	Universal Media Type	

Media Weight (Upper Nibble)

0x0	Light Weight
0x1	Normal Weight
0x2	Heavy Weight
0x3 – 0xE	Reserved for Media Weight identifiers
0xF	Universal Media Weight

NOTE: As of 7/1/2012 Media Weight is not sent (i.e. always 0x0) except in the case of Plain media type with Heavy weight on select engines. Future engines and options should support sending and receiving both media type and weight in N1.

N2 Input Tray # (Compatibility Level 1)

0x00	For devices without input trays (i.e. duplex option)
0x01 – 0xFF	Specific Bin # from which to pick media

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command performs an identical function to SID and in addition notifies the device of the media type that it is to process. The parameter byte N1 identifies the specific media type to be processed. If an input option is capable of detecting the media type of which it is notified and determines that it has a different media type, the input option should not pick. Instead, the device should set the Media Type Mismatch IR condition.

If an input option is notified that it is to process the universal type, the input option should pick the media no matter what type it senses. In the case where an option performs a action based on media type, the option should perform its default action when the universal type is received.

The media type specified by this command will be remembered by the option until one of the following events occur:

- A new SIDX is received. (The new media type specified by the SIDX takes effect.)
- A SID is received. (The media type is set to Universal.)
- A SCR, SR, HR, or IRC is received. (The media type is set to Universal.)

1.11.44Pass-Through Control (PTC)

Op Code	Mnemonic	Command Name
0xA4	PTC	Pass-Through Control

Command Structure: OpCode, 0x05, N1, N2, N3, N4

Link Pacing:
- Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Parameters:

N1 Direction
This byte is unused. It is included for forward compatibility with future function.

N2 Motor
This byte is unused. It is included for forward compatibility with future function.

N3,N4 to define a specific machine transport speed
N3 = Most significant byte
N4 = Least significant byte
Default Units: LSB=0.0625 (1/16) mm/sec, approx 0.0025 "/sec

Response: None

Description:

The printer sends this command to instruct a device to enable its pass-through control. This command is used on selected options to replace the staging tracking function used in previous color paper feed options. When the option received this command, it will either activate the pass-through rollers at the given speed, or de-activate the pass-through rollers.

1.11.45Image Location Notification (ILN)

Op Code	Mnemonic	Command Name
0xB5	ILN	Image Location Notification

Command Structure: *0xB5*, 0x03, N1, N2

Parameters: N1 - High Byte of location of Image from Second Transfer
N2 – Low Byte of location of Image from Second Transfer
Units: 1/256 mm

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command sends a distance in 1/256 mm precision to the tray option. This command tells the option the location of the image when the printer's first paper path sensor is made. The location indicates the distance the image is away from the second transfer at the time when the printer's first paper path sensor was made.

1.11.46Set Control Parameters (SCP)

Op Code	Mnemonic	Command Name
0xB8	SCP	Set Control Parameters

Command Structure: OpCode, Length Field, N1, N2, …

Parameters: N1 Subcommand
Values: 0x00 = Set Pick Parameters.
0x01 = Set Cut Adjust (left/back)
0x02 = Set Cut Adjust (right/front)

0x02 - 0xFF = Reserved

Subcommand: Set Pick Parameters.

Subcommand Parameters: N2

N2, to define the value of the pick distance.
Default Units: millimeters

Valid Range - One byte unsigned integer
N2 = 0 to 255 Range

N3 (Compatibility level 20 or greater)
N3, to determine whether the option can use the feed path sensor a.k.a trailing edge sensor to determine when to stop the pick.
0x00 The use feed path sensor in the pick sequence is not allowed.
0x01 The use feed path sensor in the pick sequence is allowed.

Subcommand: Set Cut Adjust
Subcommand Parameters: N2, N3, N4, N5

N2, N3, N4, N5, defines the cut adjust. (N2 = MSB,,,N5=LSB)
Units: millimeters*128

Valid Range - Four byte signed integer
N2, N3, N4, N5 = 0x00000000 to 0xffffffff Range

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

Set Pick Distance:

This command is to be used by the engine to communicate to paper-feed options the distance to feed beyond the device exit point when picking i.e. **Pick Page** command is received or retrying a pick i.e. **Pick Retry** command is received. The option should use the most recent pick distance set by the engine. If new pick distance is set while a pick is in progress the option should use the new pick distance immediately and if the newly received pick distance is less than the distance fed since the pick started the option should stop the pick as soon as possible.

After a power-on reset or receiving a **System Configuration Reset**, **Hard Reset** or **Soft Reset** command, if the device is commanded by the engine to pick and has not received the ***Set Pick Distance*** subcommand, the device should feed to its exit point.

Set Cut Adjust:

This command is to be used by the engine to communicate to paper-feed options the distance to adjust the normal cut location (For Options that support cutting media). Positive values will move the cut farther down the page from the reference location. The engine will send this command at power-on-reset (POR) or when changed by the RIP or User or Engine so the option is not required to save it through any power cycle (hot plug, POR, hibernate…)

There are now two "Cut Adjust" values: the original which has been renamed (Left/back) and a new right/front value. "Cut Adjust Right", when used in conjunction with Left/back, allows the adjustment of skew in addition to the normal cut adjustment. Both values should be the same except when trying to adjust cut skew.

1.11.47Get Control Parameters (GCP)

Op Code	Mnemonic	Command Name
0xB9	GCP	Get Control Parameters

Command Structure: OpCode, Length Field, N1 ...

Parameters: N1 Control Parameter Requested
0x00 = Cut Error
(Units: 1/16 mm)

0x01 – 0xFF = Reserved

Link Pacing: Engine to Device: Standard Communications Syntax

Response Syntax: Multi-Byte Response Length Field, N1, N2

Response Byte Definitions:

Length Field Number of bytes in the response including the count byte
Values: Range: 0x01 to 0xFF

For cut error requests (Command data N1 = 0x00):
N1, N2 Cut error value
(Two byte signed value, N1 Most Significant Byte)
Values: 0x0000-0xFFFF

Description: Command to get general control parameters from an option device to the engine to be used in normal operation. This is currently specified to

retrieve the cut error introduced to the leading edge of the media by the cutting device mechanism from a previous cut operation.

1.11.48Pick Time Error Notification (PTEN)

Op Code	Mnemonic	Command Name
0xBA	PTEN	Pick Time Error Notification

Compatibility Levels: 22 and above

Command Structure: *0xBA*, 0x03, N1, N2

Parameters: N1 - High Byte of Pick Time Error
N2 – Low Byte of Pick Time Error
Units: milliseconds
Range: 0x8001 to 0x7FFF (-32,767 ms to 32,767 ms)

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command sends a time difference in milliseconds to the tray option. This command tells the option the difference between the pick time the source tray actually measured and the pick time reported to the engine in the previous QDPT. This command sends a signed quantity. A negative number means the page is ahead of where it should be. This command is used when source tray does not have the capability to correct for the pick time error itself but must rely on the tray above to do so.

1.11.49Query Health Check Data (QHCD)

Op Code	Mnemonic	Command Name
0xBB	QHCD	Query Health Check Data

Compatibility Levels: 23 and above

Command Structure: *0xBB*, 0x02, N1

Parameters: N1 – Health Check Test Requested

Link Pacing: - Engine to Device: Standard Communications Syntax

Response Syntax: Multi-Byte Response Length Field, N1 …Nn

Response Byte Definitions:

Length Field	Number of bytes in the response including the length byte *Values:* Range: 0x01 to 0xFF
For Length > 1 N1…Nn	Health Check Test Data (device dependent)

Description:

This command is used to query the 'Health Check' data from the option for the specified health check test requested.

If the option does not support a health check test, then it is to respond with length of 1 and no following data. No errors will be posted.

The response to a health check test is device dependent. For specific definitions, values, and meanings of the Health Check Test Data bytes; see the device dependent response.

1.11.50Query Number of Covers (QNC)

Op Code	Mnemonic	Command Name
0xBC	QNC	Query Number of Covers

Compatibility Levels: 27 and above

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response Byte Definitions:

Length Field	Number of bytes in the response including this byte *Values:* 0x01-0FF

N1 Device's Specific Covers Status (Open or Closed)

Bit 7 (MSB)	0	Cover #1	- Cover Not Available
	1		- Cover Available
Bit 6	0	Cover #2	- Cover Not Available
	1		- Cover Available
Bit 5	0	Cover #3	- Cover Not Available
	1		- Cover Available

Bit 4	0	Cover #4	- Cover Not Available
	1		- Cover Available
Bit 3	0	Cover #5	- Cover Not Available
	1		- Cover Available
Bit 2	0	Cover #6	- Cover Not Available
	1		- Cover Available
Bit 1	0	Cover #7	- Cover Not Available
	1		- Cover Available
Bit 0	0	Cover #8	- Cover Not Available
	1		- Cover Available

Description:

The printer issues this command to a device to query the device to determine the number of trays the device contains. This command is issued to both input and output devices; to input devices to learn how many input sources the device has, and to output devices to learn how many output destinations the device has. For devices that do not have input sources or output destinations, such as the duplex option, "no trays" is returned by the device.

1.12 MISC COMMANDS

1.12.1 Media Type Notification (MTN)

Op Code	Mnemonic	Command Name
0x??	MTN	Media Type Notification

DELETED – Reserving Option to Restore this Command in the Future

1.12.2 Media Size Notification (MSN)

Op Code	Mnemonic	Command Name
0x91	MSN	Media Size Notification

Command Structure: OpCode, Length Field, N1
0x91, 0x02, N1
0x91, 0x03, N1, N2

Parameters: N1 Media Size (to be received)
00 Media Size Identified in Subsequent Parameter Byte
01 - FF Media Size Identifier (see table below)

N2 Media Width in mm (only applicable if N1 = 0)

Media Size	Actual Paper Size
0x01	8.5" x 11"
0x02	8.5" x 14"
0x03	B5 (7.2" x 10.1")
0x04	A4 (8.3" x 11.7")
0x05	Exec (7.25" x 10.5")
0x06	A5 (5.83" x 8.27")
0x07	Custom
0x08	Unknown
0x09	7.5" envelope
0x0A	#9 envelope (8.9")
0x0B	#10 envelope (9.5")
0x0C	DL envelope (8.66")
0x0D	C5 envelope (9.02")
0x0E	B5 envelope (9.84")
0x0F	Other envelope
0x10	Reserved (A3)
0x11	Reserved (Ledger)
0x12	Reserved (B4)
0x13	* Statement (5.5” x 8.5”)
0x14	* Folio (8.5" x 13")
0x15	A6 (4.13" x 5.83")
0x16	Officio

* Note for legacy color finishers (C75x, C76x), statement and folio medias used enumerations 0x10 and 0x11 respectively. The values above represent the values currently used by the RIP code.

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used by the printer to notify the device of the media size that it is to process. The parameter byte is the specific paper size to be processed.

Note: It is anticipated that finisher options that perform hole-punching operations will require this command.

1.12.3 Operate Device Control Panel (ODCP)

Op Code	Mnemonic	Command Name
0x??	ODCP	Operate Device Control Panel

DELETED – Reserving Option to Restore this Command in the Future

1.12.4 Invoke Low Power State (ILPS)

Op Code	Mnemonic	Command Name
0x1E	ILPS	Invoke Low Power State

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

This command is used by the printer to instruct the device to enter its energy conservation or *Low Power State* (turn off power to optical sensors, motors, etc).

The device must still communicate with the printer, and monitor for device IR conditions (for example: cover open, etc) during the *Low Power State*.

The device is normally restored to full power via the **Restore Full Power State** command. However, after a **System Configuration Reset**, **Hard Reset**, or **Intervention Required Clear** command, the device should also restore itself back to the *Full Power State*.

1.12.5 Restore Full Power State (RFPS)

Op Code	Mnemonic	Command Name
0x1F	RFPS	Restore Full Power State

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response: None

Description:

The printer sends this command to instruct a device in the *Low Power State* to restore itself to *Full Power State* and thus be prepared to process print jobs. The device restores power to all its electrical hardware it had removed for the *Low Power State*.

The printer can send this command to a device when the device is already in the *Full Power State* and not be considered an Invalid Paperport Protocol; the device should simply remain in the *Full Power State*.

Note: When a device powers on it is to automatically be in the *Full Power State*. Also, whenever a device receives a **System Configuration Reset, Hard Reset** or **Intervention Required Clear** command it should restore itself to the *Full Power State*.

1.12.6 Tray Lock Notification (TLN)

Op Code	Mnemonic	Command Name
0x88	TLN	Tray Lock Notification

DELETED – Reserving Option to Restore this Command in the Future

1.12.7 Divert Media Not Into Device (DMNID)

Op Code	Mnemonic	Command Name
0x20	DMNID	Divert Media Not Into Device

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

The printer issues this command to a device to instruct it to position the printer's paper path's exit deflector or its own paper path's input deflector such that media will not enter into the device.

Devices With An "Output Pass Thru Paperpath":

This command is sent from the printer to a device (such as a modular output stacker) to notify the output device to position the printer's paper path's exit deflector or its own paper path's input deflector such to route media Not into the device's "output pass thru paperpath" (ie, the media will be routed to the device below this device).

Devices Without An "Output Pass Thru Paperpath":

This command is not normally sent as a part of normal steady state processing since for these devices they are to implicitly restore the deflector at the trailing edge of the interpage gap. However, a few abnormal situations arise that require the printer to ensure its paper path's exit deflector or the device's paper path's input deflector is in the "restored" position (ie, positioned such the media will Not be routed into the device's paper path). Such an example is an "Invalid Manual Feed" condition. The printer will only send this command to these type devices when the "abnormal" sheet of which the printer wants to route to its output bin leading edge has reached the printer's exit sensor; ie, at the leading edge of the Interpage Gap. When the device receives this command, it restores the deflector to its normal position (ie, not to receive the media).

Note : For side-mounted options such as the HCOS and HCOF, it is anticipated that the engine WILL issue the DMNID command as necessary.

1.12.8 Communications Error Response (CER)

Op Code	Mnemonic	Command Name
0xFF	CER	Communications Error Response

Description:

This op code is reserved since it is the quiese state of the communications bus. Note: The printer will never send this command.

1.12.9 Device Maintenance Operation (DMO)

Op Code	Mnemonic	Command Name
0x90	DMO	Device Maintenance Operation

Command Structure: OpCode, Length Field, N1, N2
0x90, 0x03, N1, N2

Parameters:

N1 Input / Output Tray #
0x00 For devices without trays (example: duplex option)
0x01 - 0xFF Specific Bin # (Input or Output)

N2 Maintenance Operation
0x00 - 0xFF see below

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

The print engine uses this command to instruct a device to perform some routine maintenance required by the device. The print engine can also send this command to instruct a device to terminate a previous requested maintenance operation. The control of execution of the device's maintenance operations is given to the print engine to allow the print engine to control when such device maintenance activities are conducted (for task management, for energy management, etc).

The Input / Output Bin parameter byte, N1, identifies the specific bin or tray the device is to perform the maintenance operation. For input or output devices that only have one tray (or bin), the print engine requests maintenance on that tray (or bin) by sending N1 = 0x00 or 0x01. For input & output devices or devices that do not have a tray (or bin), that require routine maintenance, the print engine sends N1 = 0x00. If the selected device does not have the tray # specified, the device should raise an *Invalid Paperport Protocol - IR* condition.

The printer can send this command to a device at any time. If the device is unable to perform the requested maintenance for some reason (for example: the device is in its IR state, or the device's *Input Source Ready* is negated; ie, the device is busy, etc), the device performs No Operation.

The print engine determines when a device has completed its execution of the maintenance operation requested by polling the device with the **Query Device Quiesce** command. During the execution of the maintenance test, the device's response to this query is *device mechanism not quiesce.* Once the device has completed executing the maintenance operation requested and all moving parts are stopped, it responds to this query with *device mechanism quiesce* to notify the print engine that the maintenance test has been completed. If the device detects any Intervention Required conditions or any Equipment Check conditions during the execution of the requested maintenance operation, the device terminates the maintenance operation or continues it to completion depending upon what is best for the specific device and technology. The device reports the Intervention Required or Equipment Check condition to the print engine via the **Query Device Status 1** response.

It is the device's responsibility to determine under what abnormal conditions (if any) it should terminate the requested maintenance. For example: an input option may choose to terminate its requested maintenance if it detects the present of its input tray (ie, it has been re-installed for a maintenance operation that requires it to be removed). For details of the specifics with regards to abnormal termination of requested maintenance operations, refer to the individual device's functional spec.

A more detailed description of the Maintenance Operation parameter, N2 is given below:

Value	Name	Device's Action
N2 = 0x00	Reserved	No Maintenance Operation Performed
N2 = 0x01	Standard Clean Operation	Device performs their Standard cleaning operation. For details including timings, see specific device's functional specification.

Value	Name	Device's Action
N2 = 0x02	Extended Clean Operation	Device performs their Extended cleaning operation (generally longer and more through). for details including timings, see specific device's functional specification.
N2 = 0x03 - 0xFD		No Maintenance Operation Performed
N2 = 0xFE	Cancel Maintenance Operation	Device cancels ALL maintenance operation for the logical tray specified by parameter N1.
N2 = 0xFF	Maintenance Operation to be Performed Defined in Next Byte, Nm	The specific maintenance operation the print engine is requesting is contained in a subsequent Nm byte.

1.12.10Special Development Command (SDC)

Op Code	Mnemonic	Command Name
0xFE	SDC	Special Development Command

Command Structure: OpCode, Length Field, N1, N2, ... N253

Parameters: N1, N2, ... N253 Range: 0x00 - 0xFF
Definition: TBD by individual device

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: Device Dependent

Description:

This command is ***reserved*** for development engineering purposes only. Its intent is to allow a means for development engineers to embed needed functions within a device's microcode for development engineering purposes (ie, to aid in engineering experiments, testing, code downloads, etc) of which invocation of that special development engineering function is controlled by this command.

Specific purposes and uses of this command is device and development engineering specific. Therefore, consult the device's specifications for details of the syntax and usage of this command by a particular device.

Warning:
The Paperport II specification will not attempt to control this command and ensure that every device supports and/or complies with some other device's definition and use of this command. Therefore, it is recommended that this command be removed from a device for production.

SDC Command User's Request:
To assist in all development engineers and device programmers to have some level of understanding of the use of this command by the various development engineers and device programmers, it is

requested that any user of this command provide a Description, including syntax, and purpose of their use of this command to the Paperport II specification control board. This application note will be added to the "SDC Application Document" to allow others visibility of your application of the **Special Development Command**.

Note: This command is intended exclusively for use by manufacturing or development of the Paperport Devices. The command should not be incorporated within an Engine code formal release. Implementation of this command is solely at the discretion of the respective Device.

Exception Usage:

Although the SDC command is intended for development purposes only, the following exception has been adapted to support improved gap performance while maintaining compatibility with original level devices: the N1 parameter field value of 0x31 (Request Actual Pick Feed Time Cmd) is required of all ***Mono*** Smart Option Input Trays.

Command Op Code		0xFE	
Length Field		0x03	
Parameters	N1	0x31	Request Actual Pick Feed Time Cmd
	N2	0x00	For Input devices without a Tray
		0x01-FF	Input Tray Bin Number
Response			
Length Field		0x03	
N1,N2	0x0000-0xFFFF		Selected Device's Feed Time in milliseconds (N1 is the Most Significant Byte)

This command is intended to provide information similar to the Query Device Pick Time(QDPT) cmd, Paperport Spec 1.11.10 / 1.15.15. The response to this SDC is to be the actual last measured time between the actuation of the pick motor and detection of the pass-through sensor for that respective paper feed while the QDPT response is a algorithmic determination of the projected feed time, based on potentially several prior measurements and the requested machine speed.

The input Tray device will clear this response value after completing the SDC request, and subsequent SDC commands prior to a pick operation will report 0x00. Based on current Tray architecture it is estimated that the resolution would be on the order of 10-20 milliseconds

1.12.11Special Manufacturing Command (SMC)

Op Code	Mnemonic	Command Name
0xFD	SMC	Special Manufacturing Command

Command Structure: OpCode, Length Field, N1, N2, ... N253

Parameters: N1, N2, ... N253 Range: 0x00 - 0xFF
Definition: TBD by individual device

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is ***reserved*** for manufacturing purposes only. Its intent is to allow a means for manufacturing to embed needed functions within a device's microcode for testing the device during the manufacturing process of which invocation of that special manufacturing test / function is controlled by this command.

Specific purposes and uses of this command is device and manufacturing specific. Therefore, consult the device's specifications for details of the syntax and usage of this command by a particular device.

Warning:
The Paperport II specification will not attempt to control this command and ensure that every device supports and/or complies with some other device's definition and use of this command.

SDC Command User's Request:
To assist in all development engineers, device programmers, and manufacturing engineers to have some level of understanding of the use of this command, it is requested that any user of this command provide a Description, including syntax, and purpose of their use of this command to the Paperport II specification control board. This application note will be added to the "SMC Application Document" to allow others visibility of your application of the **Special Manufacturing Command**.

Note: This command is intended exclusively for use by manufacturing or development of the Paperport Devices. The command should not be incorporated within an Engine code formal release. Implementation of this command is solely at the discretion of the respective Device.

1.12.12Read Device's Nvram (RDN)

Op Code	Mnemonic	Command Name
0x93	RDN	Read Device's Nvram

Command Structure: OpCode, Length Field, N1, N2, N3

Parameters: N1,N2 Device's Nvram Address
Range: 0x0000 - 0xFFFF

N3 Number of Data Bytes To Be Read From Device's Nvram
Range: 0x01 - 0xFF

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used only for devices that contain a nonvolatile storage capability, such as Nvram, in which the printer system (Engine or Rip) needs to store information related to the device. For example: A mailbox option may have nvram to allow the printer system to store the printer system's mailbox names in the device's nvram.

Addressing :
The addressing is modeled the same as addressing static ram beginning at address 0x0000. Any offsets to the device's physical address, or paging, etc; the device translates the N1,N2 logical address to the device's nonvolatile memory's physical address.

A portion of the device's nonvolatile memory may be reserved for the device's use only. If the Engine attempts to read / write this "device reserved" address space, the device will raise a *Paperport II Protocol Error*. Refer to each device's functional spec for details.

Data:
The printer system is allowed to store any information in the device's nonvolatile memory in any format. The device does not examine the contents of the data; therefore, any data integrity needed by the printer system must be performed by the printer system.

The data will be retrieved in sequential order given the address (N1,N2) assigned and the number of bytes to be read (N3); ie, N1,N2 = 0x0100 with a request to read N3 = 0x04 bytes will result in data being read and returned to the printer system from Address 0x0100, 0x0101, 0x0102, 0x0103 respectively.

Protocol:
Below describes the protocol required to actually read data from the device's nonvolatile memory.

1. Engine sends **Read Device's Nvram** command with desired address & # bytes to read
2. Engine queries device for completion of the reading operation by sending the device the **Query Read / Write Nvram Complete** command until it receives a response back from the device indicating the *Nvram Read / Write Complete*. Until the device completes reading all the bytes requested, it returns a *Nvram Read / Write Not Complete* response.
3. Engine instructs the device to begin transmitting the # bytes read via the **Transmit Requested Read Data** command.
4. The device returns the data read from its nonvolatile memory (see response to **Transmitt Requested Read Data** command for details).

If this command is received by a device that does not contain nonvolatile memory available to the printer system should respond with a *Paperport II Unsupported Command Error*.

1.12.13Write Device's Nvram (WDN)

Op Code	Mnemonic	Command Name
0x94	WDN	Write Device's Nvram

Command Structure: OpCode, Length Field, N1, N2, N3, N4, ... Nm

Parameters:

N1,N2	Device's Nvram Address Range: 0x0000 - 0xFFFF
N3	Number of Data Bytes To Be Read From Device's Nvram Range: 0x01 - 0xFF
N4, ... Nm	Data Bytes To Be Written Range: 0x00 - 0xFF

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used only for devices that contain a nonvolatile storage capability, such as Nvram, in which the printer system (Engine or Rip) needs to store information related to the device. For example: A mailbox option may have nvram to allow the printer system to store the printer system's mailbox names in the device's nvram.

Addressing :
The addressing is modeled the same as addressing static ram beginning at address 0x0000. Any offsets to the device's physical address, or paging, etc; the device translates the N1,N2 logical address to the device's nonvolatile memory's physical address.

A portion of the device's nonvolatile memory may be reserved for the device's use only. If the Engine attempts to read / write this "device reserved" address space, the device will raise a *Paperport II Protocol Error*. Refer to each device's functional spec for details.

Data:

The printer system is allowed to store any information in the device's nonvolatile memory in any format. The device does not examine the contents of the data; therefore, any data integrity needed by the printer system must be performed by the printer system.

The data will be retrieved in sequential order given the address (N1,N2) assigned and the number of bytes to be read (N3); ie, N1,N2 = 0x0100 with a request to read N3 = 0x04 bytes will result in data being read and returned to the printer system from Address 0x0100, 0x0101, 0x0102, 0x0103 respectively.

Protocol:
Below describes the protocol required to actually read data from the device's nonvolatile memory.

1. Engine sends **Write Device's Nvram** command with desired address, # bytes to read followed by the data bytes.
2. Engine queries device for completion of the writing operation by sending the device the **Query Read / Write Nvram Complete** command until it receives a response back from the device indicating the *Nvram Read / Write Complete*. Until the device completes writing all the bytes requested, it returns a *Nvram Read / Write Not Complete* response.

If this command is received by a device that does not contain nonvolatile memory available to the printer system should respond with a *Paperport II Unsupported Command Error*.

Optra S Printer Implementation / Limitation Notes:

1 Due to Engine ram constraints the **Write Device's Nvram** command will only send a maximum of 4 data bytes.

1.12.14Transmit Requested Read Data (TRRD)

Op Code	Mnemonic	Command Name
0x12	TRRD	Transmit Requested Read Data

Command Structure: OpCode

Parameters: None

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see section 1.15.25

Description:

After the engine has requested data from the device's nonvolatile memory and the device has indicated it has extracted that data, the engine uses this command to instruct the device to transmit the data back to the Engine. The number of bytes transmitted back to the Engine will be the number of bytes requested to be read via the **Read Device's Nvram** command. For more information with respect to the protocol for reading a device's nonvolatile memory and the transmitting of that data, refer to the **Read Device's Nvram** command and **Transmit Requested Read Data** command sections respectively.

If this command is received by a device that does not contain nonvolatile memory available to the printer system should respond with a *Paperport II Unsupported Command Error*.

1.12.15Restore Device's Nvram Data Area (RDNDA)

Op Code	Mnemonic	Command Name
0x95	RDNDA	Restore Device's Nvram Data Area

Command Structure: OpCode, Length Field, N1

Parameters: N1 Initialization Parameter
0x00 Reset Critical Parameters
0x01 Reset Non-Critical Parameters
0x02 - 0xFF Reserved

Link Pacing: - Engine to Device: Standard Communications Syntax
- Broadcast: Broadcast Communications Syntax

Response:

Description:

Some devices contain Nvram of which a portion is allocated to the printer and another portion is allocated to the device for its use (referred to as the Device's Nvram Data Area). Whenever the printer requires the device to initialize or reset the Device's Nvram Data Area, it sends the device this command with the Initialization Parameter indicating the type reset desired.

Reset Critical Parameters

The device upon receiving this request, resets its Device's Nvram Data Area critical parameters. Critical parameters are those parameters required to operate the device properly; ie, if their contents were to become corrupted, the device would no longer function properly. An example of a critical parameter may be a parameter used by the device's system card to control the voltage value to some hardware that was set thru adjustment at the factory for that particular hardware components.

Reset Non-Critical Parameters

The device upon receiving this request, resets its Device's Nvram Data Area non-critical parameters. Non-critical parameters are those parameters the device stores in nvram to help manage the device's operation but are not critical to the device's operation; ie, if their contents were to become corrupted, the device would still operate. An example of a non-critical parameter may an error log kept by the device.

The determination of which parameters are critical parameters and which ones are non-critical parameters is device dependent and specific; therefore, consult the device's specification and documentation for a description of its Device's Nvram Data Area.

Protocol:
Below describes the protocol required to actually re-initialize or reset the Device's Nvram Data Area:

1. The engine sends the **Restore Device's Nvram Data Area** command with the desired Initialization Parameter.
2. The engine then queries the device for completion of the initialization of the Device's Nvram Data Area by sending the device the **Query Read / Write Nvram Complete** command until it receives a response back from the device indicating the *Nvram Read / Write Complete*. Until the device completes initializing all its nvram data area parameters, it returns a *Nvram Read / Write Not Complete* response.

Notes:

1. The same "rules" that exist for the Read Device's Nvram and Write Device's Nvram commands exist for this command (see those command sections for details).
2. When the device receives this command, it will only initialize its Device's Nvram Data Area. The device's nvram area allocated to the printer system will Not be initialized by the device as a result of this command. The printer system can initialize its nvram area using the **Write Device's Nvram** command.
3. Devices that do not have a Device's Nvram Data Area; ie, have nvram but it is 100% allocated as printer system nvram data area, should ignore this command when received.
4. If the device receives this command with an Initialization Parameter it does not support, the device should raise a *Paperport II Invalid Command Parameter Error*.
5. If the device does not have nvram receives this command, the device should raise a *Paperport II Unsupported Command Error*.

1.12.16Locate Output Bin (LOB)

Op Code	Mnemonic	Command Name
0x96	LOB	Locate Output Bin

Command Structure: OpCode, Length Field, N1

Parameters: N1 Locate Bin Parameter
0x00 Reserved
0x01 Locate at Output Bin #1
0x02 Locate at Output Bin #2
:
0xFF Locate at Output Bin #255

Link Pacing: - Engine to Device: Standard Communications Syntax

*Res*ponse: None

Description:

Whenever the printer desires to position the device's output bin to a particular bin location independent of a print job, for example: to allow a user to locate the device's output bins for ease of print job pick-up & removal, the printer sends the device this command.

Notes:

1. The printer must ensure the printer system's paperpath is quiesed; ie, no sheets are in the paperpath, before sending this command to the device.
2. If the device is already at the requested output bin location, the devices performs no operation.
3. No feedback is provided by the device as when the output bin is in position. If the printer desires to know this, it can query the device for "device quiesed".
4. Any IR conditions or Equipment Check conditions detected by the device during its movement to locate the output bin will be reported and handled as any other IR or Equipment Check condition (ie, as normal).
5. As with printing and homing operations, during the movement to locate the output bin all output bin non-critical status information will not be detected & reported by the device to avoid false status changes due to the mechanical movement. Any non-critical status changes that do occur during the movement will be detected after the output bin has stopped and will be reported back to the printer at that time.
6. If the device receives this command while already moving the output bin for this command, it will stop the device's hardware at the next output bin.
7. If the device receives a **Query Device Minimum Gap** command while moving the output bin for this command, the device will "hold off" the sheet associated with the Query Device Minimum Gap command with a gap response = 0xFFFF and stop the device's hardware at the next output bin.
8. If the device receives this command when the device is not quiesed (ie, media is still being processed), the device will raise a *Paperport II Command Protocol Error*.
9. If the device receives this command with a parameter value it does not support, for example: locate to bin #8 for a device that only has 5 output bins, the device will raise a *Paperport II Invalid Command Parameter Error*.
10. If the device does not have an output bin, for example: any input device, that receives this command, the device will raise a Paperport II Unsupported Command Error.

1.12.17Submit Logical Page to Queue

Op Code	Mnemonic	Command Name
0x98 0x97	SLPQ	Submit Logical Page to Queue

NOTE: All new output options should support both of these op-codes to maintain comapatibiity with older mono engines. Older mono engines use the 0x97 op code that was later used for the Track Speed command (see section 1.10.8) which is used only for input options on some color printers. Therefore, this command should not be broadcast to all options.

Command Structure: Opcode, Length Field, N1, N2, N3, N4, N5, N6
0x98, 0x08, N1, N2, N3, N4, N5, N6, N7

Parameters:

N1: Sub-Command
- 0x00 Submit New Page
- 0x01 Update Last Page Sent
- 0x02 Remove Last Submitted Page by Page ID (Listed in N2).
- 0x03 – 0xFF Reserved

N2: Page ID
- 0x00 Reserved
- 0x01 – 0xFF Page ID of sheet destined for bin destination listed in N3.

N3: Bin Destination
- 0x00 – 0xFF

N4: Tray Offset Position (on Mono, this is often referred to "Jog Position")
- 0x00 Default Position (non-aligned position)
- 0x01 Position #1 (Left / Front) Position of Two Position.).
- 0x02 Position #2 (Right / Rear Position of Two Position).
- 0x03 – 0xFF Reserved

N5: Accumulator Function
- 0x00 Do Not Accumulate
  (When sheets are going directly to the output bin. No finishing operations).
- 0x01 Accumulate
  (When sheets are being accumulated to have a finishing operation performed.)
- 0x02 – 0xFF Reserved

N6: Finishing Function
- 0x00 No Action - Reserved
- 0x01 Sheet Will Be Flushed
- 0x02 Sheet Will Be Stapled
- 0x03 Sheet Will Be Primed
- 0x04 – 0xFF Reserved

N7: Hole-Punch Function
- 0x00 No Hole Punch
- 0x01 NA 3-Hole Punch

0x02 ISO 2-Hole Punch
0x03 US 2-Hole Punch
0x04 ISO 4-Hole Punch
0x05 – 0xFF Reserved

N8: Staple Position Function (This byte will only be included if the option supports multiple staple positions. See section 5.3.5.1.3)

0x00 No Staple
0x01 Top Left (aka Portrait Single)
0x02 Top Dual (aka Portrait Dual Top)
0x03 Left Dual (aka Portrait Dual Left)
0x04 – 0x0F Reserved
0x11 Top Right (aka Landscape Single)
0x12 (previously Landscape Dual Top)
0x13 (previously Landscape Dual Left)
0x14 Top Left Diagonal
0x15 Top Right Diagonal
0x16 – 0x1F Reserved for Landscape Staple Patterns

Device Response: None.

1.12.18Initiate Calibration Process (ICP)

Op Code	Mnemonic	Command Name
0xB0	ICP	Initiate Calibration Process

Command Structure: OpCode, Length Field, N1, N2
0xB0, 0x03, N1, N2

Parameters:

N1 Calibration to be Performed
0x00 Transparency Sensor Calibration
0x01 Paper Path Sensor Calibration
0x02 – 0xFF Reserved

N2 Bin / Tray Number
0x00 Single Bin / Tray Units
0x01 Bin 1 / Tray 1
0x02 – 0xFF Bin / Tray 2 to Bin / Tray 255

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

The print engine may send this command after a device has indicated to the print engine that a calibration is necessary. The device will indicate whether a calibration is necessary by its response to the **QMSCS**, **Query Media Sensor Calibration Status**, or other such command. It is recommended that each value for N1 have associated with it a unique query command to determine calibration status. A unique command for each is necessary since a device may contain multiple pieces of hardware that require calibration. This will also serve to keep the query command short since a parameter will not need to be passed. A short command is beneficial since the print engine will have to continuously poll the device until the calibration is complete. See the description of the **QMSCS** command for information on how to determine calibration status during a calibration process.

1.12.19Request to DownLoad Data (RDLD)

Op Code	Mnemonic	Command Name
0xA1	RDLD	Request to DownLoad Data

Command Structure: OpCode, Length Field, N1
0xA1, 0x02, N1

Parameters: N1 Job Type
0x00 – 0xFF Reserved

Link Pacing: - Engine to Device: Standard Communications Syntax

Response Syntax: Multi-Byte Response

Response Bytes: Length Field Number of Response Bytes Including Length Field
Values: 0x04

N1 Job Accepted?
Values: 0x00 – Job Accepted
0x01 – Job NOT Accepted
0x02 – 0xFF Reserved

N2 Block Size in Bytes
Values: Range: 0x01 – 0xFF

N3 Time-out in Milliseconds
Values: Range: 0x01 – 0xFF

Description:

This command allows the print engine to determine if the device is capable of accepting a block download of data bytes. The device will respond by either accepting or not accepting the job of type specified in command parameter N1. The device will also specify the number of bytes it is capable of receiving in one data block. In addition, the device will specify a time-out in milliseconds that will allow the device sufficient time to receive and process the data block. Under certain circumstances a device may not be able to not accept the first **RDLD** command but the device might be able to accept subsequent **RDLD** commands. One example of this is if the device needs to jump to a different section of code before having the ability to accept the data block. Once the device accepts the download request, the device is prepared to accept **DLDB**, **DownLoad Data Block** commands until the print engine sends the **DLDC**, **DownLoad Data Complete**. The device will accept data blocks of the size specified by the second response byte and echo those bytes within the time-out value specified by the third response byte.

1.12.20DownLoad Data Block (DLDB)

Op Code	Mnemonic	Command Name
0xA2	DLDB	DownLoad Data Block

Command Structure: OpCode, Length Field, N1, N2, N3, N4, …
0xA2, n, N1, N2, N3, N4, … N(n-1)

Parameters:

N1 Data Byte 1
Values: Range: 0x00 – 0xFF

N2 Data Byte 2
Values: Range: 0x00 – 0xFF

N3 Data Byte 3
Values: Range: 0x00 – 0xFF

N4 Data Byte 4
Values: Range: 0x00 – 0xFF

Link Pacing: - Engine to Device: Standard Communications Syntax

Response Syntax: Single-Byte Response

Response Bytes: N1 Block Processed Successfully?
Values: 0x00 – Success
0x01 – Failure
0x02 – 0xFF Reserved

Description:

Once a device has accepted a request to download data via the **RDLD**, **Request to Download Data**, command, this command can be sent by the print engine to begin dowloading data blocks of the size specified in the response to the **RDLD** command. The device will continue to process data blocks until the print engine ends the dowload via the **DownLoad Data Complete**, **DLDC**, command.

1.12.21DownLoad Data Complete (DLDC)

Op Code	Mnemonic	Command Name
0xA3	DLDC	DownLoad Data Complete

Command Structure: OpCode

Note: **This is not a multi-byte command even though its op-code is greater than 0x80.**

Parameters: None

Link Pacing: - Engine to Device: Standard Communications Syntax

Response Syntax: Single-Byte Response

Response Bytes: N1 Download Processed Successfully?

Values: 0x00 – Success
0x01 – Failure
0x02 – 0xFF Reserved

Description:

Once a device accepts a request to download data and it has processed all the data blocks that the print engine whishes to send, the print engine should end the process by sending a **DLDC**, **Download Data Complete**, command. The device will send a single-byte response indicating whether the process was successful. If the device had to jump to a code segment separate from the normal code segment it executes, this command may or may not send the device back to its original code segment. If the device does not return to executing its normal application code, the print engine will have to send an **EAE**, **Enable Application Execution**, command.

1.12.22Enable Application Execution (EAE)

Op Code	Mnemonic	Command Name

Op Code	Mnemonic	Command Name
0xA0	EAE	Enable Application Execution

Command Structure: OpCode

Note: **This is not a multi-byte command even though its op-code is greater than 0x80.**

Parameters: None

Link Pacing: - Engine to Device: Standard Communications Syntax

Response Syntax: Single-Byte Response

Response Bytes: N1 Time fore the device can process subsequent commmands.
Units: Milliseconds
Values: Range: 0x01 – 0xFF

Description:

The print engine uses this command to tell the device to start executing its application code. It may be needed when a device has a special sector of code such as a boot sector. Both sections of code should be able to process this command. The boot sector would jump to the application code when this command is received. The application sector would accept the command and do nothing since application code is already being executed.

1.12.23Fan Activation Notification (FAN)

Op Code	Mnemonic	Command Name
0xB4	FAN	Fan Activation Notification

Command Structure: OpCode, Length Field, N1

Parameters: N1 Data Byte 1
Values: 0 = Off
1 = On
2 – 255 Reserved.

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

The print engine uses this command to tell the device to turn its fan (if equipped) on or off.

1.12.24Query Serial Port Debug Data (QSPDD) (Compatibility Level >= 8)

Op Code	Mnemonic	Command Name
0xA6	QSPDD	Query Serial Port Debug Data

Command Structure: OpCode, Length Field, N1
Parameters: N1 SubCommand
0x00 Number of debug bytes
0x01 Transmit debug data
0x02 End transmitting debug data
0x03 – 0xFF Reserved

Link Pacing: - Engine to Device: Standard Communications Syntax

Response Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, ...

Response Byte Definitions:

Length Field Number of bytes in the response including the count byte
Values: Range: 0x01 to 0xFF

N1 Debug Byte #1
Values: Range: 0x01 to 0xFF

N2 Debug Byte #2
Values: Range: 0x01 to 0xFF

N3 Debug Byte #3
Values: Range: 0x01 to 0xFF

N4 Debug Byte #4
Values: Range: 0x01 to 0xFF

N5 Debug Byte #5
Values: Range: 0x01 to 0xFF

N6 Debug Byte #6
Values: Range: 0x01 to 0xFF

N7 Debug Byte #7
Values: Range: 0x01 to 0xFF

Description:

The option devices' keep a circular debug buffer in RAM and outputs its contents on the serial port.

This command provides a means for the engine to extract this data when needed. Since the option can only transmit 8 bytes at a time, the engine will have to send the command sufficient times to retrieve all the data. When the device receives the QSPDD command with the sub-command set to "Number of debug bytes" it will return the number of debug bytes available. When the device receives this command with the sub-command set to "Transmit debug data" it will send the first 7 bytes of debug data along with the length field. The device upon receiving each subsequent command with the same sub-command is to send the next 7 debug bytes in the buffer. The engine will send the QSPDD command with the sub-command set to "End transmitting debug data" to indicate the completion of the data transfer. The option's response to this will have a byte count of 0x01 to indicate no debug data is being sent. The engine may end the data transmission sequence at any time. The option will hold off posting new debug data to the buffer after receiving the first QSPDD command until it receives the QSPDD command with the "End transmitting debug data" sub command. For any of the following conditions the device will reset the counter pacing the data transmissions: QSPDD command with sub-command set to "End transmitting debug data" Engine raised Link Reset, System Configuration Reset command, Hard Reset command or Soft Reset command.

1.12.25Lighting Notification (LIN)

Op Code	Mnemonic	Command Name
0xB6	LIN	Lighting Notification

Command Structure: OpCode, Length Field, N1, N2, …

Parameters: N1 Subcommand

Values:
- 0 = Set Global Intensity
- 1 = Turn Off
- 2 = Turn On
- 3 = Set Light to Flashing
- 4 = Set Global Simple Flashing Rate
- 5 = Set Global Synchronized Flashing Rate
- 6 – 255 = Reserved

Subcommand: Set Global Intensity
Subcommand Parameters: N2

N2 Light Intensity Enumeration

Values:
- 0 = Bright
- 1 = Dim
- 2-255 = Reserved

Subcommand: Turn Off, Turn On, Set Light to Flashing
Subcommand Parameters: N2

N2 Light Selection

Values:
- 0 = All Lights

1 = Light 1
…
255 = Light 255

Subcommand: Set Global Simple Flashing Rate
Subcommand Parameters: N2, N3

N2 On Time
Values: 0-255 in 20mSec increments
0 = 0mSec, 255 = 5100 mSec

N3 Off Time
Values: 0-255 in 20mSec increments
0 = 0 mSec, 255 = 5100 mSec

Subcommand: Set Global Synchronized Flashing Rate
Subcommand Parameters: N2, N3, N4

N2 On Time
Values: 0-255 in 20mSec increments
0 = 0mSec, 255 = 5100 mSec

N3 Off Time
Values: 0-255 in 20mSec increments
0 = 0 mSec, 255 = 5100 mSec

N4 Delay
Values: 0-255 in 10mSec increments
0 = 0mSec, 255 = 2550 mSec

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

The LIN command is used to tell the device how to control lighted devices present to the option. The length of this command is variable depending on the subcommand specified.

The engine shall not send the LIN command to the option if when a QDC Lighting was sent, the option responded that it did not support Lighting, or had no Lights.

The “Set Global Intensity” subcommand sets the global Light intensity of all the Lights in the option. The N1 and N2 parameters are sent to the option. The N2 byte specifies an enumeration for what light intensity all lights in the option should be set to when turned on. The global light intensity in the option should be changed by the option as soon as it can process the LIN command.

The “Turn On” and “Turn Off” subcommands turn on or off individual lights in the option. The N1 and N2 parameters are sent to the option. The light should be turned on or off by the option as soon as it can process the LIN command.

The “Set Light to Flashing” subcommand sets the status of the LED selected to indicate that it shall be flashed. If “Simple Flashing” is enabled, then the LED shall be allowed to flash as soon as it can be processed by the option. If “Synchronized Flashing” is enabled, then the LED shall flash as soon as the point in time known as Lighting Sync is reached after a Lighting Ping (LIP) command is received.

The “Set Global Simple Flashing Rate” subcommand notifies the option of the On Time and Off Time of the period for the driving of the light(s); this inherently includes the duty cycle information. The flashing for the lights are not synced up by the LIP command in this mode. The flashing for the selected lights shall take effect as soon as the option can process the LIN command. Only one flashing style shall be utilized for the option at any one given time; when the “Set Global Simple Flashing Rate” subcommand is received and processed, any other flashing modes or settings shall be overwritten.

The “Set Global Synchronized Flashing Rate” subcommand notifies the option of the On Time and Off Time of the period for the driving of the light(s); this inherently includes the duty cycle information. It also specifies the Delay from receiving the LIP command until starting the period of the flashing for the lights. Only one flashing style shall be utilized for the option at any one given time; when the “Set Global Synchronized Flashing Rate” subcommand is received and processed, any other flashing modes or settings shall be overwritten.

If an unsupported LIN subcommand is specified to an option, it should ignore the specified LIN subcommand.

1.12.26Lighting Ping (LIP)

Op Code	Mnemonic	Command Name
0xB7	LIP	Lighting Ping

Command Structure: OpCode

Parameters: None

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is used to sync the beginning of the period of flashing for lights for the LIN “Set Global Synchronized Flashing Rate” subcommand. It can be sent to the broadcast address, or

addressed to individual options. This command shall not be broadcast to the options if all options connected to the printer are not a minimum of paperport compatibility level 10.

If an option does not support “Set Global Synchronized Flashing Rate”, but recognizes the Lighting Ping (LIP) command, it should respond as normal to the command, but not act on the command.

When a LIN command and “Set Global Synchronized Flashing Rate” subcommand is received by an option, the option shall wait the specified Delay from the rising edge of the Stop bit on the paperport, after the LIP is received, until starting the beginning of the period for flashing Lights. The point in time at the end of the Delay and start of the period for flashing of lights shall be called Lighting Sync. Note that if a LIP is received in the middle of a Synchronized Flashing sequence, the flashing shall continue as normal, until the Sync point is reached, whereupon the flashing period shall start over from the beginning.

Figure 4-1 shows a Ping, and Sync sequence for the “Synchronized Flashing” lighting mode.

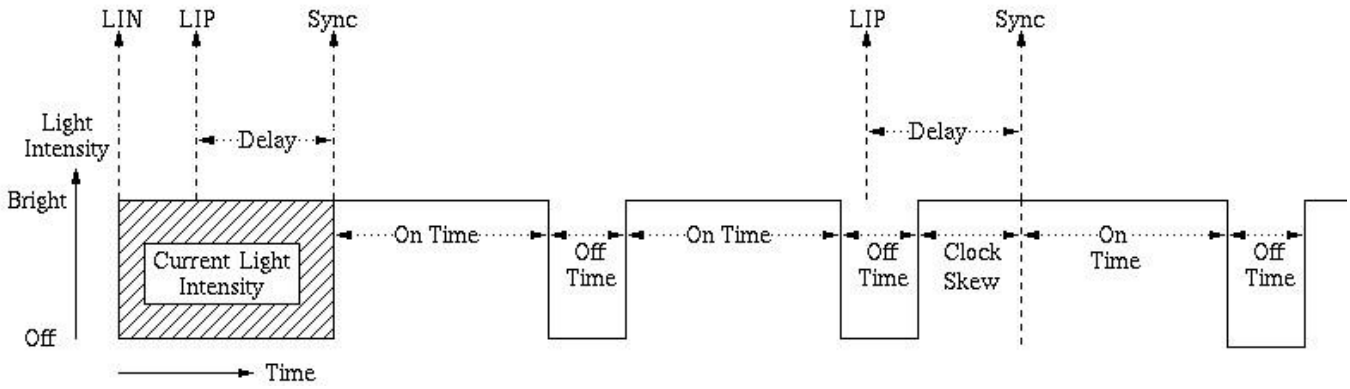


FIGURE 4-1 **Illustration of “Synchronized Flashing” lighting mode and Ping, Sync sequence**

1.13 DIAGNOSTIC TESTS COMMANDS

The printer uses these commands to solicit a device to perform a diagnostic test selected at the printer's control panel and determine the results of executing the diagnostic tests selected.

1.13.1 Diagnostic Test (DT)

Op Code	Mnemonic	Command Name
0x89	DT	Diagnostic Test

Command Structure: OpCode, Length Field, N1, N2, ... Nm

Parameters:

N1 Diagnostic Test

00	Specific Diagnostic Test in Subsequent Parameter Byte
01 - FF	Specific Diagnostic Test

N2 Diagnostic Test

Nm Diagnostic Test

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: None

Description:

This command is only used when the printer is in CE Diagnostics mode.

The printer sends this command to request a device to execute a specific diagnostic test (usually selected by a service person via the printer control panel). The parameter byte identifies the specific diagnostic test to be executed.

The printer is required to ensure the device is idle (i.e., <u>not</u> still processing previously sent commands or managing sheets in its paper path).

There are two types of Diagnostic Tests that the user can select: (1) A test of which the test simply executes and then terminates upon completion of executing the diagnostic test, and (2) A test of which the test is a solicitation of status information of the device's hardware operation and the test operator must determine the termination of the test. The protocol required by the engine and device are described below for both of these type diagnostics tests.

<u>Test Execution and Device Terminates Diagnostic Test Upon Completion:</u>

For diagnostic tests of which the device automatically terminates the diagnostic test upon its completion (example: execute system card's BAT test), the following diagnostic test protocol is used. After the printer issues this command to request the diagnostic test to be executed, the printer begins querying the device with the **Query Device Status** command to ensure an IR condition has not arisen, and the **Query Diagnostic Test Complete** command to be notified when the diagnostic test is completed. When the printer is notified that the diagnostic test is completed, the printer will post the operator panel message allowing the Service Person to "continue" to another diagnostic test. When the Service Person requests to "continue" to another diagnostic test, the printer must send an **Intervention Required Cleared** command to the device to allow the device to "reset" itself after the execution of the previous diagnostic test.

If the device fails the diagnostic test, it will notify the printer via the **Query Device Status 1** command and the various Query IR commands with an *Intervention Required Absolute Immediate Stop, Equipment Check* status. The specific failure can be determined via the **Query Equipment Check Status** command. If the device passes the diagnostic test, the **Query Device Status 1** command's response will indicate no IR condition (ie, "clear" status).

Test Execution and User Terminates Diagnostic Test:
For diagnostic tests of which device hardware status or operational information is provided (such as sensors or switches) via the diagnostic test data message, the following diagnostic test protocol is used. The printer invokes the specific test by sending the device the **Diagnostic Test** with parameter N1 selecting the specific test. The device executes that test. The printer continues to send the device **Query Diagnostic Test Complete** commands to the device until the device reports diagnostic test complete. The device indicates diagnostic test complete when it has the entire initial (ie, first) test data message assembled and ready to report back to the printer. At this time, the printer sends the device **Query Diagnostic Test Data Message** command to receive the diagnostic test data message and display it on the printer's control panel. The printer continues to send the device **Query Diagnostic Test Data Message** commands thus receiving any updates to the diagnostic test message (ie, for the example of sensors: indication if the sensor value has changed). In the mean time the device is constantly executing the "diagnostic test" selected and updating its diagnostic test data message to reflect any changes. The printer continues this querying for updated diagnostic test data until the operator hits the 'Continue' button on the printer. At this time, the printer issues an **Intervention Required Cleared** command to the device to allow additional diagnostic tests to be performed, if desired.

For either of these two categories of diagnostic tests, if the device detects an IR condition during the test, it will terminate execution of the diagnostic test and notify the printer via the **Query Device Status 1** command that an *Intervention Required Absolute Immediate Stop, IR Condition* status has occurred. The specific IR condition can be determined by the engine via the **Query IR Absolute Immediate Stop** command.

Once the first diagnostic test is selected, the device identifies itself as being in the "diagnostic state" of which it can only be exited by a power-on reset or the printer issuing a **System Configuration Reset**, **Hard Reset, Soft Reset** or **Intervention Required Clear** command.

1.13.2 Query Diagnostic Test Complete (QDTC)

Op Code	Mnemonic	Command Name
0x21	QDTC	Query Diagnostic Test Complete

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.20

Description:

This command is only used when the printer is in CE Diagnostics mode and the printer has requested the device to execute a diagnostic test via the **Diagnostic Test** command to determine when the device has completed the execution of the diagnostic test requested. See the *Diagnostic Test* command section above for details.

1.13.3 Query Diagnostic Test Data Message (QDTDM)

Op Code	Mnemonic	Command Name
0x25	QDTDM	Query Diagnostic Test Data Message

Command Structure: OpCode

Link Pacing: - Engine to Device: Standard Communications Syntax

Response: see Section 1.15.21

Description:

The printer sends this command to a device after invoking a Diagnostic Test to a device of which the value or state of the device's hardware (such as sensors or switches) is to be reported by the device instead of a simple "pass/fail" or "test completed/not completed" response. This command provides a means for the device to report this specific test data message information back to the printer to be displayed on the printer's control panel.

See the *Diagnostic Test* command section above for more details.

DEVICES RESPONSE SET DESCRIPTION

The following sections describe the various Responses to the query commands. First the *Structure* (or syntax) of the Responses is described. Followed by a description of the actual query responses; first the *device independent* responses (ie, all devices provide the same response), and then the *device dependent* responses (ie, those that are unique for each device type).

1.14 RESPONSE STRUCTURE

All Query Responses fall into either one of two categories: a *Single Byte Response* or a *Multibyte Response*. Single byte responses are as their name implies; one byte long. Likewise, multibyte responses are as their name implies; several bytes long. The syntax for each response type is listed below:

Single Byte Response:	Response Byte
MultiByte Response:	Length Field, Response Byte$_1$, Byte$_2$, ... Byte$_n$ (Length Field = #bytes in Response including itself)

1.15 DEVICE-INDEPENDENT RESPONSES

Described below are the device independent responses; ie, status responses that are the same for all device types.

1.15.1 Query Reset Complete Response

This is the Query Reset Complete (QRC) Response:

Syntax: Single Byte, Not Bit Encoded

Values:		
	0x00	Reset Complete
	0x80	Reset Not Complete
	0x01-0x7F	Reserved
	0x81-0xFF	Reserved

1.15.2 Query Device Type Response

This is the Query Device Type (QDT) Response:

Syntax: Single Byte, Not Bit Encoded

Values:

0x00	Nop Response
0x01	Smart Input Tray
0x02	Feeder
0x03	Envelope Feeder
0x04	Duplexer
0x05	Modular Output Stacker
0x06	Output Sorter
0x07	Mailbox
0x08	Sorter / Mailbox
0x09	High Capacity Output Stacker (Legacy Top-Mounted)
0x0A	Modular Mailbox Stacker
0x0B	Modular High Capacity Output Stacker
0x0C	Modular Stapler
0x0D – 0x1F	Reserved
0x20	SR Stapler
0x21	SR Offset Stacker
0x22 – 0x2F	Reserved
0x30	New Side-Mounted High-Capacity Output Stacker (HCOS)
0x31	New Side-Mounted High-Capacity Output Finisher (HCOF)
0x32	New High-Capacity Input Tray (HCIT)
0x33	SR Horizontal Transport Unit (HTU)
0x34-0x40	Reserved
0x41	Color EP Smart Input Tray
0x42	Color EP High Capacity Smart Input Tray
0x43	Color EP Envelope Feeder
0x44	Color EP Duplexer
0x45	Color Special Media Tray
0x46	Cutter Feeder Attachment (CFA) – Partner A
0x47	Input Roll (PIR) – Partner B
0x48-0x4F	Reserved
0x50	G2 Smart Input Tray
0x51	RFID Tray
0x52-0xFF	Reserved

Definitions of the various device types above is given in the *Appendix A - Definition of Terms* section. The physical attributes (or model) assumed by the engine for a given *device type* is shown in *Appendix D - Device Type(s) Physical Model*.

1.15.3 Query Paper Path Characteristics Response

This is the Query Paper Path Characteristics (QPPC) Response:

Syntax: Multi-Byte Response Length Field, N1, N2

Response Byte Definitions:

Length Field Number of bytes in the response including the count byte
Values: Range: 0x01 to 0xFF

For Sensors' Distribution request:
N1 Number of sensor in the PassThru Paperpath.
Values: 0x00-0x3F
N2 Number of sensor in the Feed Paperpath.
Values: 0x00-0x3F

For Requests other than Sensors' Distribution
N1, N2 Value of the Paperpath Characteristics Requested
(requested by printer via command's N1 parameter value)
(N1 Most Significant Byte)
Values: 0x0000-0xFFFF

Notes: If the printer requests a paperpath characteristic that does not exist in the device (for example: the distance between the 4th & 5th sensors of a device that only has 4 passthru sensors, or the distance from the 1st sensor to the top of the device of a device without any passthru sensors), the device returns a *"zero response"*, N1,N2 = 0x0000 (full response: 0x03, 0x00, 0x00) or a *"null response"*, length = 0x01 (full response: 0x01) to indicate to the engine that it does not have such a paperpath characteristic.

1.15.4 Query Device Software Revision (QDSR) Response

This is the Query Device Software Revision (QDSR) response:

Syntax: Length, N1, N2, N3, … Nm
0x04, N1, N2, N3
0x08, N1, N2, N3, N4, N5, N6, N7

0x09, N1, N2, N3, N4, N5, N6, N7, N8

Parameters:

N1 = Firmware Application Code Major Revision Level,
N2 = Firmware Application Code Minor Revision Level
N3 = Application Code Compatibility Level (0,1,5,7)

N4 = Application Code Extended version A
N5 = Application Code Extended version B
N6 = Application Code Extended version C
N7 = Application Code Build number

N8 = Application Code Compatibility Level 2 (7+)

Parameters N4-N7 are optional, but if any are included in the response, they all must be included in the response.

N8 is provided for new options. If included, it will supercede N3 when attached to an engine that recognizes this data byte (option compatibility levels > 7).

Range: 0x00 to 0xFF for all parameters except N3 and N8

NOTES :

1. For the C750 options (duplexer, HCOF, HCIT, and 500-sheet tray), the compatibility levels started over at level 0. The C750 options derived from Optra T options electronics (5-bin mailboxer and stacker) begin at the compatibility level already in existence at start of production. No compatibility checks are currently in the C750 engine firmware.
2. Color Options that support in-system flash programming via PJL files submitted as print jobs shall set the MSB of the Compatibility Level to 1. The paperport commands required to support this feature include RDLD and EAE.
3. For backward compatibility with OptraS, T61x-T64x models, options must have N3 = 7.

1.15.5 Query Device Boot Software Revision (QDBSR) Response

This is the Query Device Boot Software Revision (QDBSR) response:

Syntax: Length, N1, N2, N3, N4, N5, N6, N7
0x08, N1, N2, N3, N4, N5, N6, N7

Parameters:

Length = 8:

N1 = Firmware Boot Code Major Revision Level,

N2 = Firmware Boot Code Minor Revision Level
N3 = Boot Code Compatibility Level

N4 = Boot Code Extended version A
N5 = Boot Code Extended version B
N6 = Boot Code Extended version C
N7 = Boot Code Build number
Parameters N4-N7 are optional, but if any are included in the response, they all must be included in the response.

Range: 0x00 to 0xFF for both N1-N7

NOTES :

1. For the C750 options (duplexer, HCOF, HCIT, and 500-sheet tray), the compatibility levels started over at level 0. The C750 options derived from Optra T options electronics (5-bin mailboxer and stacker) begin at the compatibility level already in existence at start of production. No compatibility checks are currently in the C750 engine firmware.
2. Color Options that support in-system flash programming via PJL files submitted as print jobs shall set the MSB of the Compatibility Level to 1. The paperport commands required to support this feature include RDLD and EAE.

1.15.6 Query Device Status 1 (QDS1) Response

This is the *general status* Query Device Status (QDS) response.

The *general status* basically provides the printer with 4 pieces of information concerning the device: (1) Identifies if there is any sheet in the device's "PassThru" paperpath by indicating the status of its "PassThru" sensors, (2) Identifies if the device is "ready" to accept or send media, (3) Identifies if the device is raising that it has a status change the engine should know about, but the status change is not of a critical nature (ie, one that prohibits the device from operating), and (4) Identifies if the device is capable of processing print jobs; ie, is Not in an *Intervention Required* state.

For a general description of *IR / Error Processing* see section 0.

Syntax: Single Byte, Bit Encoded

Values:	Bit 7 (MSB)	0	No Media Present at "Paperpath" Sensor #1
		1	Media Present at "Paperpath" Sensor #1
	Bit 6	0	No Media Present at "Paperpath" Sensor #2
		1	Media Present at "Paperpath" Sensor #2

Bit 5	0	Reserved (always 0)
Bit 4	0	Page Not In Output Device
	1	Page In Output Device
Bit 3	0	Input Source Not Ready
	1	Input Source Ready
Bit 2	0	No Non-Critical Status Change Condition
	1	Non-Critical Status Change Condition
Bit 1	0	No Intervention Required Condition - Homing Allowed
	1	Intervention Required Condition - Homing Allowed
Bit 0	0	No Intervention Required Absolute Immediate Stop Condition
	1	Intervention Required Absolute Immediate Stop Condition

Media Present at "Paperpath" Sensor #1, or 2

This response byte accommodates a device that has up to 2 sensors in its paper path(s). For devices that have less than 2 sensors in the paper path, they should always indicate *Media Not Present at Paperpath Sensor* for the sensors that don't exist.

The definition of the sensors are as follows:
Sensor #1 is the sensor nearest the printer in the PassThru Paperpath.
Sensor #2 is the sensor away from the printer in the PassThru Paperpath if that sensor exist otherwise it's the sensor nearest the printer in the Feed Paperpath if that sensor exist.

Example of sensor in the Feed Paperpath is the Trailing Edge sensor in G2 Smart Input Trays – See Appedix C # G.

Whenever the device detects media under Sensor #1, it asserts *Media Present at "Paperpath" Sensor #1*. Likewise, whenever the device detects media under Sensor #2, it asserts *Media Present at "Paperpath" Sensor #2*. Conversely, whenever the device detects media is no longer present at one of the sensors, it negates the appropriate sensor's bit.

At any given time, more than just one sensor can have media present; thus, requiring more than one bit to be asserted. Take the example of the sensors physical placement such that the distance from sensor #1 to sensor #2 is 6 inches. A letter size sheet (8.5" x 11") being fed from the tray below to the printer will have this device's sensor response bits do something like this: first = No Media Present Under Any Sensor, then as the sheet reaches sensor #2 and the response byte will indicate *Media Present at "Pass Thru" Sensor #2*, then as the sheet feeds further it will reach sensor #1 and the response byte will indicate *Media Present at "Pass Thru" Sensor #1 & 2*, then finally the trailing edge of the sheet will clear sensor #2 and the response byte will indicate *Media Present at "Pass Thru" Sensor #1*, then the trailing edge will eventually clear sensor #1 and the response byte will indicate *Media Not Present at "Pass Thru" Sensor #1, or 2* at which time the printer can know the sheet no longer is in this device's "Pass Thru" paper path.

The device must "debounce" their paper present sensors to avoid providing the printer will erroneous "Paperpath" sensor information.

Note: In the future, for devices that may have more than 2 sensors in their paper path, a **Query Device Status 2** command will be defined to provide status for sensors #3-4, QDS3 for sensors #5-6, etc.

For the C750 HCIT, a "virtual sensor" shall be reported as sensor #1. This is a sensor meant to mimic the real pass-thru sensor mounted near the top-most roller on the 500-sheet tray options. Sensor 2 in the QDS1 command shall be a logical OR of the 2 real sensors in the HCIT.

Page In Output Device

Input Devices:

Input devices always negates this bit = 0 (ie, Page Not In Output Device).

Input / Output Devices (example: duplex option):

The combination Input/Output device (for example a duplex option) asserts this bit to indicate to the print engine that the sheet is in the Input/Output device; ie, this Input/Output device associated with the previous issued **Select Output Device** command has successfully received the sheet inside the device. THIS BIT DOES NOT INDICATE THAT THE PAGE IN THE INPUT/OUTPUT DEVICE IS READY TO BE FED BACK INTO THE PRINTER. IT SIMPLY MEANS THE PAGE IS LOCATED IN THE INPUT/OUTPUT DEVICE. To feed the sheet from the Input/Output device and back into the printer, the printer must select the Input/Output device as the input source via the **Select Input Device** command. Then when the Input/Output device asserts the *Input Source Ready* bit (see below for details), the printer may issue the **Pick Page** command at which time the Input/Output device picks the media, feeds it into the printer, and negates the *Input Source Ready* bit. The Input/Output device negates the *Page In Output Device* bit when it receives the **Select Input Device** command.

The use of this signal is illustrated in the "Engine / Device Handshaking Timing / Event Diagram - Duplex Printing" found in section 6.0.

Output Devices:

The Output device (for example a output stacker) asserts this bit to indicate to the print engine that a sheet has successfully reached its output destination (ie, tray or bin). Upon receiving this signal the printer will query the device to the *Page ID* of the sheet. After returning the *Page ID* as the response to the **Query Page Complete ID** command the device negates this bit. For additional details, see the **Query Page Compete ID** command in section 1.11.12.

For output options such as the finishers or staplers that have accumulators, this bit shall be asserted once the sheet is completely in the accumulator. If the sheet is routed past the accumulator, this bit shall be asserted when the sheet reaches the output tray.

The use of this signal is illustrated in the "Engine / Device Handshaking Timing / Event Diagram - Smart Input Tray & Smart Output Device" found in section 6.0.

Input Source Ready

Input Devices:

This bit is ALWAYS associated with the "selected input source". The printer selects an input source by issuing the **Select Input Device** command. Once the printer has selected the input device, the device asserts this bit to indicate to the printer that the device is ready to have the media (ie, paper, envelope, transparency, etc) picked from the associated input device (ie, paper tray, envelope feeder, feeder, duplexer, etc). The *Input Source Ready* bit remains asserted until the input device receives the **Pick Page** command at which time the input device picks the media, feeds it into the printer, and negates the *Input Source Ready* bit. The device re-asserts the *Input Source Ready* bit when it is ready to pick another sheet from its input device.

The use of this signal is illustrated in the "Engine / Device Handshaking Timing / Event Diagram - Smart Input Tray & Smart Output Device" found in section 6.0.

Input / Output Devices (example: duplex option):

This bit is ALWAYS associated with the "selected input source". The printer selects an input source by issuing the **Select Input Device** command. Once the printer has selected the input / output device, the device asserts this bit to indicate to the printer when the device is ready to have the media picked from the associated input device. The *Input Source Ready* bit remains asserted until the input device receives the **Pick Page** command at which time the input device picks the media, feeds it into the printer, and negates the *Input Source Ready* bit.

The use of this signal is illustrated in the "Engine / Device Handshaking Timing / Event Diagram - Duplex Printing" found in section 6.0.

Output Devices:

Output devices always negates this bit = 0 (ie, Input Source Not Ready).

Non-Critical Status Change

This bit is normally negated indicating to the printer that the device does not have any "non-critical status change" to report to the printer. A "non-critical status change" is one that an operator intervention condition exists at the device, but it is (or may) still capable of processing media for printing. Examples of such "non-critical status change" IR conditions are: input tray almost empty, output bin almost full, an inventory change has occurred at the device (for example: media capacity change in a tray, media size change in a tray, etc), etc. When the device detects the "non-critical status change" it notifies the printer by asserting this bit. The printer can send the **Query Non-Critical Status Change** command to learn of the specific "non-critical status change". See section 0 for a hierarchy of the query commands associated with IR processing. Also, see the query commands (section 0) and query command responses (section 0) for detailed descriptions of the various IR conditions and how they are handled.

The device negates this bit when the "non-critical status change" has successfully been communicated to the printer; ie, when the device returns the specific "non-critical status change" response to the **Query Non-Critical Status Change** command.

Intervention Required Condition - Homing Allowed

This bit is normally negated indicating to the printer that the device does not have any intervention required conditions of which printer system homing can occur; ie, the device is capable of processing media for printing. When the device detects an intervention required condition and determines it can allow homing by the printer system, it asserts this bit to alert the printer of the IR condition. For a better description of "homing", please refer to Appendix A. The printer can send the **Query Intervention Required Condition - Homing Allowed** command to learn the specific IR condition. See section 0 for a hierarchy of the query commands associated with IR processing. Also, see the query commands (section 0) and query command responses (section 0) for detailed descriptions of the various IR conditions and how they are handled.

The device negates this bit when it processes the **Query Intervention Required Condition - Homing Allowed** command to indicate to the printer that the IR condition has been resolved; ie no longer exists.

Intervention Required Absolute Immediate Stop

This bit is normally negated indicating to the printer that the device does not have any intervention required conditions of which the device must require the printer system to immediately stop processing media for printing. When the device detects an intervention required condition and determines it can not allow any further processing or movement of the media, it asserts this bit to alert the printer of the IR condition. The printer can send the **Query Intervention Absolute Immediate Stop** command to learn the specific IR condition. See section 0 for a hierarchy of the query commands associated with IR processing. Also, see the query commands (section 4.0) and query command responses (section 5.0) for detailed descriptions of the various IR conditions and how they are handled.

The device negates this bit when it processes the **Query Intervention Required Absolute Immediate Stop** command to indicate to the printer that the IR condition has been resolved; ie no longer exists.

Note: For a device to be capable of processing print jobs, both the *Intervention Required Condition - Homing Allowed* and the *Intervention Required Absolute Immediate Stop* bits must be negated.

Each of these bits are negated; ie, cleared, collectively whenever a **System Configuration Reset**, **Hard Reset**, **Soft Reset** or **Intervention Required Clear Notification** command is received. If the status condition exists upon completion of the processing of these commands, the respective bit is set to alert the printer of the device's status.

1.15.7 Query Non-Critical Status Change Response

When a device notifies the printer of a "Non-Critical" status change via the *general status* Query Device Status 1 (QDS1) response, the printer sends the **Query Non-Critical Status Change** command to determine the specific change of status. The details of the Query Non-Critical Status Change response are given below.

For a general description of *IR / Error Processing* see section 0.

Syntax: Multi-Byte Response Length Field, N1, N2 ...

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 Non-Critical Status Byte #1
Values: 0x00 Clear Response (ie, No Status Change)
0x01 Inventory Change with Device's Tray (or Bin)
0x02 Inventory Change with Device's Stapler (Original mono dev)
0x81 Inventory Change with Device's Stapler
0x03 Inventory Change with Device's Accumulator (Original mono dev)
0x82 Inventory Change with Device's Accumulator
0x83 Reserved
0x84 Reserved
0x85 Inventory Change with Device's Hole Puncher
0x86 Inventory Change with Device's 2nd Stapler
0x03 - 0xFF Reserved

Inventory Change with Device's Tray (or Bin)

This value is returned when the input or output device identifies an Inventory Change with one of its trays (bins). The engine determines the specific Inventory Change information by sending the device the **Query Device Inventory** command. See the specific device's Query Device Inventory responses (section 1.16) for details of the type inventory changes that are reported.

Note: With this value, the specific tray (bin) with the Inventory Change is identified by parameter N2 (below).

Inventory Change with Device's Stapler (or 2nd Stapler)

This value is returned when the output device identifies an Inventory Change with its stapler. The engine determines the specific Inventory Change information by sending the device the **Query Device Inventory** command. See the specific device's Query Device Inventory responses (section 1.16) for details of the type inventory changes that are reported.

Note: With this value, no parameter N2 is returned.

Inventory Change with Device's Accumulator

This value is returned when the output device identifies an Inventory Change with its Accumulator. The engine determines the specific Inventory Change information by sending the device the **Query Device Inventory** command. See the specific device's Query Device Inventory responses (section 1.16) for details of the type inventory changes that are reported.

Note: With this value, no parameter N2 is returned.

Inventory Change with Device's Hole Puncher

This value is returned when the output device identifies an Inventory Change with its Hole Punch Chad Bin. The engine determines the specific Inventory Change information by sending the device the **Query Device Inventory** command. See the specific device's Query Device Inventory responses (section 1.16) for details of the type inventory changes that are reported.

Note: With this value, no parameter N2 is returned.

N2	Tray (of Bin) Reporting Status Change		
	Values:	0x00	No Tray
		0x01	Tray #1
		0x02	Tray #2
		.	
		.	
		.	
		0xFF	Tray # 255

The device clears this status when the "non-critical status change" has successfully been communicated to the printer; ie, when the device returns the specific "non-critical status change" response to the **Query Non-Critical Status Change** command.

Notes:

1. Devices receiving this query, but have not indicated to the engine that a "non-critical status change" exists should report a *clear status*; ie, N1 = 0x00 (0x02, 0x00 response).
2. Input & Output devices that have trays (or bins) should report a status change on a "tray basis"; ie, separately provide a "non-critical status change" signal for status changes for a particular tray (or bin) individually.
3. For devices that do not have a tray (or bin) can either send N2 = No Tray (0x03, N1, 0x00 response) or omit parameter N2 (0x02, N1 response).
4. A "Supply Empty" (for example: "out of paper" or "out of staples") is reported by the device indicating a non-critical status change. The printer determines it is a "supply empty" condition via the **Query Device Inventory** command with a response of capacity being zero.
5. A device can report to the printer a "Supply Almost Empty" condition as a non-critical status change identified specifically via an encoded N1 value (to be defined when needed) to this query. The printer can determine the device's specific capacity via the **Query Device Inventory** command.

6. An "Output Bin Full" condition is reported as a non-critical status change for the specific tray of which the printer learns the tray is full via the **Query Device Inventory** command with a response of capacity equals full. Note: The device must still be able to accept into its output tray (or bin) some additional sheets: a minimum number of sheets equal to the # of sheets that could be in the whole printer system's paperpath.
7. A device can report to the printer an "Output Bin Almost Full" condition as a non-critical status change identified specifically via an encoded N1 value (to be defined when needed) to this query. The printer can determine the device's specific capacity via the **Query Device Inventory** command.
8. Any device's change status information that does not pertain to a tray (or bin) is to be added to parameter N1 as they become defined.
9. Device's that have an operator panel of which the device can be taken off-line can report this status change via parameter N1 (not defined at this point since no device has been identified with this feature).

1.15.8 Query IR Condition - Homing Allowed Response

When a device notifies the printer of an Intervention Required Condition - Homing Allowed status via the *general status* Query Device Status 1 (QDS1) response, the printer sends the **Query Intervention Required Condition - Homing Allowed** command to determine the specific Intervention Required condition raised. The details of the Query Intervention Required Condition - Homing Allowed response are given below.

For a general description of *IR / Error Processing* see section 0.

Syntax: Multi-Byte Response Length Field, N1, N2 ...

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 — IR Condition - Homing Allowed Status Byte #1

Values:

Bit 7 (MSB)	0	Reserved (always 0)
Bit 6	0	No Paper Jam
	1	Paper Jam
Bit 5	0	Reserved (always 0)
Bit 4	0	Reserved (always 0)
Bit 3	0	No Tray Pulled Out
	1	Tray Pulled Out
Bit 2	0	No Tray Pushed Back In
	1	Any Tray Pushed In (after 1 indicated pulled out)
Bit 1	0	Reserved (always 0)
Bit 0	0	Reserved (always 0)

Paper Jam

Normally this bit is negated to indicate that no paper jam has been detected by the device. When the device detects a paper jam but it is capable of continuing to feed the media to allow the printer system (engine or other devices) to continue feeding the media until it "homes" its mechanism, the device raises this bit (along with the *Intervention Required Condition - Homing Allowed* bit in the Query Device Status 1 response). An for example of "homing" is a D-Roll pick technology: once it initiates a pick operation, it must continue to feed media until the D-Roll reaches "home" to position the D-Roll such that no damage will occur to the D-Roll when the tray is removed and reinstalled.

Devices should attempt to support this form of paper jam notification back to the engine since it allows the most freedom to the rest of the printer system's mechanisms.

The device negates this bit when the *Paper Jam* condition has successfully been communicated to the printer; ie, when the device returns the specific "IR Condition - Homing Allowed" response to the **Query Intervention Required Condition - Homing Allowed** command.

The engine can determine the specific number of sheets in the device and their respective location by sending the device the **Query Paper Position Status** command.

Tray Pulled Out

Normally this bit is negated to indicate that none of the device's trays (or bins) have been pulled from the device. The device asserts this bit whenever Any tray has been pulled out to notify the engine that one of its trays has been pulled (along with the *Intervention Required Condition - Homing Allowed* bit in the Query Device Status 1 response). This applies to both input and output devices that have trays or bins.

The device negates this bit when the *Tray Pulled* condition has successfully been communicated to the printer; ie, when the device returns the specific "IR Condition - Homing Allowed" response to the **Query Intervention Required Condition - Homing Allowed** command, and no other *Tray Pulled* condition exists that has not been reported to the printer.

Devices must hold off reporting a tray pushed in status until after it ensures the engine has received the previous (ie, this) tray pulled status.

The engine can determine the specific tray pulled out by sending the device the **Query Tray Pulled Location** command.

For devices reporting *Tray Pulled* - The device should enter a "pseudo IR state" and continue to allow steady state printing from the trays that are still installed. If all the device's trays are pulled out then it can enter a "full IR state" since it is physically incapable of processing any print jobs.

Any Tray Pushed In (after 1 indicated pulled out)

The device asserts this bit whenever Any tray has been pushed in, but only after at least 1 tray has been reported as pulled out to notify the engine that one of its trays has been pushed back in (along with the *Intervention Required Condition - Homing Allowed* bit in the Query Device Status 1 response). This applies to both input and output devices that have trays or bins.

The device negates this bit when the *Any Tray Pushed In* condition has successfully been communicated to the printer; ie, when the device returns the specific "IR Condition - Homing Allowed" response to the **Query Intervention Required Condition - Homing Allowed** command.

Devices must hold off reporting a tray pushed in status until after it ensures the engine has received the previous (ie, this) tray pulled status (to avoid a situation of the printer receiving a

tray pushed back in status and never having seen the fact that the tray had been pulled out in the first place).

The engine can determine the specific tray pushed in by sending the device the **Query Tray Pulled Location** command.

For devices reporting *Any Tray Pushed In*, the engine will send the **Intervention Required Clear Notification** command to instruct the device to exit its IR state ("pseudo" or "full") and become ready to process print jobs once again.

Each of these bits are negated; ie, cleared, collectively whenever a **System Configuration Reset**, **Hard Reset**, **Soft Reset** or **Intervention Required Clear Notification** command is received. If the status condition exists upon completion of the processing of these commands, the respective bit is set to alert the printer of the device's status.

Notes:

1. Devices receiving this query, but have not indicated to the engine that a "IR Condition - Homing Allowed" status should report a *clear status*; ie, N1 = 0x00 (0x02, 0x00 response).

1.15.9 Query IR Absolute Immediate Stop Response

This is the Query IR Absolute Immediate Stop (QIRAIS) response which provides the specific IR condition that requires the printer system to immediately stop processing print jobs.

A *Clear Response*, value of this response = 0x00, indicates the device does Not have any IR conditions requiring the printer system to immediately stop processing print jobs.

For a general description of *IR / Error Processing* see section 0.

Syntax: Multi-Byte Response Length Field, N1, ... Nm

Response Byte Definitions:

Length Field	Number of bytes in the response including this byte		
	Values:	Range: 0x01 to 0xFF	
N1	Intervention Required Condition		
	Values:	Bit Encoded	
	Bit 7 (MSB)	0	No Equipment Check
		1	Equipment Check
	Bit 6	0	No Paper Jam
		1	Paper Jam
	Bit 5	0	No Cover Open

	1	Cover Open
Bit 4	0	No Cover Closed
	1	Any Cover Closed After One Indicated Open
Bit 3	0	No Aux Power Supply Status Change
	1	Aux Power Supply Status Change
Bit 2	0	No HW IR
	1	HW IR
Bit 1	0	No Media Mismatch
	1	Media Mismatch
Bit 0	0	Reserved (always 0)

Equipment Check

Normally this bit is negated to indicate the device has not detected any equipment check conditions and is capable of processing media (bit 7 = 0).

If at any time the device detects an Equipment Check condition, it should shut down its mechanism and then assert this bit to alert the print engine of its equipment check condition (along with the *IR Immediate Absolute Immediate Stop* bit in the Query Device Status 1 response). An Equipment Check is any hardware or software condition that may prohibit the device from functioning properly (such as: bad Ram memory detected, paper feed motor not working properly, invalid Paperport II communications protocol detected, etc) that may required a service person to correct. The print engine can identify the specific Equipment Check condition by sending the **Query Equipment Check Status** command to the device. This bit will remain asserted indicating an equipment check condition until the **Intervention Required Clear Notification** command is received at which time this bit will be negated by the device and the device will attempt to exit its *Intervention Required* (equipment check) state. If the equipment check condition still exists upon exiting the *Intervention Required* state, the device re-initiates the equipment check status via asserting this bit along with the respective bits in the Query Device Status 1 response bytes.

Paper Jam

Normally this bit is negated to indicate that no paper jam has been detected by the device. When the device detects a paper jam and is must request the printer system to stop immediately, the device raises this bit to alert the printer (along with the *IR Immediate Absolute Immediate Stop* bit in the Query Device Status 1 response). An for example of a paper jam requiring the printer to stop immediately may be one of which to continue to feed the media would damage the mechanism.

The device negates this bit when the *Paper Jam* condition has successfully been communicated to the printer; ie, when the device returns the specific "IR Absolute Immediate Stop" response to the **Query IR Absolute Immediate Stop** command.

Cover Open

Normally this bit is negated to indicate that none of the device's covers are open and that the end user can not access portions of the device while it is processing media. When the device

detects that any of its covers may be open, it shuts down its mechanism and then asserts this bit to notify the printer that the device has a cover open IR condition (along with setting the *Intervention Required Absolute Immediate Stop* bit in the Query Device Status 1 response).

The device negates this bit when the *Cover Open* condition has successfully been communicated to the printer; ie, when the device returns the specific "IR Absolute Immediate Stop" response to the **Query IR Absolute Immediate Stop** command, and no other *Cover Open* condition exists that has not been reported to the printer.

Devices must hold off reporting an "Any Cover Closed" status until after it ensures the engine has received the previous (ie, this) cover open status.

The engine can determine the specific cover that is opened by sending the device the **Query Cover Open Location** command.

Any Cover Closed After One Indicated Open

The device asserts this bit whenever Any cover has been closed, but only after at least 1 cover has been reported as open to notify the engine that one of its covers has now been closed (along with the *IR Immediate Absolute Immediate Stop* bit in the Query Device Status 1 response).

The device negates this bit when the *Any Cover Closed* condition has successfully been communicated to the printer; ie, when the device returns the specific "IR Absolute Immediate Stop" response to the **Query IR Absolute Immediate Stop** command.

Devices must hold off reporting a cover closed status until after it ensures the engine has received the previous (ie, this) cover open status (to avoid a situation of the printer receiving a cover closed status and never having seen the fact that the cover was opened in the first place).

The engine can determine the specific tray pushed in by sending the device the **Query Cover Open Location** command.

For devices reporting *Any Cover Closed* with no other covers open being reported, the engine will send the **Intervention Required Clear Notification** command to instruct the device to exit its IR state and become ready to process print jobs once again.

Aux Power Supply Status Change

Any device that has an auxiliary power supply uses this bit to report back to the engine any changes in its status. At POR or during Reset processing, the device "assumes" its auxiliary power supply is On and thus reports a status change if it detects it is off. For a definition of Aux Power Supply, see the **Query Aux Power Status** command and Appendix-A.

Devices that do not have an auxiliary power supply always reports this bit = 0; *No Aux Power Status Change*.

HW IR

The device asserts this bit whenever it has a hardware IR condition other than the ones identified by the other bits in this byte. The hardware IR condition is defined as a hardware condition of which the device can not operate properly and one that the user can correct. For example: A mailbox output device may detect that its mailbox unit (ie, hardware) is no longer properly aligned to the printer and requires the user to be notified to correct the alignment before printing can continue.

When the device detects that any of its "hardware IR conditions" exist (which are device dependent and specific), it shuts down its mechanism and then asserts this bit to notify the printer that the device has a *HW IR* condition (along with setting the *Intervention Required Absolute Immediate Stop* bit in the Query Device Status 1 response).

The device negates this bit when the *HW IR* condition has successfully been communicated to the printer; ie, when the device returns the specific "IR Absolute Immediate Stop" response to the **Query IR Absolute Immediate Stop** command.

Media Mismatch

This bit is used exclusively by input devices that have media sensors installed (i.e. detect transparency vs. non-transparency). In the case where the option was sent a SID or SIDX command and the media detected does not match the commanded type, this bit is asserted. This bit will be cleared after this

The device negates this bit when the *Media Mismatch* condition has successfully been communicated to the printer; ie, when the device returns the specific "IR Absolute Immediate Stop" response to the **Query IR Absolute Immediate Stop** command, and no other *Media Mismatch* condition exists that has not been reported to the printer.

Each of these bits are negated; ie, cleared, collectively whenever a **System Configuration Reset**, **Hard Reset**, **Soft Reset** or **Intervention Required Clear Notification** command is received. If the status condition exists upon completion of the processing of these commands, the respective bit is set to alert the printer of the device's status.

Notes:

1. Devices receiving this query, but have not indicated to the engine that a "IR Absolute Immediate Stop" status should report a *clear status*; ie, N1 = 0x00 (0x02, 0x00 response).

1.15.10Query Equipment Check Status Response

This is the Query Equipment Check Status (QEQCKS) response. This response identifies the specific *Equipment Check* condition that created the Intervention Required Absolute Immediate Stop status alerting the print engine that an Equipment Check was detected by the device. An Equipment Check condition is any hardware or software condition detected by the device that

indicates it can not operate properly and may require service (examples: memory check indicates bad Ram, unable to detect feedback from a motor, invalid Paperport protocol detected from the print engine, etc.).

Syntax: Single Byte Response N1

Response Byte Definitions:

N1 Equipment Check Service Number

Values:

0x00	No Equipment Check Detected
0x01 to 0x39	Common (to all devices) Equipment Check
0x40 to 0xFE	Unique (device dependent) Equipment Check
0xFF	Reserved

Common (to all devices) Equipment Check Values

Values:

0x01	Failed BAT Test (Bad System Card)
0x02	Invalid Paperport II Command Parameter
0x03	Invalid Paperport II Command Protocol
0x04	Invalid / Unsupported Paperport II Command
0x05	Device Detected Communications Error
0x06	Reserved for communication error common to all devices
0x07	Reserved for communication error common to all devices
0x08	Reserved for communication error common to all devices
0x09	UART Framing Error Detected by Device
0x0A	UART Overrun Error Detected by Device
0x0B	UART Parity Error Detected by Device
0x0C	Other UART Error Detected by Device
0x0D	Multiple UART Errors Detected by Device
0x0E	Invalid Echo
0x0F	Reserved for Engine Detected Equipment Check
0x10 – 0x2F	Reserved for Equipment Check Value Common to all Devices
0x30	Motor one error
0x40	Motor two error
0xD0	Motor three error
0x50	Hardware error

Unique (device dependent) Equipment Check Values

Values:

0x40 to 0xFE	See Individual Device's Functional Spec

Notes:

1. All devices that are capable of reporting any of the type equipment check conditions listed in the "Common (to all devices) Equipment Check Values" category should use the N1 value listed; so all devices will report the same N1 value for that "common" type error.
2. As Devices define their Equipment Check conditions they should identify if their equipment check is of a "general type" equipment check that all devices should report the same; and thus, register in the Paperport II spec as a "Common (to all devices) Equipment Check Values".
3. Equipment Check conditions that are unique to a device are categorized as "device dependent" equipment checks and should return a N1 value in the dedicated range set aside for device dependent equipment check conditions. Note: These "Equipment Check" values for N1 are not common; ie, the same number can be used for different devices (eg, tray, duplexer, stacker, etc) may use the save N1 value to indicate different equipment check conditions / situations. Example: N1 = 0x40 may mean Service Error 100 - DC Motor Failure on the duplex option and N1 = 0x40 may mean Service Error 053 - Bad Tray Size Sensing Hardware on an input tray.
4. For Equipment Check, N1, values in the range 0x40 to 0xFE, the print engine must associate this response data to the device queried and determine from that context what the specific equipment check condition is. For these specific "Equipment Check" conditions and their associated N1 values, see each individual device's functional specifications for details.

The Equipment Check status is not cleared upon transmitting the status response to this query; ie, consecutive **Query Equipment Check Status** commands should receive the same status response. The Equipment Check status is cleared whenever a **System Configuration Reset**, **Hard Reset**, **Soft Reset** or **Intervention Required Clear Notification** command is received. If the status condition exists upon completion of the processing of these commands, the respective status is set again to alert the printer of the device's status.

1.15.11 Query Number Trays Response

This is the Query Number Trays (QNT) response:

Syntax: Multi-Byte Response Length Field, N1, ...

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: 0x01-0xFF

N1 Number of Trays

Values:	0x00	No Trays
	0x01-0xFF	01 to 255 Trays Installed on Device

1. Devices that do not have any "trays" (ie; input sources for input devices or output destinations for out devices), simply reply with zero trays; 0x00.

1.15.12Query Paper Position Status Response

This is the Query Paper Position Status (QPPS) response. It is used by the printer to determine the number of sheets inside a device that the operator must clear for paper jam clearance, and their specific location. Whenever the printer issues this query, the device is to report its current paper position status.

Note: This device does not have to be the device that raised the IR condition.

Syntax: Multi-Byte Response Length Field, N1, …
0x02, N1
0x04, N1, N2, N3

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: 0x01-0FF

The specific response is dependent upon the printer's request as specified via command's N1 parameter value:

A. Printer Requested Number of Sheets in Device

Response:	N1	Number of Sheets in Device's Paperpath
	0x00	No Sheets in the Device's Paperpath
	0x01 - 0xFF	1 to 255 Sheets in Device's Paperpath

B. Printer Requested Position of Sheet in Device's Paperpath
(Response Length = 2)

Response:	N1	Position of Sheet Requested in Device's Paperpath
	0x00	Invalid Response - Sheet requested does not exist inside device's paperpath
	0x01 - 0xFF	Identifies the position along the device's paperpath for the sheet requested Note: Specific values and their meanings are device dependent. Reference the device's functional specification for details for a particular device.

(Response Length = 4)

Response:	N1	Jam Location of offending sheet in Device's Paperpath
	0x00 - 0xFF	Identifies the sensor within the device that called out a paper, homing or motor jam.

Note: Specific sensor enum values and their meanings are device dependent, and are tabulated in a separate document. The QPPS N1 response value may be used to build the YYY part of the Jam error code.

N2	Area of Sheet Requested in Device's Paperpath		
Bit 7 (MSB)	0	Area #1	- Area 1 clear.
	1		- Area 1 needs to be cleared.
Bit 6	0	Area #2	- Area 2 clear.
	1		- Area 2 needs to be cleared.
Bit 5	0	Area #3	- Area 3 clear.
	1		- Area 3 needs to be cleared.
Bit 4	0	Area #4	- Area 4 clear.
	1		- Area 4 needs to be cleared.
Bit 3	0	Area #5	- Area 5 clear.
	1		- Area 5 needs to be cleared.
Bit 2	0	Area #6	- Area 6 clear.
	1		- Area 6 needs to be cleared.
Bit 1	0	Area #7	- Area 7 clear.
	1		- Area 7 needs to be cleared.
Bit 0	0	Area #8	- Area 8 clear.
	1		- Area 8 needs to be cleared.

N3	Area of Sheet Requested in Device's Paperpath		
Bit 7 (MSB)	0	Area #9	- Area 9 clear.
	1		- Area 9 needs to be cleared.
Bit 6	0	Area #10	- Area 10 clear.
	1		- Area 10 needs to be cleared.
Bit 5	0	Area #11	- Area 11 clear.
	1		- Area 11 needs to be cleared.
Bit 4	0	Area #12	- Area 12 clear.
	1		- Area 12 needs to be cleared.
Bit 3	0	Area #13	- Area 13 clear.
	1		- Area 13 needs to be cleared.
Bit 2	0	Area #14	- Area 14 clear.
	1		- Area 14 needs to be cleared.
Bit 1	0	Area #15	- Area 15 clear.
	1		- Area 15 needs to be cleared.
Bit 0	0	Area #16	- Area 16 clear.
	1		- Area 16 needs to be cleared.

Note : The option should clear areas as they become clear (sensors uncovered). If the option cannot detect if the area is cleared (sensors uncovered) the bit should remain set until the appropriate door is opened/Closed.

Note : Each area bit corresponds to a specific jam animation. Only one bit should be set at any given time, this should be the offending jam.

The status response is reset whenever a **System Configuration Reset**, **Hard Reset**, **Soft Reset** or **Intervention Required Clear Notification** command is received. If the status condition exists upon completion of the processing of these commands, the respective response is set again to alert the printer of the device's status.

1.15.13Query Cover Open Location Response

This is the Query Cover Open Location (QCOL) response. It is used by the printer to determine the locations of the device's covers reported as open. Whenever the printer issues this query, the device is to report its current cover open location status.

Syntax: Multi-Byte Response Length Field, N1, ...

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: 0x01-0FF

N1 Device's Specific Covers Status (Open or Closed)

Bit 7 (MSB)	0	Cover #1	- Cover Not Open
	1		- Cover Open
Bit 6	0	Cover #2	- Cover Not Open
	1		- Cover Open
Bit 5	0	Cover #3	- Cover Not Open
	1		- Cover Open
Bit 4	0	Cover #4	- Cover Not Open
	1		- Cover Open
Bit 3	0	Cover #5	- Cover Not Open
	1		- Cover Open
Bit 2	0	Cover #6	- Cover Not Open
	1		- Cover Open
Bit 1	0	Cover #7	- Cover Not Open
	1		- Cover Open
Bit 0	0	Cover #8	- Cover Not Open
	1		- Cover Open

Notes:
1. For covers listed in a response byte that the devices does not have, the device should always report that cover as "not open".

2. For devices that may have more than 8 covers, this response can be extended via the length field and adding N2, N3, ... all of which have the same syntax as N1 (ie, N2 is for covers #9-16, N3 for covers #17-24, ...)

The status response is reset whenever a **System Configuration Reset**, **Hard Reset**, **Soft Reset** or **Intervention Required Clear Notification** command is received. If the status condition exists upon completion of the processing of these commands, the respective response is set again to alert the printer of the device's status.

1.15.14Query Tray Pulled Location Response

This is the Query Tray Pulled Location (QTPL) response. It is used by the printer to determine which specific tray(s) or bin(s) the device reported as "pulled out" (ie, not installed for operation). Both input and output devices provide this response if they have trays, bins, etc that store a supply (paper, stables, etc) or receive media into an output destination that can be removed making them temporarily inoperable. Whenever the printer issues this query, the device is to report its current tray pulled location status.

Syntax: Multi-Byte Response Length Field, N1, ...

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: 0x01-0FF

N1 Device's Specific Tray Pulled Status (Pulled or Pushed In)

Bit 7 (MSB)	0	Tray #1	- Tray Pushed In
	1		- Tray Pulled
Bit 6	0	Tray #2	- Tray Pushed In
	1		- Tray Pulled
Bit 5	0	Tray #3	- Tray Pushed In
	1		- Tray Pulled
Bit 4	0	Tray #4	- Tray Pushed In
	1		- Tray Pulled
Bit 3	0	Tray #5	- Tray Pushed In
	1		- Tray Pulled
Bit 2	0	Tray #6	- Tray Pushed In
	1		- Tray Pulled
Bit 1	0	Tray #7	- Tray Pushed In
	1		- Tray Pulled
Bit 0	0	Tray #8	- Tray Pushed In
	1		- Tray Pulled

Notes:
1. For trays listed in a response byte that the devices does not have, the device should always report that the tray is "pushed in".
2. For devices that may have more than 8 trays (or bins), this response can be extended via the length field and adding N2, N3, ... all of which have the same syntax as N1 (ie, N2 is for trays #9-16, N3 for trays #17-24, ...)

The status response is reset whenever a **System Configuration Reset**, **Hard Reset**, **Soft Reset** or **Intervention Required Clear Notification** command is received. If the status condition exists upon completion of the processing of these commands, the respective response is set again to alert the printer of the device's status.

1.15.15Query Device Pick Time Response

This is the Query Device Time (QDPT) response:

Syntax: Multi-Byte Response Length Field, N1, N2

Response Byte Definitions:

Length Field	Number of bytes in the response including this byte	
	Values:	0x03
N1, N2	Device's Pick Time	(N1 is Most Significant Byte)
	Values:	0x0000 - 0xFFFF where 0x0001 = 1 msec

The device's "pick time" is defined to be the time it is expected the device to take from receiving the **Pick Page** command to pick the sheet from the requested input bin and feed its leading edge to (and make) the paper path's "Pass Thru" paper sensor at the process speed specified by the most recent **Select Process Speed** command. For devices that do not have a "Pass Thru" paper sensor, they return their pick time as the time it takes the to feed the leading edge of the media to the bottom of the device above it.

The device's response is dependent upon the state of the printer system. If this query command is received immediately after a POR, Set Configuration Reset command, Hard Reset command, Soft Reset command, or IR Clear command; the device returns its "Minimum" pick time; ie, the minimum amount of time it will take for the device to pick a sheet and feed it to the first "Pass Thru" sensor for the previously received **Printer Characteristics Notification** (ie, printer model) and **Select Process Speed** to ensure the printer can manage feeding sheets and interpage gaps even if during this time an operator has altered the supplies capacity in the tray thus changing the pick time. If either or neither of these commands have been received by the device, the device will use its POR defaults for these parameters. For any other time that the Query Device Pick Time command is sent (ie, other than immediately after POR, "reset" command, or IR Clear command), the device returns its currently required pick time.

1.15.16Query Device Minimum Interpage Gap Response

This is the Query Device Minimum Interpage Gap (QDMG) response:

Syntax: Multi-Byte Response Length Field, N1, N2

Response Byte Definitions:

Length Field	Number of bytes in the response including this byte		
	Values:	0x03	
N1, N2	Device's Minimum Gap Between Sheets Specified (N1 is Most Significant Byte)		
	Values:	0x0000 - 0xFFFE	where 0x0001 = 1 mm
		0xFFFF	Device Not Ready

Output Devices:
The "minimum gap" is the minimum distance needed between the previous sheet and the sheet identified in the **Query Device Minimum Interpage Gap** command.

Input/Output Devices:
Such an example of an input/output device is the duplex option. The "minimum gap" is the minimum distance needed between two subsequent sheets to be feed into the device back to back (example: 2-4-1 sequence).

The "minimum gap" is to be specified based on the previously received **Printer Characteristics Notification** (ie, printer model) command, **Select Process Speed** command, and **Select Output Device** command (ie, the sheet's characteristics), and Submit Page command (SLPQ); along with the **Query Device Minimum Interpage Gap** command (ie, sheet's destination and "finishing" operation). For any of these commands not received by the device, the device will use its POR defaults for those parameters.

Device Not Ready (0xFFFF) Response:

If the device is not ready to receive the sheet to be picked, it returns this gap response. The engine will not pick the sheet, but instead poll the device via this command for a gap response suitable for picking the sheet (ie, a valid interpage gap ranging from 0x0000 to 0xFFFE mm).

An example of when a device may need to utilize the Device Not Ready response is a device that must move a paper stack elevator into position to accept the next sheet, and the time required to make such a mechanical movement is unpredictable; thus, making it unlikely the device could return a dependable interpage gap to the engine. In this case, the device can

utilize the Device Not Ready response "holding off" the next sheet to be picked until the elevator is in a position that allows for a predictable, dependable interpage gap to be returned to the engine. In most cases the gap response after the device has indicated a Device Not Ready response and it is now ready to accept the next sheet will be very short and very likely 0x0000 (no interpage gap required) since during the Device Not Ready "hold off" duration the gap between the current sheet and the next sheet has become sufficiently large enough unto itself.

1.15.17Query Page Complete ID Response

This is the Query Page Complete ID (QPCID) response. The device identifies via this response the *Page ID* of the sheet of which it just reported page complete via the *"Page In Output Device"* signal in the **Query Device Status** general status response. The Page ID value specified in this response is to be the same Page ID for this sheet assigned by the **Select Output Device** command.

Syntax: Multi-Byte Response Length Field, N1
0x02, N1

Response Byte Definitions:

Length Field Number of bytes in the response including the count byte
Values: Range: 0x01 to 0xFF

N1 Page ID
Values: Range: 0x01 to 0xFF

Definition of *Page Is Complete*: The page has safely and successfully reached its output destination (ie, bin).

1.15.18Query Device Quiesce Response

This is the Query Device Quiesce (QDQ) Response:

Syntax: Single Byte, Not Bit Encoded

Values:			
	0x00	Device Mechanism Quiesce	
	0x80	Device Mechanism Not Quiesce	
	0x81	Device Processing - Not Quiesce	Compatibility Level 5
	0x01-0x7F	Reserved	
	0x82-0xFF	Reserved	

Device Mechanism Quiesce is defined as 100% of the device's mechanism is no longer moving; ie, no movement of drive systems, pick mechanisms, deflector mechanisms, motors, etc.

Device Mechanism Not Quiesce is returned if any of the device's motors are running.
Device Processing – Not Quiesce is returned as a handshaking method to indicate that recalculation of critical variables and NVRAM activity is ongoing but device's mechanism is no longer moving.

1.15.19Query Debug Message Response

This is the Query Debug Message (QDM) response:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, ...

Response Byte Definitions:

Length Field	Number of bytes in the response including the count byte *Values:* Range: 0x01 to 0xFF
N1	Debug Byte #1 *Values:* Range: 0x01 to 0xFF
N2	Debug Byte #2 *Values:* Range: 0x01 to 0xFF
N3	Debug Byte #3 *Values:* Range: 0x01 to 0xFF
N4	Debug Byte #4 *Values:* Range: 0x01 to 0xFF
N5	Debug Byte #5 *Values:* Range: 0x01 to 0xFF
N6	Debug Byte #6 *Values:* Range: 0x01 to 0xFF
N7	Debug Byte #7 *Values:* Range: 0x01 to 0xFF
N8	Debug Byte #8 *Values:* Range: 0x01 to 0xFF

For the specific definitions, values, and meanings of the Debug Bytes; see the device's Functional Specification.

After sending the QDM response, the device can "clear" its debug bytes at its discretion. Many devices choose to clear them as part of **System Configuration Reset**, **Hard Reset**, **Soft Reset** and **Intervention Required Clear** processing.

Optra s Printer Implementation / Limitation Notes:

1 Device's response is limited to 3 bytes: Length, DataByte1, DataByte2
Engine will send the command consecutively sufficient times to obtain the complete response from the device. The device upon each subsequent command is to send the next 2 response bytes.

Printer Transmissions	Device's QDM Response
1	length=03, D1, D2
2	length =03, D3, D4
3	length=03, D5, D6
4	length=03, D7, D8

2 During the 4 sets of printer transmissions (see note #1 above), the engine may intermingle other Paperport II commands.

3 For any of the following conditions the device resets its counter "pacing" the 8 byte transmission (see note #1 above) to zero: Engine raised Link Reset, Device raised Link Reset, System Configuration Reset command, Hard Reset command or Soft Reset command.

4 All smart options except the Finisher (ie, Mailbox) option only report back 8 debug message bytes. The Finisher reports back 16 debug message bytes. Notes 1 - 3 above apply the same for the Finisher option, except on a 16 byte boundary instead of the 8 byte boundary indicated.

1.15.20Query Diagnostic Test Complete Response

This is the Query Diagnostic Test Complete (QDTC) Response used by the device to indicate when it has completed the execution of the previous selected diagnostic test (selected via the **Diagnostic Test** command):

Syntax: Single Byte, Not Bit Encoded

Values:

0x00	Diagnostic Test Complete
0x80	Diagnostic Test Not Complete
0x01-0x7F	Reserved
0x81-0xFF	Reserved

1.15.21Query Diagnostic Test Data Message Response

This is the Query Diagnostic Test Data Message (QDTDM) Response used by the device to report the resultant data for a specific diagnostic test to be displayed on the printer's op panel.

Syntax: Length Field, N1, D1, D2, D3, ... Dn

Length Field Number of bytes in the response including the length field
Values: Range: 0x01 to 0xFF

N1 Number of Valid Data Bytes
Values: 0x00 - 0xFF
Note: Only valid data bytes are to be displayed on the printer's operator panel; invalid bytes are ignored)

D1 Diagnostic Test Data Message Byte #1
Values: 0x00 - 0xFF

D2 Diagnostic Test Data Message Byte #2
Values: 0x00 - 0xFF

: :
: :

Dn Diagnostic Test Data Message Byte #n
Values: 0x00 - 0xFF

For specific definitions, values, and meanings of the Diagnostic Test Data Bytes; see the device's Functional Specification. For a diagnostic test of which there is no diagnostic test data, the device returns a "clear" response (ie, all zeros: N1 = 0x00, D1 thru Dn = 0x00) when the engine sends this command.

Optra S Printer Implementation / Limitation Notes:

1 Device's response is limited to 3 bytes: Length, DataByte1, DataByte2
Engine will send the command consecutively sufficient times to obtain the complete response from the device. The device upon each subsequent command is to send the next 2 response bytes.

Printer Transmissions	Device's QDM Response
1	length=03, N1, D1
2	length =03, D2, D3
3	length=03, D4, D5
4	length=03, D6, D7

2 During the 4 sets of printer transmissions (see note #1 above), the engine may intermingle other Paperport II commands.

3 For any of the following conditions the device resets its counter "pacing" the 8 byte transmission (see note #1 above) to zero: Engine raised Link Reset, Device raised Link Reset, System Configuration Reset command, Hard Reset command or Soft Reset command.

1.15.22Query Aux Power Status Response

This is the Query Aux Power Status (QAPS) Response:

Syntax: Single Byte, Not Bit Encoded

Values:		
	0x00	Auxiliary Power On
	0x01	Auxiliary Power Not On
	0x02-0xFF	Reserved

Devices that support this command and do Not have an auxiliary power supply should always report *"Auxiliary Power On"*.

1.15.23Query Operation Complete Response

This is the Query Operation Complete (QOC) Response:

Syntax: Multi-byte Response Length Field, N1, N2, N3

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 Status Response
0x00 Operation Not Complete
0x80 Operation Complete with All Finisher Operations Executed Successfully
0x81 Operation Complete without All Finisher Operations Executed Successfully
0x01 - 0x7F Reserved
0x82 - 0xFF Reserved

Note : N2 has a different set of possible values for different options.

N2 Exception Response (For MONO Stapler Option)

0x00 No Exception Response
0x01 Other
0x02 Out of Staples
0x03 Paper Jam
0x04 Stapler Jam
0x05 Priming Failed
0x06 No Media In Stapler
0x07 Accumulator Full – Sheets Were Flushed
0x08 Media Size Changed – Sheets Were Flushed
0x09 - 0xFE Reserved
0xFF Exception Response given in next byte

N2 Exception Response (For COLOR Finisher Option)
- 0x00 No Exception Response
- 0x01 Other
- 0x02 Out of Staples
- 0x03 Single Sheet Staple Job Requested
- 0x04 Flushed Accumulator due to Too Many Sheets
- 0x05 Device Instructed by Printer to Flush Print Job
- 0x06 Bin Change Requested
- 0x07 Paper Jam
- 0x08 Stapler Jam
- 0x09 - 0xFE Reserved
- 0xFF Exception Response given in next byte

N3 Exception Response byte #2 (not implemented)

Status Response

The device reports back to the printer via this byte if it has completed the "finishing operation" for this print job and if the "finishing operation" was executed successfully or not. Note: the device actually may only detect if the "finishing operation" was not executed successfully and thus by deduction return a response that the "finishing operation" was completed successfully due to none of the detectable failure conditions were detected. Thus, there is some room for error in this status response of which the device in good faith will report that the "finishing operation" has been executed successfully when in fact it has not since the device may not be explicitly capable of detecting that.

When the device needs to report to the printer "not complete", it may either return: 0x02, 0x00 or 0x03, 0x00, 0x00.

Exception Response

The device reports back to the printer via this byte the specific reason it did not execute the "finishing operation" requested.

The device will only report the last reason why the "finishing operation" was not executed successfully; ie, the device will not accumulate and report multiple reasons, if they exist, for why the "finishing operation" was not executed.

The exception response = 0x01, *Other*, is used by the device when none of the other specific exception responses listed apply or when the device an not explicitly identify the cause for not executing the "finishing operation".

Notes:

1. Status bytes N1 and N2 are reset (ie, N1=0x00 & N2=0x00) by the device once the **Query Operation Complete** response byte is transmitted back to the printer.
2. Devices that support this command and do Not have a "finishing" operation should always report *"Operation Complete"*.

1.15.24Query Nvram Read / Write Complete Response

This is the Query Nvram Read / Write Complete (QNRWC) Response:

Syntax: Single Byte, Not Bit Encoded

Values:		
	0x00	Nvram Read / Write Complete
	0x80	Nvram Read / Write Not Complete
	0xFF	Nvram Read / Write Aborted
	0x01-0x7F	Reserved
	0x81-0xFE	Reserved

Nvram Read/Write Complete or Not Complete response is returned by the device for the Query Nvram Read/Write Command to indicate to the print engine if the device has (or has not) completed one of the following requested operations:

1. Read Device's Nvram
2. Write Device's Nvram
3. Restore Device's Nvram Data Area

Note: At this time no devices support the 0xFF, Nvram Read / Write Aborted response.

1.15.25Transmit Requested Read Data Response

This is the Transmit Requested Read Data (TRRD) Response used by the device to transmit the data from the device's Nvram requested by the Engine via the previous **Read Nvram Data** command.

Syntax: Length Field, D1, D2, D3, ... Dn

Length Field	Number of bytes in the response including the length field
Values:	Range: 0x01 to 0xFF
D1, D2, ... Dn3	Data Byte 1, 2 ... n respectively
Values:	0x00 - 0xFF

Optra s Printer Implementation / Limitation Notes:

Optra s Printer Implementation / Limitation Notes:

1 Due to Engine ram constraints the **Read Device's Nvram** command will only request a maximum of 4 data bytes; thus, the Device's response is limited to a maximum of 5 bytes:
Length, DataByte1, DataByte2, DataByte3, DataByte4

1.15.26Query Device Characteristics Response

1.15.26.1 N1=0x90 -- Lighting System

This is the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x90 to query the device characteristics of its lights. The device shall respond with the number of lights present, the supported level of control for lights, and the location of each light. The maximum number of lights is equal to the maximum number of bins/trays: 128 (as defined by the Query Device Characteristics command) + 2 + 16 + 14 (doors or covers) = 160. The +2 is to allow for 1 light to be below the bottom bin/tray and for 1 light to be above the top bin/tray and the +16 is to allow up 16 beacon lights at custom locations to be defined by the implementation.

Syntax: Multi-Byte Response Length Field, N1, N2, N3, …..

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xA2

N1 Number of Lights Present
Values:
0x00 0 Lights Present
0x01 1 Lights Present
0x02 2 Lights Present
0x03 3 Lights Present
…
0xA0 160 Lights Present

N2 Control Level Supported
Values:
0x00 Control Not Supported
0x01 Control Level 1
Supports Turn On, Turn Off, and Set Global Intensity
0x02 Control Level 2
Supports Control Level 1, Set Light to Flashing, and Set Global Simple Flashing Rate
0x03 Control Level 3
Supports Control Level 2 and Set Global Synchronized Flashing Rate

0x04-0xFF Reserved

N3, N4, …, Nx Location of Light 1, 2, …, x

Values:

0x00 Light is located below the bottom most bin/tray
0x01 Light is located at bin/tray #1
0x02 Light is located at bin/tray #2
0x03 Light is located at bin/tray #3
…
0x80 Light is located at bin/tray #128

0xA0 Custom location - Beacon Light #1
0xA1 Custom location - Beacon Light #2
0xA2 Custom location - Beacon Light #3
0xA3 Custom location - Beacon Light #4
….
0xAF Custom location - Beacon Light #16

0xB0 Beacon illuminates device below's door
0xB1 Beacon is located at door or cover #1
0xB2 Beacon is located at door or cover #2
0xB3 Beacon is located at door or cover #3
…
0xB8 Beacon is located at door or cover #8
0xB9 Beacon illuminates device above's door
0xBA Beacon illuminates device to the left's door
0xBB Beacon illuminates device to the right's door
0xBC Beacon illuminates device to the front's door
0xBD Beacon illuminates device do the back's door

0xFF Light is located above the top most bin/tray

1.15.26.2 N1=0x91 -- Diverter Mechanism

This is the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x91 to query the type of diverter mechanism. The response to this command will indicate if the option has a motor reversing actuated diverter or if it has diverter that is actuated by a solenoid controlled wrap spin clutch mechanism. The engine will use this information to deliver the necessary gaps needed between pages. In the future the response to this query may be expanded to include any necessary diverter timing data which may be useful to the engine.

Syntax: Multi-Byte Response Length Field, N1

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: 0x02

N1 Type of Diverter Mechanism
Values:
0x00 Motor Reversing
0x01 Solenoid Controlled

1.15.26.3 N1=0x92 -- In-System Programming

This is the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x92 to query if an option supports in-system programming. In the future the response to this query may be expanded to include details related to the protocols needed to successfully program an option. For example the rate at which programming data maybe sent to the option could be specified here.

Syntax: Multi-Byte Response Length Field, N1

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: 0x02

N1 Indicates if in-system programming is supported
Values:
0x00 In-System Programming is not supported
0x01 In-System Programming is supported

1.15.27Query Dot XX Error Code Response

This is the Query DotXX Error Code (QDXX) Response:

Command Syntax: See section: 1.11.22

Response: Multi-Byte Response Length Byte, N1

Response Byte Definitions:

Length Field Number of bytes in the response including the count byte
Values: 0x02

N1 Error Code
Values():

0x19-0x63	(25-99) – dot XX error code for paper jams
0x01-0x63	(1-99) – dot XX error code for equipment checks

The dot XX error codes for paper jams starts at 25. This will allow the error codes 00-24 to be used by the engine code. The above response values are defined for mono options only at this time. The range of valid values for color options is TBD.

1.16 DEVICE-DEPENDENT RESPONSES

Described by section for each unique device type are the responses to commands that are unique for that device type.

1.16.1 Trays Options

The following sections give the specific unique responses by command for Trays Options including the High-Capacity Input Tray (HCIT) .

1.16.1.1 Query Device Characteristics Response

1.16.1.1.1 QDC Response for Smart Tray

This is the Query Device Characteristics (QDC) Response for a Smart Tray:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5
0x06, N1, N2, N3, N4, N5

Response Byte Definitions:

Length Field	Number of bytes in the response including this byte *Values:* Range: 0x01 to 0xFF
N1, N2, N3, N4, N5	Device Characteristics for Tray # requested by printer via command)

N1 Device Characteristics

Values:	Bit Encoded	
Bit 7 (MSB)	0	Pick / Feed Times Not Consistent
	1	Pick / Feed Times Consistent
Bit 6	0	Media Size Is Changeable
	1	Media Size Is Not Changeable
Bit 5	0	No Input Tray Empty (Paper Out) Detection (engine algorithm must detect paper out condition)
	1	Input Tray Empty (Paper Out) Detection
Bit 4	0	Tray Not Lockable
	1	Tray Lockable
Bit 3	0	No Media Size Sensing
	1	Media Size Sensing
Bit 2	0	Transparency Not Supported
	1	Transparency Supported
Bit 1	0	Paper Not Supported
	1	Paper Supported
Bit 0	0	Envelope Not Supported
	1	Envelope Supported

Pick / Feed Times Consistent

If the device's pick & feed times from sheet to sheet are very consistent, it sets this bit to notify the print engine an "adaptive feed algorithm" for managing sheet's interpage gap can be utilized. If the device's pick & feed times from sheet to sheet are not consistent, it negates this bit to notify the print engine the best feed algorithm to use is a fixed time or gap. Device developers who need help in quantifying if their device provides a consistent pick & feed times should contact the Print Engine development group at Lexmark.

Media Size Changeable

At times a print job will select a specific input tray along with a specific media size. Normally, if the device's tray selected for the print job does not contain the selected size media, the printer system will prompt the user to change the media size in the device to that requested by the print job. However, some devices due to their techonology may not be able to allow the user to change the size of the media loaded in the input tray. For example, the device may require a "homing" of its mechanism when media is loaded; or in order for the user to physically set the device's media size sensing hardware to the new size, it requires the device be removed from the printer system, etc. Thus, some devices may need to indicate to the print engine that it can or can not support the printer system prompting the user to dynamically change the size media while a print job is in process. This bit allows the device to indicate such device characteristics to the print engine. Thus, devices that can support media size changes at any time notify that characteristic to the print engine by negating this bit. While devices that can not support media size changes at any time (but only a prescribed time; such as, installation, idle, etc) set this bit to indicate this characteristic to the print engine.

Paper Out Detection Capability

The input device negates this bit if it can Not detect a paper out condition; thus, requiring the printer to utilize an algorithm to detect a paper out condition for an input tray. The input device asserts this bit if it can detect and report paper out conditions for the input tray.

Tray Lockable

If the tray can be locked via the **Tray Lock Notification** command, the device asserts this bit to indicate this capability. If the device can not support locking this tray, then it negates this bit.

Media Size Sensing

If the device supports automatic media size sensing for this tray, the device asserts this bit to indicate this capability. If the device can not automatically determine the media's size, then it negates this bit (and the print engine must determine the installed media's size by some other means; via the operator panel or some other way).

Transparency Media Supported

If the device supports feeding transparencies from this tray, the device asserts this bit to indicate this capability. If the device can not support feeding transparencies from this tray, then it negates this bit.

Paper Media Supported

If the device supports feeding paper from this tray, the device asserts this bit to indicate this capability. If the device can not support feeding paper from this tray, then it negates this bit.

Envelope Media Supported

If the device supports feeding envelopes from this tray, the device asserts this bit to indicate this capability. If the device can not support feeding envelopes from this tray, then it negates this bit.

N2 Number of Reportable Levels for Empty

Values:

0x00	No Media Capacity Detection Capability
0x01	N/A
0x02	2 Levels of Detection (0% and 100%)
0x03	3 Levels of Detection (0%, 0%<L_2%<100%, 100%)
0x04	4 Levels of Detection (0%, 0%<L_2%<L_3%, L_2%<L_3%<100%, 100%)
:	:
:	:

Note: The accuracy of the multi-level detection is device dependent. For specific details, reference the specific device's functional specification.

N3 Early Warning Level Reported

Values:

0x00	No Early Warning Detection Capability
0x01 - 0x63	1% to 99% Is When Early Warning Reported
0x64 - 0xFF	reserved

N4, N5 Tray's Maximum Media Capacity
Values: Range: 0x0001 to 0xFFFF
where 0x0001 = Capacity of 1 sheet, and
0xFFFF = Capacity of 65,535 sheets

Notes:
1. If the printer solicits the characteristics of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.

1.16.1.1.2 QDC Response for Smart Tray's Media Length Zone Sensor

This the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x93 to query the device characteristics of its media zone sensor:

Note that this QDC response uses command parameter N2 also.

Syntax: Multi-Byte Responses 2, N1
5,N1,N2,N3,N4

Response Byte Definitions:

For command parameter N2 = 0 : (number of zones)

Syntax: Multi-Byte Response 2, N1

Length Field = 2

N1 Number of Zones
Values : 0x01 to 0xFF

For command parameter N2 = 0x01 – 0xFF (zone specific data, where command data N2 indicates which zone, N2 = 0x01 – 0xFF)

Length Field = 5

N1 LSByte of Minimum length for selected Zone in mm
N2 MSByte of Minimum length for selected Zone in mm
N3 LSByte of Maximum length for selected Zone in mm
N4 MSByte of Maximum length for selected Zone in mm

1.16.1.2 Query Device Inventory Response

1.16.1.2.1 QDI Response for Smart Tray

This is the Query Device Inventory (QDI) Response for a Smart Tray:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5 ...

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2, N3, N4 — Inventory For requested Tray # (Compatibility Level <= 7)
N1, N2, N3, N4,N5,N6 — Inventory For requested Tray # (Compatibility Level >= 8)

N1 — Tray's Current Paper Capacity
Values:
0x00 - 0x64 0 to 100% Full
0x65 - 0xFF reserved

N2 — Tray's Current Paper Size Setting
Values:
0x00 Unknown Media Size (Compatibility Level >= 8)
0x02 A4 Media Size
0x04 Legal Media Size
0x06 B5 Media Size
0x08 A5 Media Size
0x0A Executive Media Size
0x0C Letter Media Size
0x0E Custom Media Size
0x10 Statement Media Size (Compatibility Level >= 8)
0x11 Folio Media Size (Compatibility Level >= 8)
0x12 A6 Media Size(Compatibility Level >= 8)
0x13 Oficio Media Size (Compatibility Level >= 8)
All Other Values < 0x80 reserved for non-envelope sizes
0x80 Envelope (Compatibility Level >= 8)
0x81 No. 9 Envelope (Compatibility Level >= 8)
0x82 No. 10 Envelope (Compatibility Level >= 8)
0x83 DL Envelope (Compatibility Level >= 8)

0x84 C5 Envelope (Compatibility Level >= 8)
0x85 B5 Envelope (Compatibility Level >= 8)
0x86 to 0xFF reserved for envelope sizes

N3 Tray's Current Paper Orientation Setting (relative to feed direction)
Values: 0x00-0xFF Reserved (not defined at this time)

N4 Tray Characteristics
Values: Bit Encoded

Bit 7 (MSB)	0	Reserved; Always 0	
Bit 6	0	Tray Not Locked	Reserved; Always 0
	1	Tray Locked	
Bit 5	0	Page is NOT Transparency	
	1	Page is Transparency	
Bit 4	0	Reserved; Always 0	
Bit 3	0	Reserved; Always 0	
Bit 2	0	Reserved; Always 0	
Bit 1	0	Reserved; Always 0	
Bit 0	0	Reserved; Always 0	

Note: Existing Mono trays do not return N4.

N5,N6 Tray's Current Measured Media Length (Compatibility Level >= 8)
N5 = Most significant byte
N6 = Least significant byte
Units: 0.0625 (1/16) mm
Values: 0x0000 = Measured media length not supported.
0x0001 = Measured media length unknown.
0x0002 to 0xFFFF = 0.125 mm to 4095.9375 mm

Notes:

1. The device always returns the inventory of the tray requested independent if an inventory change has occurred or not.
2. If the printer solicits the inventory of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.
3. The device does not directly alert the printer it has reached its "Early Warning" condition (for example: Input Tray Almost Empty for input devices or Output Bin Almost Full for output devices); but instead simply raise a Non-Critical Status in the Query Device Status 1 response when the supply has reached the "early warning" threshold level (ie, the value identified by the Query Device Characteristics response). The printer upon querying the inventory of the device and detecting the device's supply level matches that of the "early warning" level reported in the Query Device Characteristics response can post to the printer's operator panel a message alerting the customer of the early warning condition.
4. If a tray has size sensing hardware it may choose not to support reporting measured media length in N5,N6.

5. If a tray has size sensing hardware it may not be able to detect sizes listed as N2 >= 0x10.
6. If a tray does not have size sensing hardware it may choose to infer values for N2 from the media length that it measures.
7. If a tray has not yet had a chance to measure the media it should return N5,N6 = 0x0001 and N2 = 0x00.

1.16.1.2.2 QDI Response for Smart Tray's Media Length Zone Sensor

This the Query Device Inventory (QDI) Response for an input tray option when the engine has sent a QDI command with parameter N1 = 0x94 to query the currently detected media length zone:

Syntax: Multi-Byte Response Length Field, N1

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 Currently selected zone

Values:		
	0x01 - 0xff	selected zone (if known)
	0x00	Unknown or unavailable

1.16.2 Feeders

The following sections give the specific unique responses by command for Smart Feeders.

1.16.2.1 Query Device Characteristics Response

This is the Query Device Characteristics (QDC) Response for a Smart Feeders:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5
0x06, N1, N2, N3, N4, N5

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2, N3, N4, N5 Device Characteristics for Tray #
(requested by printer via command)

N1 Device Characteristics

Values:	Bit Encoded	
Bit 7 (MSB)	0	Pick / Feed Times Not Consistent
	1	Pick / Feed Times Consistent
Bit 6	0	Media Size Is Changeable
	1	Media Size Is Not Changeable
Bit 5	0	No Input Tray Empty (Paper Out) Detection (engine algorithm must detect paper out condition)
	1	Input Tray Empty (Paper Out) Detection
Bit 4	0	Tray Not Lockable
	1	Tray Lockable
Bit 3	0	No Media Size Sensing
	1	Media Size Sensing
Bit 2	0	Reserved; Always 0
Bit 1	0	Paper Not Supported
	1	Paper Supported
Bit 0	0	Envelope Not Supported
	1	Envelope Supported

Pick / Feed Times Consistent

If the device's pick & feed times from sheet to sheet are very consistent, it sets this bit to notify the print engine an "adaptive feed algorithm" for managing sheet's interpage gap can be utilized. If the device's pick & feed times from sheet to sheet are not consistent, it negates this bit to notify the print engine the best feed algorithm to use is a fixed time or gap. Device developers who need help in quantifying if their device provides a consistent pick & feed times should contact the Print Engine development group at Lexmark.

Media Size Changeable

At times a print job will select a specific input tray along with a specific media size. Normally, if the device's tray selected for the print job does not contain the selected size media, the printer system will prompt the user to change the media size in the device to that requested by the print job. However, some devices due to their techonology may not be able to allow the user to change the size of the media loaded in the input tray. For example, the device may require a "homing" of its mechanism when media is loaded; or in order for the user to physically set the device's media size sensing hardware to the new size, it requires the device be removed from the printer system, etc. Thus, some devices may need to indicate to the print engine that it can or can not support the printer system prompting the user to dynamically change the size media while a print job is in process. This bit allows the device to indicate such device characteristics to the print engine. Thus, devices that can support media size changes at any time notify that characteristic to the print engine by negating this bit. While devices that can not support media size changes at any time (but only a prescribed time; such as, installation, idle, etc) set this bit to indicate this characteristic to the print engine.

Paper Out Detection Capability

The input device negates this bit if it can Not detect a paper out condition; thus, requiring the printer to utilize an algorithm to detect a paper out condition for an input tray. The input device asserts this bit if it can detect and report paper out conditions for the input tray.

Tray Lockable

If the tray can be locked via the **Tray Lock Notification** command, the device asserts this bit to indicate this capability. If the device can not support locking this tray, then it negates this bit.

Media Size Sensing

If the device supports automatic media size sensing for this tray, the device asserts this bit to indicate this capability. If the device can not automatically determine the media's size, then it negates this bit (and the print engine must determine the installed media's size by some other means; via the operator panel or some other way).

Paper Media Supported

If the device supports feeding paper from this tray, the device asserts this bit to indicate this capability. If the device can not support feeding paper from this tray, then it negates this bit.

Envelope Media Supported

If the device supports feeding envelopes from this tray, the device asserts this bit to indicate this capability. If the device can not support feeding envelopes from this tray, then it negates this bit.

N2 Number of Reportable Levels for Empty

Values:

0x00	No Media Capacity Detection Capability
0x01	N/A
0x02	2 Levels of Detection (0% and 100%)
0x03	3 Levels of Detection (0%, 0%<L_2%<100%, 100%)
0x04	4 Levels of Detection (0%, 0%<L_2%<L_3%, L_2%<L_3%<100%, 100%)
:	:
:	:

Note: The accuracy of the multi-level detection is device dependent. For specific details, reference the specific device's functional specification.

N3 Early Warning Level Reported

Values:

0x00	No Early Warning Detection Capability
0x01 - 0x63	1% to 99% Is When Early Warning Reported
0x64 - 0xFF	reserved

N4, N5 Tray's Maximum Media Capacity
Values: Range: 0x0001 to 0xFFFF
where 0x0001 = Capacity of 1 sheet, and
0xFFFF = Capacity of 65,535 sheets

Notes:
1. If the printer solicits the characteristics of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.

1.16.2.2 Query Device Inventory Response

This is the Query Device Inventory (QDI) Response for a Smart Feeders:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5 ...

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2, N3, N4 Inventory For Tray # (requested by printer via command)

N1 Tray's Current Paper Capacity
Values: 0x00 - 0x64 0 to 100% Full
0x65 - 0xFF reserved

N2 Tray's Current Paper Size Setting
Values: 0x02 A4 Paper size
0x04 Legal Paper size
0x06 B5 Paper size
0x08 A5 Paper size
0x0A Executive Paper size
0x0C Letter Paper size
0x0E Custom Paper size
All Other Values Reserved

N3 Tray's Current Paper Orientation Setting (relative to feed direction)
Values: 0x00-0xFF Reserved (not defined at this time)

N4 Tray Characteristics Reserved
Values: Bit Encoded

Bit 7 (MSB)	0	Reserved; Always 0	
Bit 6	0	Tray Not Locked	Reserved; Always 0

	1	Tray Locked
Bit 5	0	Reserved; Always 0
Bit 4	0	Reserved; Always 0
Bit 3	0	Reserved; Always 0
Bit 2	0	Reserved; Always 0
Bit 1	0	Reserved; Always 0
Bit 0	0	Reserved; Always 0

Note: N4 is reserved. At this time No Smart Feeder has any Tray Characteristics to report back to the engine; therefore, this byte will be returned as 0x00 or Not at all.

Notes:

1. The device always returns the inventory of the tray requested independent if an inventory change has occurred or not.
2. If the printer solicits the inventory of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.
3. The device does not directly alert the printer it has reached its "Early Warning" condition (for example: Input Tray Almost Empty for input devices or Output Bin Almost Full for output devices); but instead simply raise a Non-Critical Status in the Query Device Status 1 response when the supply has reached the "early warning" threshold level (ie, the value identified by the Query Device Characteristics response). The printer upon querying the inventory of the device and detecting the device's supply level matches that of the "early warning" level reported in the Query Device Characteristics response can post to the printer's operator panel a message alerting the customer of the early warning condition.

1.16.3 Envelope Feeders

The following sections give the specific unique responses by command for Smart Envelop Feeders.

1.16.3.1 Query Device Characteristics Response

This the Query Device Characteristics (QDC) Response for a Smart Envelope Feeders:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5
0x06, N1, N2, N3, N4, N5

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2, N3, N4, N5 Device Characteristics for Tray #
(requested by printer via command)

N1 Device Characteristics

Values:	Bit Encoded	
Bit 7 (MSB)	0	Pick / Feed Times Not Consistent
	1	Pick / Feed Times Consistent
Bit 6	0	Media Size Is Changeable
	1	Media Size Is Not Changeable
Bit 5	0	No Input Tray Empty (Envelope Out) Detection (engine algorithm must detect paper out condition)
	1	Input Tray Empty (Envelope Out) Detection
Bit 4	0	Tray Not Lockable
	1	Tray Lockable
Bit 3	0	No Media Size Sensing
	1	Media Size Sensing
Bit 2	0	Reserved; Always 0
Bit 1	0	Paper Not Supported
	1	Paper Supported
Bit 0	0	Envelope Not Supported
	1	Envelope Supported

Pick / Feed Times Consistent

If the device's pick & feed times from sheet to sheet are very consistent, it sets this bit to notify the print engine an "adaptive feed algorithm" for managing sheet's interpage gap can be utilized. If the device's pick & feed times from sheet to sheet are not consistent, it negates this bit to notify the print engine the best feed algorithm to use is a fixed time or gap. Device developers who need help in quantifying if their device provides a consistent pick & feed times should contact the Print Engine development group at Lexmark.

Media Size Changeable

At times a print job will select a specific input tray along with a specific media size. Normally, if the device's tray selected for the print job does not contain the selected size media, the printer system will prompt the user to change the media size in the device to that requested by the print job. However, some devices due to their techonology may not be able to allow the user to change the size of the media loaded in the input tray. For example, the device may require a "homing" of its mechanism when media is loaded; or in order for the user to physically set the device's media size sensing hardware to the new size, it requires the device be removed from the printer system, etc. Thus, some devices may need to indicate to the print engine that it can or can not support the printer system prompting the user to dynamically change the size media while a print job is in process. This bit allows the device to indicate such device characteristics to the print engine. Thus, devices that can support media size changes at any time notify that characteristic to the print engine by negating this bit. While devices that can not support media size changes at any time (but only a prescribed time; such as, installation, idle, etc) set this bit to indicate this characteristic to the print engine.

Paper Out Detection Capability

The input device negates this bit if it can Not detect a paper out condition; thus, requiring the printer to utilize an algorithm to detect a paper out condition for an input tray. The input device asserts this bit if it can detect and report paper out conditions for the input tray.

Tray Lockable

If the tray can be locked via the **Tray Lock Notification** command, the device asserts this bit to indicate this capability. If the device can not support locking this tray, then it negates this bit.

Media Size Sensing

If the device supports automatic media size sensing for this tray, the device asserts this bit to indicate this capability. If the device can not automatically determine the media's size, then it negates this bit (and the print engine must determine the installed media's size by some other means; via the operator panel or some other way).

Paper Media Supported

If the device supports feeding paper from this tray, the device asserts this bit to indicate this capability. If the device can not support feeding paper from this tray, then it negates this bit.

Envelope Media Supported

If the device supports feeding envelopes from this tray, the device asserts this bit to indicate this capability. If the device can not support feeding envelopes from this tray, then it negates this bit.

N2 Number of Reportable Levels for Empty

Values:

0x00	No Media Capacity Detection Capability
0x01	N/A
0x02	2 Levels of Detection (0% and 100%)
0x03	3 Levels of Detection (0%, 0%<L_2%<100%, 100%)
0x04	4 Levels of Detection (0%, 0%<L_2%<L_3%, L_2%<L_3%<100%, 100%)
:	:
:	:

Note: The accuracy of the multi-level detection is device dependent. For specific details, reference the specific device's functional specification.

N3 Early Warning Level Reported

Values:

0x00	No Early Warning Detection Capability
0x01 - 0x63	1% to 99% Is When Early Warning Reported
0x64 - 0xFF	reserved

N4, N5 Tray's Maximum Media Capacity

Values: Range: 0x0001 to 0xFFFF
where 0x0001 = Capacity of 1 sheet, and
0xFFFF = Capacity of 65,535 sheets

Notes:

1. If the printer solicits the characteristics of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.

1.16.3.2 Query Device Inventory Response

This is the Query Device Inventory (QDI) Response for a Smart Envelope Feeders:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5 ...

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2, N3, N4 Inventory For Tray # (requested by printer via command)

N1 Tray's Current Paper Capacity
Values: 0x00 - 0x64 0 to 100% Full
0x65 - 0xFF reserved

N2 Tray's Current Paper Size Setting
Values: 0x0E Custom Envelope size
All Other Values Reserved

N3 Tray's Current Paper Orientation Setting (relative to feed direction)
Values: 0x00-0xFF Reserved (not defined at this time)

N4 Tray Characteristics Reserved
Values: Bit Encoded

Bit 7 (MSB)	0	Reserved; Always 0	
Bit 6	0	Tray Not Locked	Reserved; Always 0
	1	Tray Locked	
Bit 5	0	Reserved; Always 0	
Bit 4	0	Reserved; Always 0	
Bit 3	0	Reserved; Always 0	
Bit 2	0	Reserved; Always 0	
Bit 1	0	Reserved; Always 0	
Bit 0	0	Reserved; Always 0	

Note: N4 is reserved. At this time No Smart Envelope Feeder has any Tray Characteristics to report back to the engine; therefore, this byte will be returned as 0x00 or Not at all.

Notes:

1. The device always returns the inventory of the tray requested independent if an inventory change has occurred or not.
2. If the printer solicits the inventory of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.
3. The device does not directly alert the printer it has reached its "Early Warning" condition (for example: Input Tray Almost Empty for input devices or Output Bin Almost Full for output devices); but instead simply raise a Non-Critical Status in the Query Device Status 1 response when the supply has reached the "early warning" threshold level (ie, the value identified by the Query Device Characteristics response). The printer upon querying the inventory of the device and detecting the device's supply level matches that of the "early warning" level reported in the Query Device Characteristics response can post to the printer's operator panel a message alerting the customer of the early warning condition.

1.16.4 Duplexer

The following sections give the specific unique responses by command for the Duplex option.

1.16.4.1 Query Device Characteristics Response

This the Query Device Characteristics (QDC) Response for the Duplex option:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4
0x04, N1, N2, N3
0x05, N1, N2, N3, N4 (Compatibility Level 7, mono)

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 Device Characteristics
Values: Bit Encoded

Bit 7 (MSB)	0	Pick / Feed Times Not Consistent
	1	Pick / Feed Times Consistent
Bit 6	0	Reserved; Always 0

Bit 5	0	Reserved; Always 0
Bit 4	0	Reserved; Always 0
Bit 3	0	Reserved; Always 0
Bit 2	0	Reserved; Always 0
Bit 1	0	Reserved; Always 0
Bit 0	0	Reserved; Always 0

Pick / Feed Times Consistent

If the device's pick & feed times from sheet to sheet are very consistent, it sets this bit to notify the print engine an "adaptive feed algorithm" for managing sheet's interpage gap can be utilized. If the device's pick & feed times from sheet to sheet are not consistent, it negates this bit to notify the print engine the best feed algorithm to use is a fixed time or gap. Device developers who need help in quantifying if their device provides a consistent pick & feed times should contact the Print Engine development group at Lexmark.

N2, N3 Minimum Gap Between First 2 Sheets Into the Duplexer for the 2-4-1-6-3-8- ... Sequence

The Minimum Gap is provided for the process speed given by the last **Printer Characteristics Notification** and **Select Process Speed** command. If either or neither of these commands have been received the device will base the Minimum Gap off its power-on reset default values for these parameters.

(N2 Most Significant Byte)

Values: 0x0000-0xFFFF where 0x0001 = 1 mm

N4 Duplexer's Physical Attachment Type (Compatibility Level 7, mono)
Physically what type of printer does the duplex attach to which thus affects the duplexer's physical interface to the printer.

Values:
0x00 Reserved
0x01 Printer w/250 Integrated Tray
0x02 Printer w/500 Integrated Tray
0x03-0xFF Reserved

1.16.4.2 Query Device Inventory Response

This is the Query Device Inventory (QDI) Response for the Duplex option:

Syntax: Multi-Byte Response Length Field, N1
0x02, 0x00

Response Byte Definitions:

Length Field Number of bytes in the response including this byte

Values: Range: 0x01 to 0xFF

N1 Inventory To Report
Values: Range: 0x00 No Inventory To Report
0x01-0xFF Reserved

1.16.5 Stackers, Finishers, and Staplers

The following sections give the specific unique responses by command for Output Stackers (including the Modular Stacker and the High Capacity Output Tray options).

1.16.5.1 Query Device Characteristics Response

This the Query Device Characteristics (QDC) Response for Output Stackers and Finishers

1.16.5.1.1 QDC Response for Device's Without Output Trays (Bins)

This the Query Device Characteristics (QDC) Response for the option when the engine has sent a QDC command with parameter N1 = 0x00 to query the device characteristics of a device without an output tray (bin):

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5,N6, N7

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 Device Characteristics for Tray #
(requested by printer via command)

N1 Device Characteristics
Values: Bit Encoded

Bit 7 (MSB)	0	Reserved; Always 0
Bit 6	0	Reserved; Always 0
Bit 5	0	Reserved; Always 0
Bit 4	0	Reserved; Always 0
Bit 3	0	Reserved; Always 0
Bit 2	0	Reserved; Always 0
Bit 1	0	Hole Punch NOT Supported
	1	Hole Punch Supported
Bit 0	0	Reserved; Always 0

Hole Punch Supported

If the output device supports hole punching the output, it sets this bit to indicate to the print engine that this function is supported and next option is required to be attached. Output devices that do not support hole punching the output negate this bit to notify the print engine.

1.16.5.1.2 QDC Response for Device's With Output Trays (Bins)

This the Query Device Characteristics (QDC) Response for the option when the engine has sent a QDC command with parameter N1 = 0x01 thru 0x80 to query the device characteristics of an output tray (bin):

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5,N6, N7

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2, N3, N4, N5, N6, N7 Device Characteristics for Tray #
(requested by printer via command)

N1 Device Characteristics
Values: Bit Encoded

Bit 7 (MSB)	0	2nd Stapler NOT Supported
	1	Device has 2nd Staplerhead unit
Bit 6	0	Jog Tray Not Supported
	1	Jog Tray Supported
Bit 5	0	Stapler Not Supported
	1	Stapler Supported
Bit 4	0	No Output Tray Full Detection (engine algorithm must detect paper out condition)
	1	Output Tray Full Detection
Bit 3	0	No Bin-Empty Sensing
	1	Bin-Empty Sensing
Bit 2	0	Fold NOT Supported
	1	Fold Supported
Bit 1	0	Hole Punch NOT Supported
	1	Hole Punch Supported
Bit 0	0	No Accumulate Job on Early Warning Level NOT Supported
	1	No Accumulate Job on Early Warning Level Supported

Device 2nd Stapler Support

If the device supports a 2nd stapler subsystem, the device asserts this bit to indicate this capability. If the device does not support this feature, then it negates this bit. (See Note 3)

Jog Tray Supported

If the output device supports jogging its output tray or stack, it sets this bit to indicate to the print engine that this function is supported. Output devices that do not support jogging the output negate this bit to notify the print engine.

Stapler Supported

If the output device supports stapling the output, it sets this bit to indicate to the print engine that this function is supported. Output devices that do not support stapling the output negate this bit to notify the print engine.

Output Tray Full Detection Capability

The output device negates this bit if it can Not detect an output tray (or bin) full condition; thus, requiring the printer to utilize an algorithm to detect an output bin full for an output tray. The output device asserts this bit if it can detect and report output tray (or bin) full conditions for the output tray.

Hole Punch Supported

If the output device supports hole punching the output, it sets this bit to indicate to the print engine that this function is supported. Output devices that do not support hole punching the output negate this bit to notify the print engine.

No Accumulate Job on Early Warning Level Supported

If the output device supports preventing accumulate jobs after an early warning level is reported, it sets this bit to indicate to the print engine that this function is supported. Output devices that do not support this feature negate this bit to notify the print engine.

N2 Number of Reportable Levels for Full

Values:

0x00	No Media Capacity Detection Capability
0x01	N/A
0x02	2 Levels of Detection (0% and 100%)
0x03	3 Levels of Detection (0%, 0%<L_2%<100%, 100%)
0x04	4 Levels of Detection (0%, 0%<L_2%<L_3%, L_2%<L_3%<100%, 100%)
:	:
:	:

Note: The accuracy of the multi-level detection is device dependent. For specific details, reference the specific device's functional specification.

N3 Early Warning Level Reported

Values:

0x00	No Early Warning Detection Capability
0x01 - 0x63	1% to 99% Is When Early Warning Reported
0x64 - 0xFF	reserved

N4, N5 Tray's Maximum Media Capacity A

See note 2 for important compatibility info

Values: Range: 0x0001 to 0xFFFF
where 0x0001 = Capacity of 1 sheet, and
0xFFFF = Capacity of 65,535 sheets

N6, N7 Tray's Maximum Media Capacity B

Values: 0x0000 reserved for compatibility
to original Optra S stackers
Range: 0x0001 to 0xFFFF
where 0x0001 = Capacity of 1 sheet, and
0xFFFF = Capacity of 65,535 sheets

Notes:

1. If the printer solicits the characteristics of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.
2. N4+N5 must be 0x01F4 (500) for compatibility with OptraS RIP. In newer architectures, if N6+N7 are reported by the option then N4+N5 are ignored.
3. N1 Bit7 has been repurposed from Tray Lockable support indicator to Device 2nd Stapler Support.

1.16.5.1.3 QDC Response for Device's Stapler

This the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x81 (0x86 for 2nd Stapler) to query the device characteristics of its stapler:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5, N6, ...

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1: Number of Reportable Levels for Full
0x00 No Media Capacity Detection Capability
0x01 N/A

0x02 2 Levels of Detection (0% and 100%)
0x03 3 Levels of Detection (0%, 0%<L2%<100%,100%)
0x04 4 Levels of Detection
(0%,0%<L2%<L3%,L2%<L3%<100%,100%)
: :
: :

N2 Early Warning Level Reported
0x00 No Early Warning Detection
0x01-0x63 1% to 99% Is When Early Warning Reported
0x64-0xFF reserved

N3,N4 Stapler's Maximum Capacity
0x0001 to 0xFFFF where 0x0001 is one staple and 0xFFFF is 65,535 staples

N5 Stapler Low Count
0x00 No count, Early Warning Not Supported.
0x01-0xFF Number of staples from low to empty.

N6 Number of Stapling Positions
0x00 Stapling Not Supported
0x01 1 Stapling Position
0x02 2 Stapling Positions
:
0xFF 255 Stapling Positions

N7 Stapling Position Patterns Supported 0-7
Values: Bit Encoded, 1 indicates support

Bit 7 (MSB)	Top Right Diagonal
Bit 6	Top Left Diagonal
Bit 5	Do Not Use (previously Landscape Dual Left)
Bit 4	Do Not Use (previously Landscape Dual Top)
Bit 3	Top Right (aka Landscape Single)
Bit 2	Left Dual (aka Portrait Dual Left)
Bit 1	Top Dual (aka Portrait Dual Top)
Bit 0	Top Left (aka Portrait Single)

N8 Stapling Position Patterns Supported 8-15
Values: Bit Encoded

Bit 7 (MSB)	0	Reserved; Always 0
Bit 6	0	Reserved; Always 0
Bit 5	0	Reserved; Always 0
Bit 4	0	Reserved; Always 0
Bit 3	0	Reserved; Always 0
Bit 2	0	Reserved; Always 0
Bit 1	0	Reserved; Always 0
Bit 0	0	Reserved; Always 0

Notes:

1. If the printer solicits the characteristics of a stapler and the device does not support a stapler, the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the stapler specified.
2. Query for Device's 2nd Stapler is only sent when if the device has a 2nd stapler installed, as indicated in the QDC Response for Device's With Output Trays (N1 = 0x01 thru 0x80) under Section 5.3.1.5.2

1.16.5.1.4 QDC Response for Device's Accumulator

This the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x82 to query the device characteristics of its accumulator:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 Number of Reportable Levels for Full
Values:

0x00	No Media Capacity Detection Capability
0x01	N/A
0x02	2 Levels of Detection (0% and 100%)
0x03	3 Levels of Detection (0%, 0%<L_2%<100%, 100%)
0x04	4 Levels of Detection (0%, 0%<L_2%<L_3%, L_2%<L_3%<100%, 100%)
:	:
:	:

Note: The accuracy of the multi-level detection is device dependent. For specific details, reference the specific device's functional specification.

N2 Early Warning Level Reported
Values:

0x00	No Early Warning Detection Capability
0x01 - 0x63	1% to 99% Is When Early Warning Reported
0x64 - 0xFF	reserved

N3, N4 Accumulator Maximum Capacity
Values: Range: 0x0001 to 0xFFFF
where 0x0001 = Capacity of 1 staple, and
0xFFFF = Capacity of 65,535 staples

1.16.5.1.5 QDC Response for Device's Jogger

This the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x83 to query the device characteristics of its jogger:

Syntax: Multi-Byte Response Length Field, N1, N2, N3

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 — Number of Jogging Positions
Values:

0x00	Jogging Not Supported
0x01	N/A
0x02	2 Jogging Positions
0x03	3 Jogging Positions
…	
0xFF	255 Jogging Positions

N2 — Jogging Position Location (for position #1)
Values:

0x00	Reserved
0x01	Left
0x02	Right
0x03 - 0xFF	Reserved

N3, N4, ... — Jogging Position Locations (for position #2, #3, ...)
[matching the total # of positions reported by parameter N2 above]

1.16.5.1.6 QDC Response for Device's Hole Puncher

This the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x85 to query the device characteristics of its Hole puncher:

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5, N6, ...

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 Patterns Supported
Values: Bit Encoded

Bit 7 (MSB)	0	Reserved; Always 0
Bit 6	0	Reserved; Always 0
Bit 5	0	Reserved; Always 0
Bit 4	0	Reserved; Always 0
Bit 3	0	US 2-hole NOT supported
	1	US 2-hole supported
Bit 2	0	NA 3-hole NOT supported
	1	NA 3-hole supported
Bit 1	0	ISO 2-hole NOT supported
	1	ISO 2-hole supported
Bit 0	0	ISO 4-hole NOT supported
	1	ISO 4-hole supported

N2 Number of Reportable Levels for Chad Bin
0x00 No Chad Bin Capacity Detection Capability
0x01 N/A
0x02 2 Levels of Detection (0% and 100%)
0x03 3 Levels of Detection (0%, 0%<L2%<100%,100%)
0x04 4 Levels of Detection (0%,0%<L2%<L3%,L2%<L3%<100%,100%)
: :
: :

N3,N4 Chad Maximum Capacity
0x0001 to 0xFFFF in tens of chads
where 0x0001 is 10 chads and 0xFFFF is 655,350 chads

N5 Early Warning Level Reported
0x00 No Early Warning Detection
0x01-0x63 1% to 99% Is When Early Warning Reported
0x64-0xFF reserved

1.16.5.1.7 QDC Response for Device's Fold Finisher

This the Query Device Characteristics (QDC) Response for options when the engine has sent a QDC command with parameter N1 = 0x87 to query the device characteristics of its Fold Finisher

Syntax: Multi-Byte Response Length Field, N1, ...

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 Patterns Supported
Values: Bit Encoded

Bit 7 (MSB)	Reserved; Always 0

Bit 6	Reserved; Always 0
Bit 5	Reserved; Always 0
Bit 4	Z-Fold Supported
Bit 3	Z-Trifold Supported
Bit 2	C-Trifold Supported
Bit 1	Booklet Supported
Bit 0	Bi-Fold Supported

1.16.5.2 Query Device Inventory Response

This is the Query Device Inventory (QDI) Response for Stacker / Finishers

1.16.5.2.1 QDI Response for Device's Output Trays (Bins)

This the Query Device Inventory (QDI) Response for options when the engine has sent a QDI command with parameter N1 = 0x01 thru 0x80 to query the device inventory of an output tray (bin):

Syntax: Multi-Byte Response Length Field, N1, N2

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2 — Inventory For Tray # (requested by printer via command)

N1 — Tray's Current Paper Capacity
Values: 0x00 - 0x64 0 to 100% Full
0x65 - 0xFF reserved

N2 — Tray Characteristics — Reserved
Values: Bit Encoded

Bit 7 (MSB)	0	Reserved; Always 0	
Bit 6	0	Tray Not Locked	Reserved; Always 0
	1	Tray Locked	
Bit 5	0	Reserved; Always 0	
Bit 4	0	Reserved; Always 0	
Bit 3	0	Reserved; Always 0	
Bit 2	0	Reserved; Always 0	
Bit 1	0	Reserved; Always 0	
Bit 0	0	Reserved; Always 0	

Note: N2 is reserved. At this time No Smart Stacker has any Tray Characteristics to report back to the engine; therefore, this byte will be returned as 0x00 or Not at all.

Notes:
1. The device always returns the inventory of the tray requested independent if an inventory change has occurred or not.

2. If the printer solicits the inventory of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.
3. The device does not directly alert the printer it has reached its "Early Warning" condition (for example: Input Tray Almost Empty for input devices or Output Bin Almost Full for output devices); but instead simply raise a Non-Critical Status in the Query Device Status 1 response when the supply has reached the "early warning" threshold level (ie, the value identified by the Query Device Characteristics response). The printer upon querying the inventory of the device and detecting the device's supply level matches that of the "early warning" level reported in the Query Device Characteristics response can post to the printer's operator panel a message alerting the customer of the early warning condition.

1.16.5.2.2 QDI Response for Device's Stapler

This the Query Device Inventory (QDI) Response for an output option when the engine has sent a QDI command with parameter N1 = 0x81 (0x86 for 2nd Stapler) to query the device inventory of its stapler:

Syntax: Multi-Byte Response Length Field, N1

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 — Stapler's Current Paper Capacity

Values:	
0x00 - 0x64	0 to 100% Full
0xFE	Stapler Removed (mono stapler)
0xFF	Stapler Removed

Notes:
1. If the device does not have a stapler, then the response Length Field should be set to 0x01.
2. The device does not directly alert the printer it has reached its "Early Warning" condition (for example: Input Tray Almost Empty for input devices or Output Bin Almost Full for output devices); but instead simply raise a Non-Critical Status in the Query Device Status 1 response when the supply has reached the "early warning" threshold level (ie, the value identified by the Query Device Characteristics response). The printer upon querying the inventory of the device and detecting the device's supply level matches that of the "early warning" level reported in the Query Device Characteristics response can post to the printer's operator panel a message alerting the customer of the early warning condition

3. Query for Device's 2nd Stapler is only sent when if the device has a 2nd stapler installed, as indicated in the QDC Response for Device's With Output Trays (N1 = 0x01 thru 0x80) under Section 5.3.1.5.2

1.16.5.2.3 QDI Response for Device's Accumulator

This the Query Device Inventory (QDI) Response for options when the engine has sent a QDI command with parameter N1 = 0x82 to query the device inventory of its accumulator:

Syntax: Multi-Byte Response Length Field, N1

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 — Accumulator's Current Capacity
Values: 0x00 - 0x64 0 to 100%
0x65 - 0xFF reserved

1.16.5.2.4 QDI Response for Device's Hole Puncher

This the Query Device Inventory (QDI) Response for an output option when the engine has sent a QDI command with parameter N1 = 0x85 to query the device inventory of its hole punch (dust box):

Syntax: Multi-Byte Response Length Field, N1

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 — Dust Box's Current Level
Values: 0x00 - 0x64 0 to 100% Full
0xFF Dust Box Removed

Notes:

If the device does not have a hole punch, then the response Length Field should be set to 0x01.

If the device cannot determine if the dust box is removed or not then the device should not return 0xFF (Dust box removed)

If the device can only distinguish if the dust box is full or not full, then the device should return 0x00 for not full and 0x64 for full.

1.16.6 Mailbox

The following sections give the specific unique responses by command for Mailbox options.

1.16.6.1 Query Device Characteristics Response

This the Query Device Characteristics (QDC) Response for a Smart Mailbox options:

1.16.6.1.1 QDC Response for Device's Output Trays (Bins)

This the Query Device Characteristics (QDC) Response for a Smart Mailbox options when the engine has sent a QDC command with parameter N1 = 0x01 thru 0x80 to query the device characteristics of an output tray (bin):

Syntax: Multi-Byte Response Length Field, N1, N2, N3, N4, N5
0x06, N1, N2, N3, N4, N5

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2, N3, N4, N5 — Device Characteristics for Tray #
(requested by printer via command)

N1 — Device Tray (Bin) Characteristics
Values: Bit Encoded

Bit 7 (MSB)	0	Tray Not Lockable
	1	Tray Lockable
Bit 6	0	Jog Tray Not Supported
	1	Jog Tray Supported
Bit 5	0	Stapler Not Supported
	1	Stapler Supported
Bit 4	0	No Output Tray Full Detection (engine algorithm must detect paper out condition)
	1	Output Tray Full Detection
Bit 3	0	Reserved; Always 0
Bit 2	0	Reserved; Always 0
Bit 1	0	Reserved; Always 0
Bit 0	0	Reserved; Always 0

Tray Lockable

If the tray can be locked via the **Tray Lock Notification** command, the device asserts this bit to indicate this capability. If the device can not support locking this tray, then it negates this bit.

Jog Tray Supported

If the output device supports jogging its output tray or stack, it sets this bit to indicate to the print engine that this function is supported. Output devices that do not support jogging the output negate this bit to notify the print engine.

Stapler Supported

If the output device supports stapling the output, it sets this bit to indicate to the print engine that this function is supported. Output devices that do not support stapling the output negate this bit to notify the print engine.

Output Tray Full Detection Capability

The output device negates this bit if it can Not detect an output tray (or bin) full condition; thus, requiring the printer to utilize an algorithm to detect an output bin full for an output tray. The output device asserts this bit if it can detect and report output tray (or bin) full conditions for the output tray.

N2 Number of Reportable Levels for Full

Values:

0x00	No Media Capacity Detection Capability
0x01	N/A
0x02	2 Levels of Detection (0% and 100%)
0x03	3 Levels of Detection (0%, 0%<L_2%<100%, 100%)
0x04	4 Levels of Detection (0%, 0%<L_2%<L_3%, L_2%<L_3%<100%, 100%)
:	:
:	:

Note: The accuracy of the multi-level detection is device dependent. For specific details, reference the specific device's functional specification.

N3 Early Warning Level Reported

Values:

0x00	No Ea**rly** Warning Detection Capability
0x01 - 0x63	1% to 99% Is When Early Warning Reported
0x64 - 0xFF	reserved

N4, N5 Tray's Maximum Media Capacity

Values: Range: 0x0001 to 0xFFFF

where 0x0001 = Capacity of 1 sheet, and

0xFFFF = Capacity of 65,535 sheets

Notes:

1. If the printer solicits the characteristics of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.

1.16.6.1.2 QDC Response for Device's Stapler

See Section 1.16.5.1.3 – Same as for Stackers / Finishers / Staplers

1.16.6.2 Query Device Inventory Response

This is the Query Device Inventory (QDI) Response for a Smart Mailbox options:

1.16.6.2.1 QDI Response for Device's Output Trays (Bins)

This the Query Device Inventory (QDI) Response for a Smart Mailbox options when the engine has sent a QDI command with parameter N1 = 0x01 thru 0x80 to query the device characteristics of an output tray (bin):

Syntax: Multi-Byte Response Length Field, N1, N2

Response Byte Definitions:

Length Field — Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1, N2 — Inventory For Tray # (requested by printer via command)

N1 — Tray's Current Paper Capacity
Values: 0x00 - 0x64 0 to 100% Full
0x65 - 0xFF reserved

N2 — Tray Characteristics — Reserved
Values: Bit Encoded

Bit 7 (MSB)	0	Reserved; Always 0	
Bit 6	0	Tray Not Locked	Reserved; Always 0

	1	Tray Locked
Bit 5	0	Reserved; Always 0
Bit 4	0	Reserved; Always 0
Bit 3	0	Reserved; Always 0
Bit 2	0	Reserved; Always 0
Bit 1	0	Reserved; Always 0
Bit 0	0	Reserved; Always 0

Note: N2 is reserved. At this time No Smart Stacker has any Tray Characteristics to report back to the engine; therefore, this byte will be returned as 0x00 or Not at all.

Notes:

1. The device always returns the inventory of the tray requested independent if an inventory change has occurred or not.
2. If the printer solicits the inventory of a tray # that the device does not support (for example: tray #4 and the device doesn't have a tray #4), the device returns an *Unsupported Tray Response*; Length byte = to itself (0x01 response) to indicate to the printer the device does not have the tray specified.
3. The device does not directly alert the printer it has reached its "Early Warning" condition (for example: Input Tray Almost Empty for input devices or Output Bin Almost Full for output devices); but instead simply raise a Non-Critical Status in the Query Device Status 1 response when the supply has reached the "early warning" threshold level (ie, the value identified by the Query Device Characteristics response). The printer upon querying the inventory of the device and detecting the device's supply level matches that of the "early warning" level reported in the Query Device Characteristics response can post to the printer's operator panel a message alerting the customer of the early warning condition.

1.16.6.2.2 QDI Response for Device's Stapler

See Section 1.16.5.2.2

1.16.6.3 Query HW IR Response

This is the Query HW IR (QHWIR) response for a Smart Mailbox option when the engine has sent the QHWIR command to the device.

Syntax: Multi-Byte Response Length Field, N1, N2

Response Byte Definitions:

Length Field Number of bytes in the response including this byte
Values: Range: 0x01 to 0xFF

N1 HW IR Condition

0x00 No HW IR Condition Exists
0x01 Mailbox Elevator Bottom Switch Made When Not Expected
0x02 Staple Jam: No Mechanical Service Required to Home Stapler
0x03 Staple Jam: Mechanical Service Required to Home Stapler
0x04 - 0xFE Reserved
0xFF HW IR Condition Defined in Next Byte

N2 HW IR Condition byte #2 (currently not used)

Mailbox Elevator Bottom Switch Made When Not Expected (N1=0x01)

The device detected that its mailbox elevator lowered down to the elevator bottom switch other than during homing; thus, the mailbox unit's alignment to the printer needs adjusting.

POWER-ON RESET (POR) / RESET PROCESSING

At power on the device must complete several things (outlined below) and then wait for the printer to bring it "on-line":

- Complete its BAT (Basic Assurance Tests)
- Initialize the serial communications interface to a device address of its Trunk Line Address
- Initialize the serial communications interface to accept a minimum set of commands (see **Hard Reset** command for details)
- Initialize the device for operation: electrically and microcode (referred to as "electrical / software initialization")
- Return a "*Reset Complete*" response to the print engine upon receipt of a **Query Reset Complete** command (when it has completed its electrical / software initialization)

For a device to be prepared to process print jobs the device must first perform an initialization (or reset) process to enable itself to process print jobs and perform the function of which the device is intended. This initialization process consists of two parts: an electrical / software initialization portion and a mechanical initialization portion. The electrical / software initialization portion consists of initializing the device's electrical hardware and software (or microcode). Electrical hardware initialization includes such things as initializing the microprocessor's control registers, serial communications hardware, enabling motor drivers, etc to ensure power on any of the electrical or electro-mechanical devices do not cause undesired results during the initialization process and to allow the device to be brought on-line by the printer. The software initialization includes such things as initializing microcode control variables, pointers, etc. This electrical / software initialization is done automatically by the device at Power-On Reset or as explicitly instructed by the printer via the **System Configuration Reset**, **Hard Reset**, **Soft Reset** and/or **Intervention Required Clear Notification** command (the printer's state or circumstance will dictate which command is sent to the device). Then following the electrical / software initialization, the device must be perform its mechanically initialization before it can begin to process print jobs. The mechanical initialization portion consists of initializing the device's mechanical mechanism (for example: homing a D-roll, or homing a deflector, etc). The device's mechanical initialization (or reset) is only performed when explicitly requested by the printer via sending the **Mechanical Reset** command to the device. When the **Mechanical Reset** command is received by the device, it initializes (or resets) the device mechanically, including any mechanical homing required by the device. The device never automatically or implicitly performs a mechanical reset. This is to ensure the printer's peak power never exceeds the printer's power supply capabilities by all devices performing their mechanical reset simultaneously.

Note: Some devices (for example: Sorters/Mailbox devices) require the mechanical reset to determine their physical characteristics and inventory. Thus, the printer at POR should send the mechanical reset to devices prior to querying for the device's characteristics and inventory.

Reset processing is initiated by the printer sending "reset" commands to the device. Four specific "reset" commands exist:

- **System Configuration Reset**
- **Hard Reset**
- **Soft Reset**
- **Mechanical Reset**

Each type reset is described in detail in the *Command Set Description section 4.0* of this spec. Also, an illustration of the printer / device command protocol for both a POR situation, and a reset command is given in the *Reset Protocols section 6.0* of this spec.

The table below defines for each type reset the operations it performs as a part of its "reset processing."

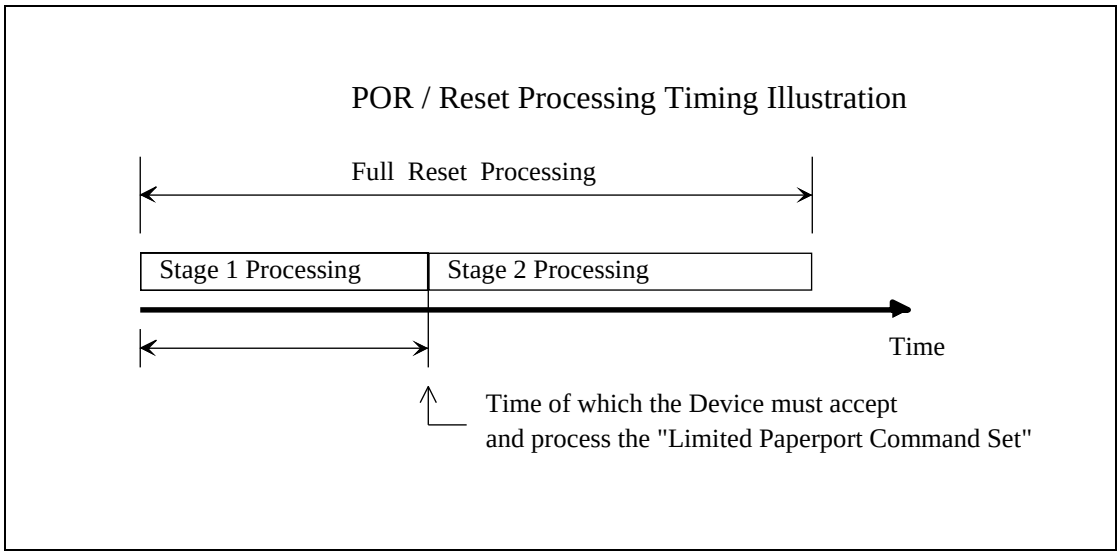
Area	POR	SCR	Hard Reset	Soft Reset	Mechanical Reset	IR Clear
Address	*Trunk*	*Trunk*	*Unchanged*	*Unchanged*	*Unchanged*	*Unchanged*
BAT Test Executed	*Yes*	*Yes*	*Yes*	*No*	*No*	*No*
Electrical Initialization	*Yes*	*Yes*	*Yes*	*Yes*	*No*	*Yes*
Software Initialization	*Yes*	*Yes*	*Yes*	*Yes*	*No*	*Yes*
Device Resets to Its Paperport II Defaults	*Yes*	*Yes*	*Yes*	*Yes*	*No*	*No*
Mechanical Homing	*No*	*No*	*No*	*No*	*Yes*	*No*

POR - Power-on Reset
SCR - System Configuration Reset command
Hard Reset - Hard Reset command
Soft Reset - Soft Reset command
Mechanical Reset - Mechanical Reset command
IR Clear - Intervention Required Clear command

Note: (1) The communication's 9th bit will be set (ie, address mode set) at the completion of transmitting the SCR, Hard Reset, and Soft Reset command's response.

(2) "Electrical" Initialization - Not all types of resets reset the same electrical hardware. The reset types identified as *Yes* above indicates they initialize the basic electrical hardware for ensure power on any of the electrical or electro-mechanical components do not cause undesired results and to allow the device to properly communicate with the printer. Other needed electrical hardware initialization, such as running BAT Tests, restoring full power, etc may only occur for certain and specific reset types.

When a device conducts POR or reset processing, the initialization process must be performed in two stages as illustrated below. During the first stage, referred to as Stage 1, the device must complete initialization to the point of which it can communicate with the printer and process a "Limited Set of Paperport II Commands" (see below for full description of this set). The Stage 1 initialization must be completed within a prescribed amount of time (see "POR / Reset Timings Requirements" table below). After Stage 1 initialization is completed, the device completes whatever "remaining" initialization is required for the type reset requested. This is referred to as the Stage 2 initialization process in the illustration below. The total time for the reset (or initialization) processing is referred to as the "Full Reset" and consists of Stage 1 + Stage 2 initialization processing. The Full Reset timing required is also specified in the "POR / Reset Timings Requirements" table below.



The table below lists the timing specifications for the various type resets:

Reset Type	Max Time for Device To Accept & Begin Processing the "Limited Paperport Command Set" (sec)	Max Time To Complete Full Reset Processing (sec)
POR	1	2
System Configuration Reset	1	2
Hard Reset	1	2
Soft Reset	0.5	1
Mechanical Reset	N/A	10
IR Clear	N/A	1

Table: POR / Reset Timings Requirements

During a power-on reset (POR), **System Configuration Reset**, **Hard Reset**, and/or **Soft Reset**; the only commands the print engine can send to the device until a *Reset Complete* response is given are listed below (commonly referred to as the *"Limited Paperport Command Set"*):

- **System Configuration Reset**
- **Assign Device Address**
- **Query Reset Complete**
- **Query Device Type**
- **Query Device Status 1**
- **Connect Communications To Next Device**

If an Intervention Required condition is reported via the response to a Query Device Status 1 command, the print engine can extend this limited set of commands to query the device for the specifics of the IR condition raised:

- **Query Non-Critical Status Change**
- **Query IR Condition - Homing Allowed**
- **Query IR Absolute Immediate Stop**
- **Query Equipment Check Status**
- **Query Paper Position Status**
- **Query Cover Open Location**
- **Query Tray Pulled Location**

POWER SUPPLY - DEVICE

Many of the devices receive their electrical power from the printer via the electrical interface (see *ELECTRICAL PHYSICAL INTERFACE section 0* for details). However, some devices power requirements are such that they must provide their own power either by having their own power supply or obtaining it from a separate auxiliary power supply. To ensure that the printer system can operate correctly it is important that All devices that do not receive their electrical power from the printer perform certain functions when the printer power on and off.

Whenever the device detects "power" on the electrical interface (ie, the printer has been powered on), it must:

- Establish power to itself
- Disconnect the serial communications Rxd (printer's Txd line) to the device "below"
- Revert to only accepting communications via its Trunk Line Address
- Perform its normal Power-On Reset (POR) processing and then wait for the printer to bring it "on-line" via querying it for device type & characteristics, assigning it a unique device address, and instructing it to connect communications to the device "below"

- Whenever the device detects no "power" on the electrical interface (ie, the printer has been powered off), it must:

- Put itself into the lowest power consumption mode possible (remotely turning its power off would be most efficient)
- Not responding to serial communications (receiving or transmitting)
- Not performing any operations within the device (mechanical)
- Monitor the electrical interface for the printer to be powered on

Devices that provide their own power can determine when to perform these requirements (above) by monitoring the "power" signals within the *ELECTRICAL PHYSICAL INTERFACE section 0*.

If the device has an external power switch for emergency on/off purposes, the operator should leave the device's power switch in the "on" position and only use it as an emergency off switch for the device. This will allow the device's power control described above to be utilized to ensure power on/off synchronization with the printer (ie, master). An operator switching power on a device after the printer has already been powered on may result in unpredictable results (noise on the serial communications bus, the device not being brought on-line by the printer, the device below not being brought on-line by the printer, etc.). Likewise, an operator switching power off a device after the printer has already been powered on may result in unpredictable results (noise on the serial communications bus, the device no longer can communicate with the printer, devices below no longer can communicate with the printer, etc.).

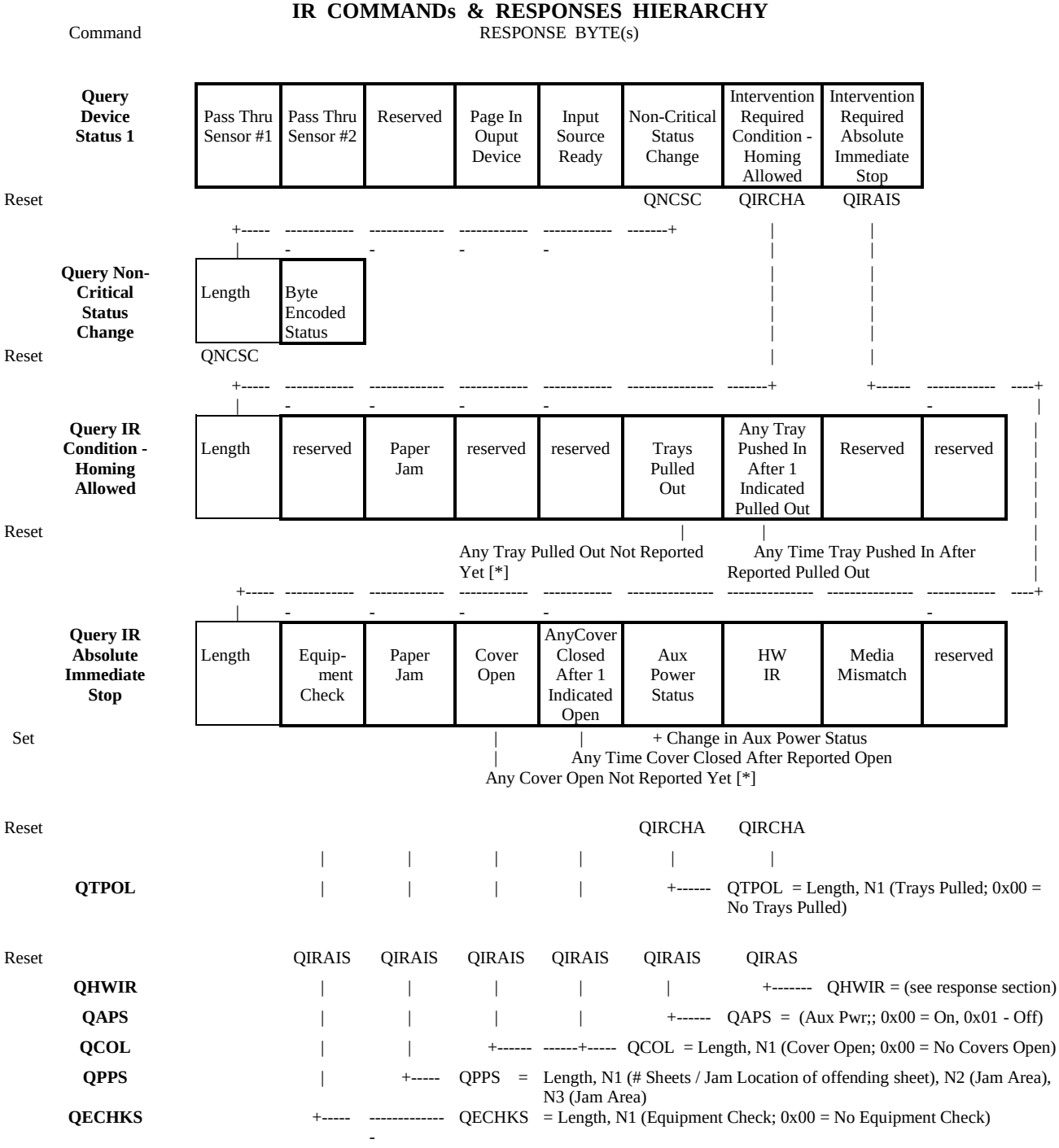
INTERVENTION REQUIRED (IR)/ ERROR PROCESSING

This section describes the printer system's Intervention Required (IR) / Error Processing. An IR condition is anything detected by the printer or device that requires an operator to intervene at the printer to resolve; for example: an out of paper condition or paper jam condition, etc. An Error condition is anything detected by the printer or device that indicates service to the printer or device may be needed (ie, intervention is required by a service person); for example: bad ram memory detected, or a motor not running. Since both the IR condition and Error condition are *intervention* conditions, the difference being who responds to the condition, the printer and device handle them in the same manner. Thus, the term IR (or Intervention Required) condition is used to refer to either of these conditions.

There are several commands and responses used by the printer system to communicate IR conditions and synchronize both printer & device(s) for processing IR conditions (see section 0 & 0, respectively). Below is an illustration of the IR condition commands & responses hierarchy.

see next page

IR COMMANDs & RESPONSES HIERARCHY



[*] Devices MUST Hold Off Reporting a Cover Closed or Tray Pushed In Until After it Ensures the Engine Has Received the Previous Cover Open or Tray Pulled Status; Respectively.

From a control or process point of view there are two categories of Intervention Required (IR) conditions. One category is where the printer system must continue processing the currently fed sheet to some physical synchronization point to allow the printer system to continue properly once the IR condition is resolved by the operator. An example of such an IR condition is a paper jam with a D-roll paper picking technology; ie, the D-roll must continue travel until it reaches its "home" position. Note: In some cases no physical synchronization of the printer system is required; thus halting the printing process can occur immediately. An example of this category IR condition would be when a paper jam occurs and all the printer's mechanical mechanisms are already in their "home" position. The other category of IR conditions are those that the printer system can not perform any physical synchronization of the printer system because it may cause harm to either the printer system and/or the operator; thus, halting the printing process must occur immediately. An example of this type IR condition would be a cover open condition; either by the printer or a device.

Both the printer and the devices detect IR conditions. Devices alert the printer of IR conditions via the *general status* response. The printer alerts devices of IR conditions via the **Printer Intervention Required Notification** command. Regardless of who detects the IR condition the printer must control the printer system during the IR state due to the possible need for mechanism "homing". It is also the printer who controls the printer system in terms of identifying when the printer & devices should exit the IR state and allow processing print jobs again. This is initiated by the printer via the **Intervention Required Clear Notification** command.

When an IR condition is detected, a device may require some movement of its mechanism to ensure it is in a proper position prior to stopping to handle the IR condition. For IR conditions detected by the device, the device should ensure its mechanism has completely stopped prior to raising the IR condition to the printer. For printer raised IR conditions or IR Conditions - Homing Allowed raised by the device, the printer issues the printer the **Query Device Quiesce** command to learn with the device's mechanism is no longer moving (and thus can prompt the operator panel to post the required operator intervention required message).

Thus, there are two categories of IR conditions and two means of IR detection (printer & device). Each of these are described, in outline form, further to provide a better understanding of IR processing.

Table 8.1 Printer Detected IR Condition - Mechanism Homing Required

Example: Paper Jam Detected by Printer and Printer's D-Roll must be "homed".

	PRINTER	DEVICE
1	◆ Detects IR Condition and Homing Required	
2	◆ Broadcasts to All Devices **Printer Intervention Required Notification** with Parameter Byte = Homing Required	
3		◆ Recognizes that the printer has detected an IR Condition and requests support for homing
4		◆ Captures ("saves") debug message bytes
5	◆ Processes sheet to "home" position and continues issuing steady state commands to device(s) to allow them to process the current sheet	◆ Continues to process "steady state" commands and process the sheet while printer mechanism "homes"
6		◆ Polls for the printer to indicate "halted" (ie, homing is complete)
7	◆ Homing is complete and the printer sends to All devices **Printer Intervention Required Notification** with the parameter byte = Print Engine Halted	
8		◆ Recognizes that the printer has "halted"; ie, completed its "homing" ◆ Completes any IR processing it is required to do; ie, finishes operation of its mechanism (for example feeding a sheet to a predefined position for easy jam clearance, or positioning a mechanism to a predefined position to allow easier access inside the device for service by the operator)
9		◆ Device completes any IR processing and its mechanism is stopped, and: - ensures mechanically is shut down - enters the IR state - begins waiting for the printer to issue that the IR has been cleared
10	◆ Printer expects all the devices are in the IR state and Quiesced.	
11	◆ Determine # of Sheets or location of jammed sheet inside device that the operator must clear by sending the device the **Query Paper Position Status** command.	◆ Return QPPS = # sheets in device or location of jammed sheet within the device that need to be removed by the operator.
12	◆ Printer prompts Operator Panel to post IR condition to alert the operator.	
13	◆ Sends **Query Debug Message** to each device to obtain their debug data	◆ Returns device's debug data to be displayed on the printer's op panel
14	◆ Waits for the operator to indicate the IR condition has been resolved (ie, cleared)	

	PRINTER	DEVICE
15	◆ Sends to All Devices **Intervention Required Clear Notification** command to instruct the devices to exit the IR State and return to "Steady State"	
16		◆ Recognize the IR Clear notification and perform IR Clear processing: - Clears IR response byte bits to allow the device to detect & report any IR conditions found during IR Clear processing - Initialize itself for processing print jobs again (which includes "resetting" the device and homing any of its mechanism - if necessary)
17	◆ Ready for "Steady State" processing; ie, print jobs	◆ Ready for "Steady State" processing; ie, print jobs

Table 8.2 Printer Detected IR Condition - Mechanism Homing NOT Required

Example: Paper Jam Detected by Printer and Printer's D-Roll already homed, or Cover Open detected by the printer.

	PRINTER	DEVICE
1	◆ Detects IR Condition and Homing NOT Required	
2	◆ Broadcasts to All Devices **Printer Intervention Required Notification** with Parameter Byte = Print Engine Halted	
3		◆ Recognizes that the printer has detected an IR condition (and homing is not requested)
4		◆ Captures ("saves") debug message bytes
5		◆ Completes any IR processing it is required to do; ie, finishes operation of its mechanism (for example feeding a sheet to a predefined position for easy jam clearance, or positioning a mechanism to a predefined position to allow easier access inside the device for service by the operator)
6		◆ Device completes any IR processing and its mechanism is stopped, and: - ensures mechanically is shut down - enters the IR state - begins waiting for the printer to issue that the IR has been cleared
7	◆ Printer expects all the devices are in the IR state and Quiesced. ◆ Printer prompts Operator Panel to post IR condition to alert the operator.	

	PRINTER	DEVICE
8	◆ Determine # of Sheets or location of jammed sheet inside the device that the operator must clear by sending the device the **Query Paper Position Status** command.	◆ Return QPPS = # sheets in device or location of jammed sheet within the device that need to be removed by the operator.
9	◆ Printer prompts Operator Panel to post IR condition to alert the operator.	
10	◆ Sends **Query Debug Message** to each device to obtain their debug data	◆ Returns device's debug data to be displayed on the printer's op panel
11	◆ Waits for the operator to indicate the IR condition has been resolved (ie, cleared)	
12	◆ Sends to All Devices **Intervention Required Clear Notification** command to instruct the devices to exit the IR State and return to "Steady State"	
13		◆ Recognize the IR Clear notification and perform IR Clear processing: - Clears the IR response byte bits to allow the device to detect & report any IR conditions found during IR Clear processing - Initialize itself for processing print jobs again (which includes "resetting" the device and homing any of its mechanism - if necessary)
14	◆ Ready for "Steady State" processing; ie, print jobs	◆ Ready for "Steady State" processing; ie, print jobs

Table 8.3 Device Detected IR Condition - Mechanism Homing Required

Example: The Device Detects an IR Condition of Which it Can Support "Homing" by the Printer; example: Paper Jam Detected by Device and printer performs "homing" (example: Printer's D-Roll must be "homed").

	PRINTER	DEVICE
1		◆ Detects IR Condition (Paper Jam) and Can Support Printer Homing (if needed)
2		◆ Captures ("saves") debug message bytes
3		◆ Sets QDS1 = Intervention Required Condition - Homing Allowed, QIRCHA = Paper Jam to notify printer of IR condition
4	◆ Broadcasts to All Devices **Printer Intervention Required Notification** with Parameter Byte = Homing Required	
5	◆ Processes sheet to "home" position and continues issuing steady state commands to device(s) to allow them to process the current sheet	◆ Continues to process "steady state" commands and process the sheet while printer mechanism "homes"
6		◆ Polls for the printer to indicate "halted" (ie, homing is complete)
7	◆ Homing is complete and the printer sends to All devices **Printer Intervention Required Notification** with the Parameter byte = Print Engine Halted	
8		◆ Recognizes that the printer has "halted"; ie, completed its "homing"
9		◆ Device completes any IR processing, and: - ensures mechanically is shut down - enters the IR state - begins waiting for the printer to issue that the IR has been cleared
10	◆ Since Printer mechanism is stopped and the Printer has ensured ALL the devices mechanisms are stopped via the **Query Device Quiesce** command.	
11	◆ Determine # of Sheets or location of jammed sheet inside the device that the operator must clear by sending the device the **Query Paper Position Status** command.	◆ Return QPPS = # sheets in device or location of jammed sheet within the device that need to be removed by the operator.
12	◆ Printer Prompts Operator Panel to post IR condition to alert operator.	
13	◆ Sends **Query Debug Message** to each device to obtain their debug data	◆ Returns device's debug data to be displayed on the printer's op panel
14	◆ Waits for the operator to indicate the IR condition has been resolved (ie, cleared)	
16	◆ Sends to All Devices **Intervention Required Clear Notification** command to instruct the devices to exit the IR State and return to "Steady State"	

	PRINTER	DEVICE
17		◆ Recognize the IR Clear notification and perform IR Clear processing: - Clears the IR response byte bits to allow the device to detect & report any IR conditions found during IR Clear processing - Initialize itself for processing print jobs again (which includes "resetting" the device and homing any of its mechanism - if necessary)
18	◆ Ready for "Steady State" processing; ie, print jobs	◆ Ready for "Steady State" processing; ie, print jobs

Table 8.4 Device Detected IR Condition - Mechanism Homing NOT Required

Example: The Device Detects an IR Condition of Which it Can Not Support "Homing" by the Printer; example: Device's Cover Open Detected by Device

	PRINTER	DEVICE
1		◆ Detects IR Condition (Cover Open) and device: - Captures ("saves") debug message bytes - Immediately shuts down itself mechanically - enters the IR state - Sets QDS1 = Intervention Required Absolute Immediate Stop, QIRAIS = Cover Open, and QCOL = Back Cover is open - begins waiting for the printer to issue that the IR has been cleared
2	◆ Broadcasts to All Devices **Printer Intervention Required Notification** with Parameter Byte = Print Engine Halted	
3	◆ Since Printer mechanism is stopped and the Printer has ensured ALL the devices mechanisms are stopped.	
4	◆ Determine # of Covers open and their location via the **Query Cover Open Location** command.	◆ Return QCOL = # covers open for the device, and their location.
5	◆ Printer Prompts Operator Panel to post IR condition to alert operator.	
6	◆ Sends **Query Debug Message** to each device to obtain their debug data	◆ Returns device's debug data to be displayed on the printer's op panel
7	◆ Waits for the operator to indicate the IR condition has been resolved (ie, cleared)	

	PRINTER	DEVICE
8	◆ Sends to All Devices **Intervention Required Clear Notification** command to instruct the devices to exit the IR State and return to "Steady State"	
9		◆ Recognize the IR Clear notification and perform IR Clear processing: - Clears the IR response byte bits to allow the device to detect & report any IR conditions found during IR Clear processing - Initialize itself for processing print jobs again (which includes "resetting" the device and homing any of its mechanism - if necessary)
10	◆ Ready for "Steady State" processing; ie, print jobs	◆ Ready for "Steady State" processing; ie, print jobs

HOT-PLUGGING DEVICES

Hot Plugging refers to attaching or removing a device from the printer while the printer is powered on. Hot Plugging is to be supported mechanically, electrically, and via the command & response set (ie, microcode) by all Paperport II Smart Options. It should be noted, Hot Plugging may increase the number of communications errors on the common communications bus.

The specific definition, implementation, and any limitations of Hot Plugging is determined by the individual printer product definition and implementation. For more details, refer to the specific printer's functional specification.

ELECTRICAL PHYSICAL INTERFACE

1.17 INTRODUCTION

The "software communications" portions of this document; ie, the command set, response set, and protocols can be applied to virtually any physical communications interface. Thus the section(s) below specify the physical communications interface for specific printer implementations.

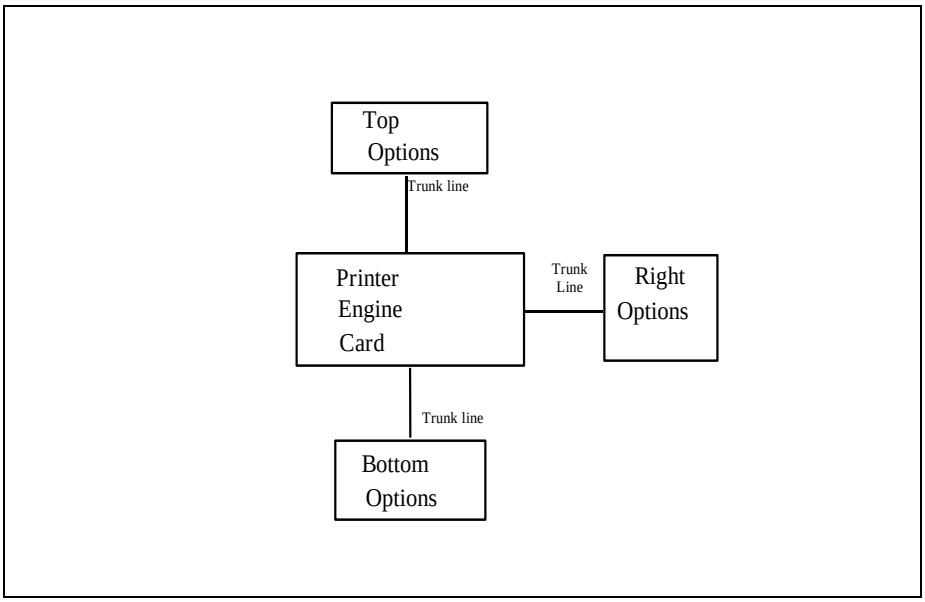
1.18 GENERAL DESCRIPTION

The interface consists of two parts:

1. Serial Communications
2. Power Signal

The printer engine card is the master of the serial interface and initiates all communications and the paper handling options are slave devices that only respond to the printer engine card. The printer is also the source of power for the low power options that do not have their own power supply. The diagram below indicates the printer engine card is in the center of a star type connection to the paper handling options. One possible connection scheme is shown, but the main point is there will be multiple trunk lines from the printer engine card to the paper handling options. Each trunk line will contain the serial communications signals and power, and all trunk lines converge to a single microprocessor on the printer engine card. Multiple paper handling options can be daisy chained on one trunk line. For example, the trunk line going to the bottom options may go to paper tray 1, then to paper tray 2, then to paper tray 3, etc.

-- see next page --



1.19 SMART OPTIONS INTERFACE

The following sections provide the specific serial communications interface for the printer product family and its smart options.

(Update Feb 2019) Note that this section of the paperport specification has gotten out-of-date, as it reflects details on a very old implementation. But at a high level, we still have an asynchronous 9-bit serial data transfer, with open-collector/open-drain drivers on each end, and protection for hot-plugging/unplugging. We also only provide +25v+/-10% power to attached options, with no more +5V, and this +25V goes away during sleep mode of the engine. Pinouts and number of pins on the auto-connects have changed over the years too on various products – this spec does NOT reflect the pinouts of the most recent printers.

1.19.1 General Description

As indicated above, the printer engine card is the master and initiator of all serial communications to the paper handling option slave devices, and the printer is the supplier of power to option devices that don't have their own power supply. Multiple trunk lines will connect to the printer engine card; current plans are for one trunk line to the option devices located below the printer, one to the option devices located at the right of the printer, one to the option devices located on top of the printer, and one to the option devices located to the side of the printer. Multiple device options can be daisy chained on any location group's trunk line. Each trunk line contains the same six signals: a twisted group of Printer TxD, GND, Printer RxD, and GND; and a twisted pair of Printer +24V and 24VReturn. The TxD and RxD signals to/from each trunk line will go to one microprocessor on the

printer engine card. The +24V and 24VReturn lines to each trunk line will come through the printer engine from the printer power supply.

1.19.2 Electrical Interface

1.19.2.1 General Description

The serial communications interface signals are Printer TxD (transmit data) and Printer RxD (receive data) and their twisted pair logic Grounds. The serial port is full duplex, meaning it can transmit and receive data simultaneously. It is also receive buffered, meaning it can begin receiving a second byte before the first byte has been read from the register; but, the first byte will be lost if it is not read before the second byte's reception is complete. The serial mode being used is an eleven bit UART. Eleven bits are sent per byte: a start bit (0), eight data bits (Least Significant Bit first), a programmable ninth data bit (set to 1 for address bytes and 0 for data bytes), and a stop bit (1). Normally the TxD and RxD signals are High (1), and the beginning of a transmission is indicated by the signal going Low (0) for a start bit. Legacy options had the baud rate at 58,252 bits/second which is 17.2 usec/bit. Therefore, a full frame of 11 bits will be 0.189 msec/frame. Each bit period is divided into sixteen samples by the microprocessor and the bit value accepted is what is present in 2 out of 3 samples during samples 7, 8 and 9 of the sixteen. Therefore, each bit sample clock period is 1.07 usec. A High level signal (above 3.85V) is read as a 1 data bit and a Low level signal (below 0.9V) is read as a 0 data bit.

As of this update (Feb 2019), baud rate is actually 57,600 baud.

1.19.2.2 Signal Driver / Receiver Characteristics

The printer TxD signal will be driven by a buffer with open-collector outputs. The signal will go from the buffer output through a 100 ohm 5% .1W series resistor before leaving the printer engine card or paper handling option card. The smart options use a 74ALS03B buffer because it has open collector outputs, allows the options to gate off the printer TxD signal to the next option, can sink a reasonable amount of current, and draws very little Vcc current. The 74ALS03B guarantees a maximum Low-level output of 0.4 V at Iol = 4 mA and Vcc = 4.5 V or 0.5V at Iol = 8 mA. The typical Low-level output is 0.25 V at Iol = 4 mA and Vcc = 5.0 V or 0.35 V at Iol = 8 mA.

The printer RxD signal will be received on the printer engine card and the printer TxD signal will be received on the paper handling options cards by a 5.1K ohm 5% .1W resistor to Vcc, then a 100 ohm 5% .1W series resistor, then a 150 pF 5% 100V NPO capacitor to Ground, then a 74XX14 hex inverting Schmitt trigger. The smart options use a 74HC14 whose positive-going threshold is 1.7 V MIN and 3.15 V MAX at Vcc = 4.5 V, and 1.97 V MIN and 3.85 V MAX at Vcc = 5.5 V, so the average of this range is 2.775 V. The negative-going threshold is 0.9 V MIN and 2.2 V MAX at Vcc = 4.5 V, and 1.1 V MIN and 2.73 V MAX at Vcc = 5.5 V, so the average of this range is 1.815 V.

On the smart options, the typical Low level of the TxD or RxD signal expected at the 74HC14 input is 0.35 V [from 0.25 ALS03 Vol + (5V / 5.1K) (100 ohm)]. This would give a typical noise immunity of 2.425 V (2.775 - 0.35). The worst case Low level is 0.52 V [from 0.4 Vol + (5.5V / 4.845K) (105 ohm)]. This would give a minimum noise immunity of 1.18 V (1.7 - 0.52). The typical High level of the TxD or RxD signal is Vcc since an open collector driver with pullup resistor is used. This would give a typical noise immunity of 3.185 V (5.0 - 1.815). The worst case High level is 3.959 V [from 4.5Vcc - (0.1mA Ioh ALS03)(5.355K) - (1uA Ii HC14)(5.46K)]. This would give a minimum noise immunity of 1.229 V (3.959 - 2.73).

1.19.2.3 Signal Termination Requirements

As noted above, each option will terminate the printer TxD signal with a 5.1K ohm 5% .1W resistor to Vcc, then a 100 ohm 5% .1W series resistor, then a 150 pF 5% 100V NPO capacitor to Ground, then input the signal to a 74HC14 input.

Each option will drive its option TxD (printer RxD) signal with a 74ALS03B open collector driver, then a 100 ohm 5% .1W series resistor, then wire OR into the printer RxD bus signal.

1.19.2.4 Timing Requirements

Paper handling options are required to respond to the printer engine within a set time. Refer to the Paperport spec *Communications Interface section 3.0* on slave device response time details.

1.19.2.5 Error Recovery

Refer to the error recovery protocol portion of the *Communications Interface section 3.0* for error response byte definition. On the hardware side, each serial transmission is started by the falling edge of the Start bit. Each bit time period is divided into sixteen samples. The value sampled on two out of three samples on samples 7, 8, and 9 is the value accepted. If a false Start bit occurred, by the 7th, 8th, and 9th sample this false Start bit would have gone away and a 1 (High) would be sampled instead of the expected 0 (Low). The microprocessor will then ignore this false Start bit, and reset its serial logic to begin looking for a valid Start bit again. On data bits, the two out of three sampling allows one sample to be invalid (due to a glitch) and the correct value to still be read.

1.19.2.6 Transmit / Receive Gating

Each paper handling option must gate the printer TxD In signal it receives before sending it on to the next paper handling option on the trunk line. This is done by connecting the output of the 74HC14 that received the printer TxD In signal to one input of a 74ALS03B. The other input of the ALS03 is connected to the gating signal that originates from the paper handling option's microprocessor port output. This gating signal, named Disable Printer TxD - (DsPrTxD-), will be active Low at the ALS03 input. In other words, when DsPrTxD- is Low (0), the output of the ALS03 will remain High (1) regardless of the level of the printer TxD In signal at the other ALS03 input. This ALS03 output is the printer TxD Out signal that is sent on to the next paper handling option down the trunk line. During Power On Reset of the paper handling option's microprocessor, the DsPrTxD- signal must be active. The paper handling option will leave the DsPrTxD signal active until its address has been established with the printer engine, then it will deactivate it for normal operation. Refer to the section on address determination protocol for more details. The trunk line printer TxD and RxD signals are normally High (1) and go Low (0) for a Start bit at the beginning of a serial transmission. Note that since the printer TxD In signal coming into a paper handling option is received by a 74HC14 inverting buffer, a second inverter must be used before this signal is input to the option's microprocessor RxD input.

The printer RxD signal is not gated by the paper handling options. Each option will use a 74ALS03B open collector driver and wire OR the output (after a 100 ohm series resistor) onto the printer RxD trunk line signal. Since the ALS03 is a NAND buffer, the paper handling option's microprocessor TxD signal will have to go through an inverter before being input to the ALS03 driver.

1.19.3 Power Signal

The power lines present in each trunk line are +24V and 24VReturn. +24V will have a -11% / +10% tolerance for a minimum of 21.4V and a maximum of 26.4V. For options with very low power requirements, Lexmark will determine if the option can use this +24V and 24VReturn for power. Therefore, anyone working on a paper handling option should consult Lexmark International to determine if their option can use the base print engine's +24V. Options that must provide their own power supply must still provide "pass through" of the +24V and 24VReturn lines to the next option down the trunk line.

1.19.4 Cable / Connector Requirements

In the new product line, the connection scheme has been defined for paper feed options above and below the printer. The tentative plan for options to the right side of the printer is to use a strain-relieved cable attached in some fashion to the top connector. The top and bottom options are to use the Molex Mini-Fit, Blind Mating Interconnect family of connectors allowing automatic connection of the options to the printer.

For stackable options such as the 500-sheet tray option, the location and gender of the connectors shall mimic the printer connector and locations on top AND bottom.

1.19.4.1 Bottom-Mounted Options (Tray Options, Duplexer, HCIT)

The Molex PN for the printer connector is 15-06-0085. This plug uses Molex 5558 series 18-24 AWG brass (tin-plated) male terminals, Molex PN 39-00-0040.

The paper handling option's mating connector will be the Molex 42474 series blind mating panel mount receptacle also with 8 circuits in a 2 x 4 arrangement. The Molex PN is 15-06-0080. This receptacle uses Molex series 5556 18-24 AWG brass (tin-plated) female terminals, Molex PN 39-00-0038.

1.19.4.2 Top-Mounted Options (Stackers, Mailboxers, HCOS, HCOF)

The connector on top of the printer for top-mounted options is Molex 15-06-0080 with the associated 39-00-0038 terminals as described above.

The connector beneath the top-mounted options is Molex 15-06-0080 with the associated 39-00-0040 terminals as described above.

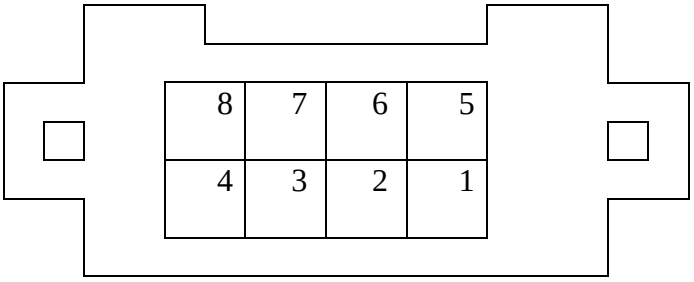
1.19.4.3 PinOut and Cabling

The smart options cabling is to be 22 AWG twisted pair for +24V and 24Return, 24AWG for printer TxD, logic Ground, printer RxD and logic Ground all twisted together.

1.19.4.3.1 8-Pin Option Connector

The 8-pin connector is currently used for top-mounted options on high-end mono products, and some color products. The signals are basically the same as the 6-pin except that a +5VDC power source is provided as well as an engine motor tach (encoder) signal in some cases.

Refer to the Molex catalog for pin positions. Looking into the front (funnel) of the printer's plug with the notch at the top, would give the following pin positions:



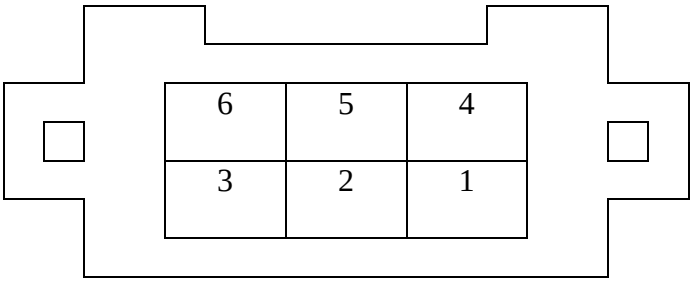
The pin outs are:

1	+24V
2	Logic Ground
3	Printer RxD
4	24Vreturn
5	Motor Tach (from Printer – bottom options only)
6	+5V
7	Logic Ground
8	Printer TxD

1.19.4.3.2 6-Pin Option Connector

The 6-pin connector is used for bottom-mounted options (duplexer and tray) and front-mounted options (envelope feeder) on high-end mono printers :

Refer to the Molex catalog for pin positions. Looking into the front (funnel) of the printer's plug with the notch at the top, would give the following pin positions:



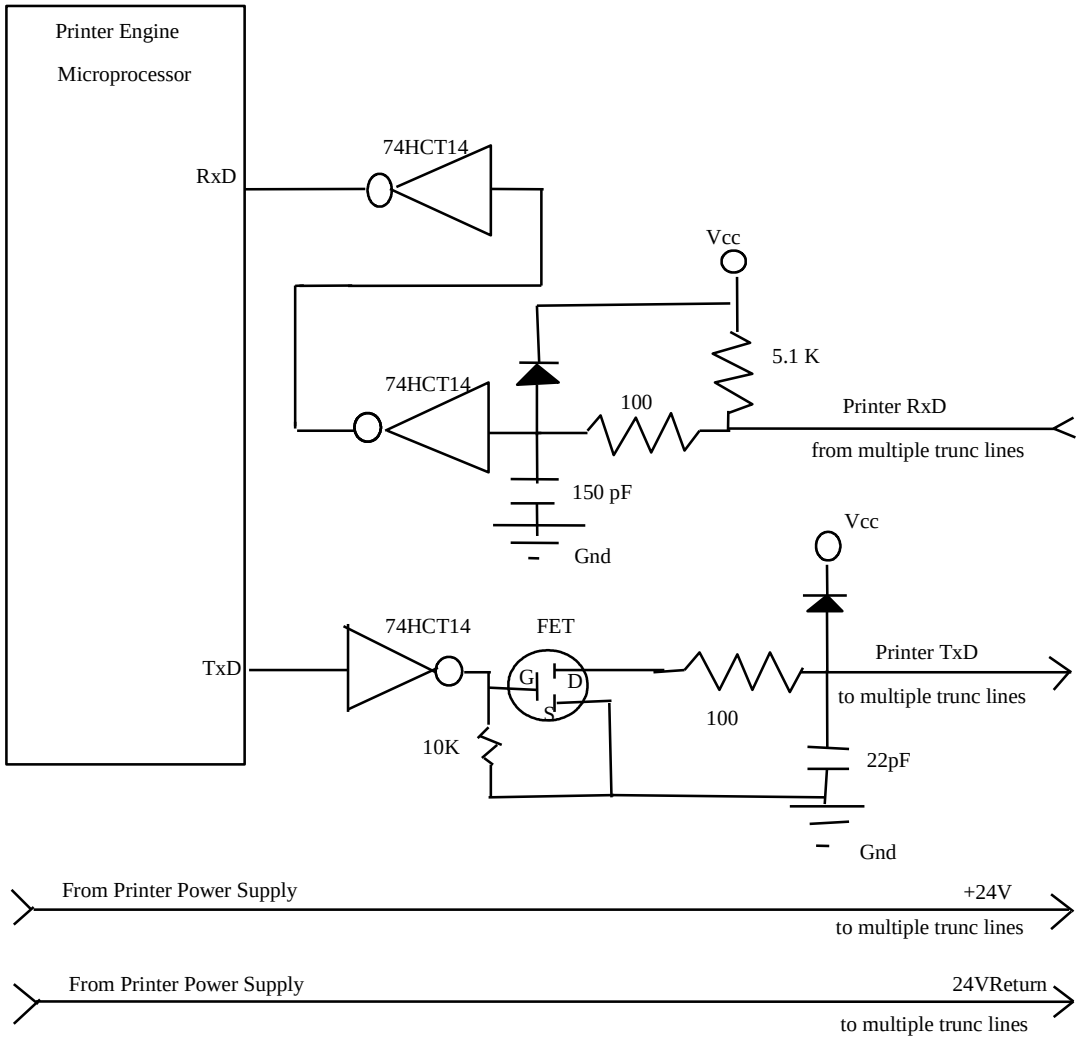
The pin outs are:

1	+24V
2	Logic Ground
3	Printer RxD
4	24VReturn
5	Printer TxD
6	Logic Ground

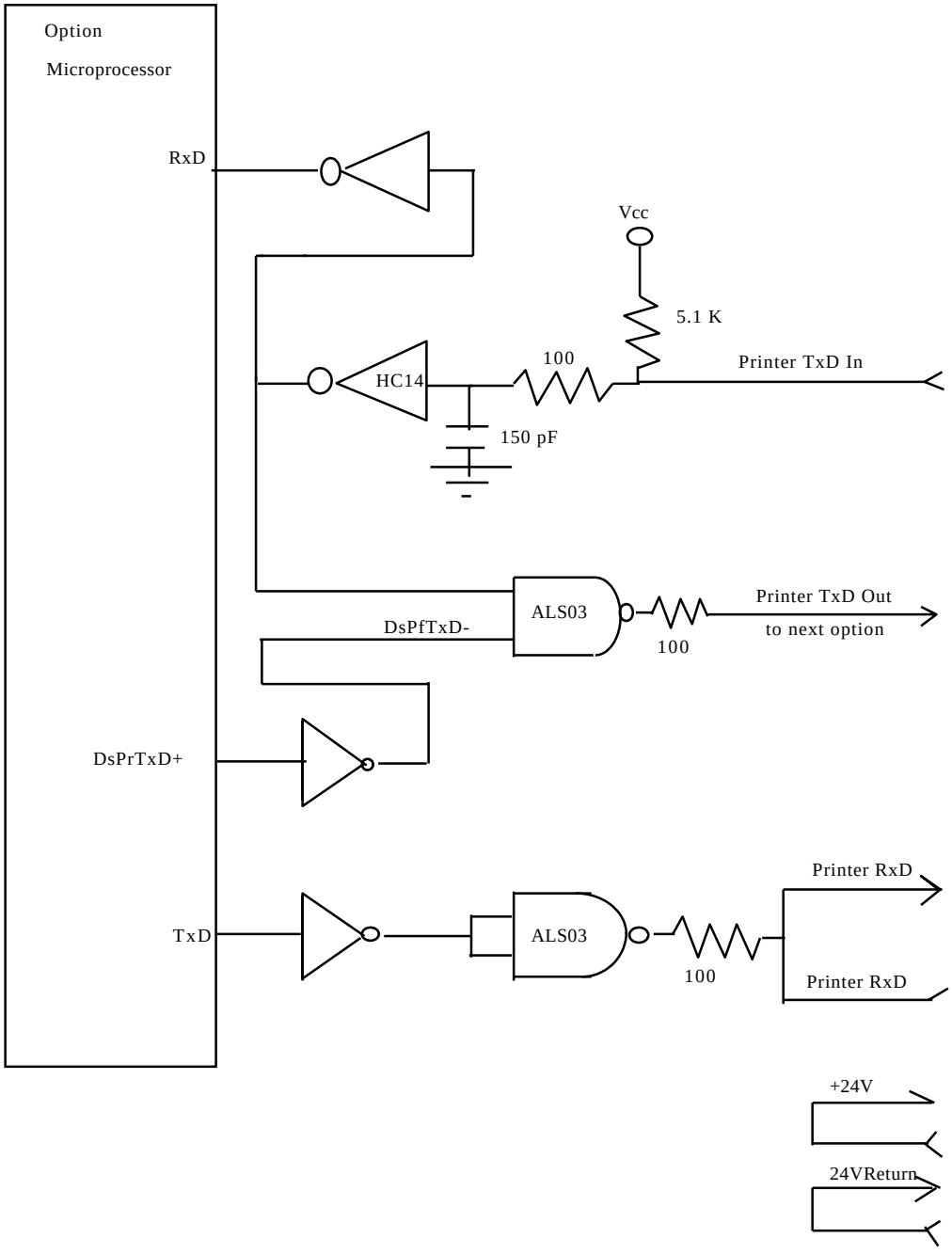
1.19.5 Printer (Master) / Device (Slave) Interface

The following diagrams summarize the printer/option interface; first from the viewpoint of the paper handling option electronics card, followed by the viewpoint of the print engine card.

The newest version of this interface also includes +5 volts as a power source from the printer to the 500-sheet tray and duplexer options. Also, there is an additional buffered tachometer signal from the printer which may be used in synchronizing paper feed into the printer. **The following diagrams reflect earlier designs but remains representative of the serial interface. These diagrams will be updated as the current designs mature.**



Printer Engine Interface



Paper Handling Option

APPENDIX A - DEFINITION OF TERMS

Throughout this document many terms will be used to describe the interface. To help with the terminology, this appendix will define some of the key terms used (listed alphabetically).

1. ACR - Architecture Change Request. A request to modify the Paperport II Architecture.
2. ACCUMULATOR - Some devices, usually output devices, may have an area within their paperpath of which the device "collects" (or accumulates) the media in order to perform a final "finishing operation" on the entire stack of media received.
3. ADDRESS - The unique identification given to a device by the printer of which the printer will use to establish communications with that device.
4. AUXILIARY POWER SUPPLY - An auxiliary power supply is a power supply source that a device may have other than the printer's power supply or its own that is used to power its system card that communicates to the Engine. For example: a device's separate auxiliary power supply to provide power to the device's motors.
5. BAT - The Basic Assurance Tests "self-tests" run by the printer and / or device at power-on or reset to check to see if it is capable of functioning correctly.
6. BROADCAST ADDRESS - The address the printer uses to communicate to All the devices at the same time. Used for broadcast communications .
7. BROADCAST COMMUNICATIONS - Communications from the master transmitted down the communications bus intended for all slave devices to receive and act upon.
8. BROADCAST COMMUNICATIONS SYNTAX - The link communications syntax used by the printer and devices for conducting Broadcast Communications.
9. COMMAND - An encoded message sent by the master to a device to request the device to perform some function.
10. COMMUNICATIONS SESSION - When the printer and device are exchanging information. See the *Link Section 0* for details.
11. DEVICE - A paper handling attachment such as a duplexer, input feeder, output stacker, etc that is controlled by the master (printer) via the paperport interface. The device manages its physical internal hardware paper path, and the printer controls the device at a "functional" level.
12. DEVICE ADDRESS - see Address
13. DUPLEXER - An option that supports the Paperport spec of which 1st acts as an output device receiving the printed output, flips the sheet over, and then acts as an input device feeding the sheet back into the printer for 2-sided printing
14. ENGINE - Synonymous with Printer, and Print Engine; *Master* in the paperport architecture
15. ENGINE To DEVICE COMMUNICATIONS - The direct communications between the master and a device once a slave device has been selected via its unique address.
16. ENVELOPE FEEDER - An input device that supports the Paperport spec that specifically only supports the feeding of envelopes into the printer. The envelope feeder may or may not have more than one input tray (bin).

17. FEEDER - An input device that supports the Paperport spec that only has 1 tray of which provides high capacity (usually 1K or more sheets) input. Generally this "tray" is a platform that moves up and down to place the media underneath the pick technology.
18. FINISHING OPERATION - Some output devices provide additional functions besides receiving and depositing the media. Some examples of such functions may be stapling the print job, punching holes in the print job, and/or binding the print job with a spiral binder, etc. These additional functions are commonly called "finishing operations" since they are the last functions done to a print job to complete its presentation for the end user.
19. HIGH CAPACITY OUTPUT TRAY (HCOT) - Output device that supports the Paperport spec that has 1 output tray (bin) and usually has a higher output capacity (> 1K sheets).
20. HOMING - A term used to describe that once a mechanism has begun its movement, for example: a D-Roll pick mechanism, it must continue to move until it reaches a "home" state; ie, a point of which required by the mechanism to avoid damage, ensure proper operation, etc. In other words, such devices can not allow movement to stop at any arbitrary point, but instead a precise physical point is required for the mechanism to terminate its movement.
21. IDLE - This is the state of the printer system when it is ready to process print jobs, but is not processing any print jobs and is not receiving any from the host computer.
22. INTERPAGE GAP - This is the space between the trailing edge of one sheet and the leading edge of the following sheet.
23. INVALID PAPERPORT PROTOCOL IR CONDITION - Whenever the device detects that the printer has sent a sequence of commands, commands, or command parameters that are not part of the Paperport Spec or not supported by this device, it indicates a design error exists by raising this error indicating the device believes it has received something from the printer that violates the Paperport spec. Example: a device receiving a **Pick Page** command when it has not received a **Select Input Device** command selecting it as an input device.
24. IR - Stands for Intervention Required. An IR is any condition or situation in which an operator is required to take action (or intervene) at the printer to resolve a given condition or situation. An example is a paper jam or load a tray that is empty.
25. IR CLEAR - The notification the printer issues to all the devices to alert them that the operator has resolved the IR condition, and the devices should finalize IR State processing and enter the Steady State to allow print job processing again.
26. IR STATE - This is the state of the printer system when an IR has been detected. The print system has previously detected the IR condition and is now taking the necessary steps to process the IR condition. After processing the IR condition, the last step of this state is to wait for notification (IR Clear) that the operator has resolved the IR condition.
27. LINK RESET - A special address value that causes the devices to return to their address mode (ie, terminate data mode) and abort current command / response processing.
28. MAILBOX - An output device that allows the operator to assign output trays (bins) to end users (some output trays may even be lockable) of which their print jobs can be outputted to their mailbox bin for pickup. IE, Everyone's output does not have to be outputted into a "common" place like it does for the printer's output bin, or an output stacker, or output sorter; but instead each individual's output is placed in their specific individual mailbox bin (tray). A mailbox device's mechanism is such that it can often be used as an output stacker via printer software controlling all the mailbox bins

(trays) logically as one big output tray. The mailbox bins (trays) may or may not be customer removable.

29. MASTER - The printer's engine mechanism controller of which manages the printer's complete physical paper path.
30. MEDIA - A general term referring to the material of which the printer will print on. It can be paper, card stock, labels, transparencies, envelopes, etc.
31. MODULAR OUTPUT STACKER - Output device that supports the Paperport spec that has 1 output tray (bin) and usually has a limited output capacity (< 1K sheets), but also has the unique feature of allowing other modular output stackers to be cascaded onto it allowing the printer system to have multiple modular output stackers (ie, multiple output bins).
32. MSB - Most Significant Bit; ie, Bit 7 of a byte.
33. NON-ECHOING COMMUNICATIONS SYNTAX - The link communications syntax used by the printer and devices for conducting command / response processing in which the device does not "echo" back to the printer the various bytes of the command.
34. NOP - The term nop refers to a condition of which the device actually performs "No Operation" pertaining to that particular command, condition, or situation.
35. NOTIFICATION - An encoded message sent by the master to a device to notify the device that some event has occurred.
36. NVRAM - Nonvolatile RAM that resides on the device's system card. Usually divided into two sections: an area available to the print engine for managing unique functions for that device, and an area available to the device, referred to as the "device data area", for the device to manage critical operational characteristics of the device's hardware. Both of these areas may contain either "critical parameters" or "non-critical parameters". Critical parameters are settings critical to the operation of the device's hardware. Non-critical parameters are settings used to control the device, but if destroyed or modified would not render the device in-operable.
37. ON-LINE - The printer and device are ready and capable of processing print jobs.
    - This term can also refer to the process of the printer bringing a device "on-line" which means the printer identifies the device, determines its unique characteristics, and finally assigns the device its unique device address. The print engine and device are now able to communicate the needed commands that allow them, together, to process print jobs.
38. OUTPUT SORTER - An output device that supports the Paperport spec that has 1 or more output trays (bins) of which multiple copies of the same document can be printed allowing each of them to separately contain sequenced output. Example: an operator desiring 3 copies of this spec might send a single print job of this spec to the printer instructing it to print 3 copies all sorted of which the operator would pick up copy #1 from tray #1, output sequenced; copy #2 from tray #2, sequenced; and copy #3 from tray #3 output sequenced.
39. OUTPUT STACKER - An output device that supports the Paperport spec that has 1 output tray (bin). Also see *Modular Output Stacker* and *High Capacity Output Tray*.
40. PAPERPORT - Hardware and software interface between the printer and its smart options
41. "PASS THRU" - Pass Thru refers to the paper path that allows the media to transport from one device through another. An Input "Pass Thru" paper path is one that allows media from an input device to transport or pass thru a device physically between it and the printer's input paper path. An Output "Pass Thru" paper path is one that allows printed media (ie, media imaged by the printer) to

transport from the printer through a device physically located between the printer and the intended output device.

42. PAGE - The term page refers to the media being processed and is often interchanged with the term "sheet".
43. POR - Stands for Power-On Reset. It is the state or processing the printer and / or device goes through when it is initially powered on.
44. PRINT ENGINE - The printer mechanism controller card that controls the printer mechanism, as well as, the printer options via the Paperport spec.
45. PRINTER SYSTEM - The print engine and all of its attached options referred to as a whole
46. RESPONSE - Encoded reply by the device to a command or notification sent by the master.
47. SHEET - The term sheet refers to the media being processed and is often interchanged with the term "page".
48. SIDE - The term side refers to the media being processed of which a particular "side" of the page is being imaged. For duplex printing it requires the processing of 2 sides to print the duplexed page.
49. SLAVE - A device that controls the paper path within its mechanical hardware and manipulates the paper per the master's instructions.
50. SMART OPTION - An optional device to the printer that supports the Paperport spec and has the ability to control & manage its own mechanism (ie, the printer is required to control its mechanism other than at a functional level).
51. SMART INPUT TRAY - An input device that supports the Paperport spec that has 1 or more input trays (bins).
52. SORTER / MAILBOX - An output device that supports the Paperport spec that has 1 or more output trays (bins) that can perform as a Sorter (see OUTPUT SORTER above), as well as, as Mailbox. A Mailbox device allows the operator to assign output trays (bins) to end users (some output trays may even be lockable) of which their print jobs can be outputted to their mailbox for pickup. IE, Everyone's output doesn't get deposited into a "common" place like it does for the printer's output bin, or output stacker, or output sorter; but instead, each individual's output is placed in their specific individual mailbox slot (tray).
53. STANDARD COMMUNICATIONS SYNTAX - The link communications syntax used by the printer and devices for conducting command / response processing in which the device does "echo" back to the printer the various bytes of the command.
54. STEADY STATE - This is the state of which the printer system processes print jobs; ie, no IR or Error conditions exist, and the printer system is actively processing a print job received from the host computer.
55. TRUNK LINE ADDRESS - Address associated with specific physical connection to the printer

APPENDIX B - COMMAND SET TABLE

This appendix is a tables of All the Paperport II commands, including undefined Op Codes. The first table lists the commands by Op Code value (ascending order) and includes parameter information. The second tables lists the commands by Command Name (alphabetically), includes the section # reference (command/response) instead of the page # reference, and Configuration Levels. For device dependent responses, the section reference is for the Smart Input Tray since it is the first device listed in the Device Dependent Responses section .

NOTE : TABLES NOT UP-TO-DATE !!!

COMMAND SET TABLE by Op Code Value (ascending order)

Command / Notification	Mnemonic	OpCode	Parameters	Pg #
SYSTEM CONFIGURATION RESET	SCR	0x00		31
HARD RESET	HR	0x01		32
SOFT RESET	SR	0x02		33
QUERY RESET COMPLETE	QRC	0x03		35
DISCONNECT COMMUNICATIONS To NEXT DEVICE	DCND	0x04		36
CONNECT COMMUNICATIONS TO NEXT DEVICE	CCND	0x05		36
QUERY DEVICE TYPE	QDT	0x06		39
CANCEL / FLUSH (Logical) PAGE QUEUE	CFPQ	0x07		80
QUERY AUX POWER STATUS	QAPS	0x08		64
QUERY OPERATION COMPLETE	QOC	0x09	N1 Status Response N2 Exception Response	65
QUERY DEVICE SOFTWARE REVISION	QDSR	0x0A		44
QUERY DEVICE STATUS 1	QDS1	0x0B		52
QUERY NON-CRITICAL STATUS CHANGE	QNCSC	0x0C		52
QUERY IR CONDITION - HOMING ALLOWED	QIRCHA	0x0D		53
QUERY IR ABSOLUTE IMMEDIATE STOP	QIRAIS	0x0E		53
QUERY NVRAM READ / WRITE COMPLETE	QNRWC	0x0F		66
QUERY EQUIPMENT CHECK STATUS	QEQCKS	0x10		54
QUERY NUMBER TRAYS	QNT	0x11		54
TRANSMIT REQUESTED READ DATA	TRRD	0x12		101
QUERY COVER OPEN LOCATION	QCOL	0x13		55
QUERY PAGE COMPLETE ID	QPCID	0x14		59
INTERVENTION REQUIRED CLEAR	IRC	0x15		62
QUERY DEBUG MESSAGE	QDM	0x16		63
JOG OUTPUT TRAY	JOT	0x17		71
PICK PAGE	PP	0x18		75
DIVERT MEDIA INTO DEVICE	DMID	0x19		76
DEVICE CONTROLS MEDIA NOTIFICATION	DCMN	0x1A		76

COMMAND SET TABLE by Op Code Value (ascending order)				
Command / Notification	**Mnemonic**	**OpCode**	**Parameters**	**Pg #**
PAGE At PRINTER's INPUT SENSOR	PISN	0x1B		77
BEGIN COMMUNICATIONS SESSION	BCS	0x1C		78
END COMMUNICATIONS SESSION	ECS	0x1D		78
INVOKE LOW POWER STATE	ILPS	0x1E		92
RESTORE FULL POWER STATE	RFPS	0x1F		92
DIVERT MEDIA NOT INTO DEVICE	DMNID	0x20		93
DIVERT MEDIA NOT INTO DEVICE	DMNID	0x20		93
QUERY DIAGNOSTIC TEST COMPLETE	QDTC	0x21		118
MECHANICAL RESET	MR	0x22		34
PICK PAGE AT SET GAP	PPASG	0x23		78
SET INTERPAGE GAP	SIG	0x24		78
QUERY DIAGNOSTIC TEST DATA MESSAGE	QDTDM	0x25		118
QUERY DEVICE QUIESCE	QDQ	0x26		60
QUERY TRAY PULLED LOCATION	QTPL	0x27		56
PICK RETRY	PR	0x28		78
QUERY HW IR	QHWIR	0x29		67
QUERRY DEVICE MINIMUM SPEED	QDSMIN	0x2A		80
QUERRY DEVICE MAXIMUM SPEED	QDSMAX	0x2B		81
PAGE AT EXIT SENSOR NOTIFICATION	PEXSN	0x2C		81
QUERY DEVICE BOOT SOFTWARE REVISION	QDBSR	0X2D		**Error! Bookmark not defined.**
QUERY DEVICE MEASURED INTERPAGE GAP	QDMIG	0x2E	None	79
QUERY PICK TIME ERROR	QPTE	0x2F	None	
QUERY MEDIA SENSOR CALIBRATION STATUS	QMSCS	0x40	None	67
ASSIGN DEVICE ADDRESS	ADA	0x80	N1 Device's Addr	37
PRINTER CHARACTERISTICS NOTIFICATION	PCN	0x81	N1 Printer Characteristics	46
QUERY DEVICE PICK TIME	QDPT	0x82	N1 Bin #	56
QUERY DEVICE MINIMUM INTERPAGE GAP	QDMG	0x83	N1 1st Sheet N2 Next Sheet	58
PRINTER INTERVENTION REQUIRED NOTIFICATION	PIRN	0x84	N1 Printer IR Type	60
SELECT INPUT DEVICE	SID	0x85	N1 Input Tray #	69
SELECT OUTPUT DEVICE	SOD	0x86	N1 Output Bin # N2 Page ID Tag	70
SELECT PROCESS SPEED	SPS	0x87	N1 Printer Process Speed	73
TRAY LOCK NOTIFICATION	TLN	0x88	N1 UnLock / Lock Tray	93

COMMAND SET TABLE by Op Code Value (ascending order)				
Command / Notification	**Mnemonic**	**OpCode**	**Parameters**	**Pg #**
DIAGNOSTIC TEST	DT	0x89	N1 Diagnostic Test#	116
QUERY DEVICE CHARACTERISTICS	QDC	0x8A	N1 Tray #	40
QUERY DEVICE INVENTORY	QDI	0x8B	N1 Tray #	43
QUERY PAPER PATH CHARACTERISTICS	QPPC	0x8D	N1 Paper Path Chacteristic Requested	41
QUERY PAPER POSITION STATUS	QPPS	0x8E	N1 Paper Status Requested	55
BROADCAST COMMUNICATIONS MODE ECHO	BCME	0x8F	N1 Echo Control	37
DEVICE MAINTENANCE OPERATION	DMO	0x90	N1 Tray # N2 Maintenance Operation	94
MEDIA SIZE NOTIFICATION	MSN	0x91	N1 Media Size	90
DEVICE OPERATION COMMAND (Formerly CLEAR / FLUSH DEVICE (Physical) PAPERPATH)	DOC	0x92	N1 Operation Parameter N2 Operation Position	72
READ DEVICE's NVRAM	RDN	0x93	N1,N2 Nvram Address N3 # Data Byte to Read	98
WRITE DEVICE's NVRAM	WDN	0x94	N1,N2 Nvram Address N3 # Data Byte to Write N4,... Data Bytes to Write	100
RESTORE DEVICE's NVRAM DATA AREA	RDNDA	0x95	N1 Initialization Parameter	102
LOCATE OUTPUT BIN	LOB	0x96	N1 Locate Bin Parameter	103
TRACK STAGING	TS	0x97	N1, N2 Track Staging Distance N3, N4 Speed Scale Factor	48
SUBMIT LOGICAL PAGE TO QUEUE	SLPQ	0x98	N1 Sub-command	
QUERY DOT XX ERROR CODE	QDXX	0x99	None	68
ENABLE APPLICATION EXECUTION	EAE	0xA0	None	109
REQUEST DOWNLOAD DATA	RDLD	0xA1	N1 Job Type	107
DOWNLOAD DATA BLOCK	DLDB	0xA2	N1….Nn Data Bytes	108
DOWNLOAD DATA COMPLETE	DLDC	0xA3	None	109
PASS THRU CONTROL	PTC	0xA4	N1…Nn Data Bytes	83
TRACK SPEED	TS	0xA5	N1, N2 Track Speed Distance N3, N4 Speed Scale Factor	46
QUERY SERIAL PORT DEBUG DATA	QSPDD	0xA6	N1 Sub-command	
INITIATE CALIBRATION PROCESS	ICP	0xB0	N1 Calibration to be performed N2 Bin / Tray Number	106

COMMAND SET TABLE by Op Code Value (ascending order)				
Command / Notification	**Mnemonic**	**OpCode**	**Parameters**	**Pg #**
INITIATE CALIBRATION PROCESS	ICP	0xB0	N1 Calibration to be performed N2 Bin / Tray Number	106
DISTANCE TO END TRACK SPEED	DETS	0xB1	N1, N2 End Track Speed Distance	50
MEDIA SENSOR ANALOG TO DIGITAL THRESHOLD OFFSET	MSADTO	0xB2	N1 Offset Value	50
SELECT INPUT DEVICE EXTENDED	SIDX	0xB3	N1 Media Type N2 Source Bin	82
FAN ACTIVATION NOTIFICATION	FAN	0xB4	N1 On/Off	110
OPERATE DEVICE CONTROL PANEL	ODCP	0x??	N1 Yes/No	91
OPERATE DEVICE CONTROL PANEL	ODCP	0x??	N1 Yes/No	91
IMAGE LOCATION NOTIFICATION	ILN	0xB5	N1, N2 Location of Image from Second Transfer	85
LIGHTING NOTIFICATION	LIN	0xB6		104
LIGHTING PING	LIP	0xB7		107
SET CONTROL PARAMETERS	SCP	0xB8	N1 Subcommand N2: Nm Parameters	84
PICK TIME ERROR NOTIFICATION	PTEN	0xBA	N1, N2 Pick Time Error	82
QUERY HEALTH CHECK DATA	QHCD	0xBB	N1 Health Check Test Requested	88
SPECIAL MANUFACTURING COMMAND	SMC	0xFD		97
SPECIAL DEVELOPMENT COMMAND	SDC	0xFE		96
COMMUNICATIONS ERROR RESPONSE	CER	0xFF		94
SINGLE BYTE COMMANDS -- Reserved	0x2C-0x7D			
MULTIBYTE COMMANDS -- Reserved	0x9A-0xFC			

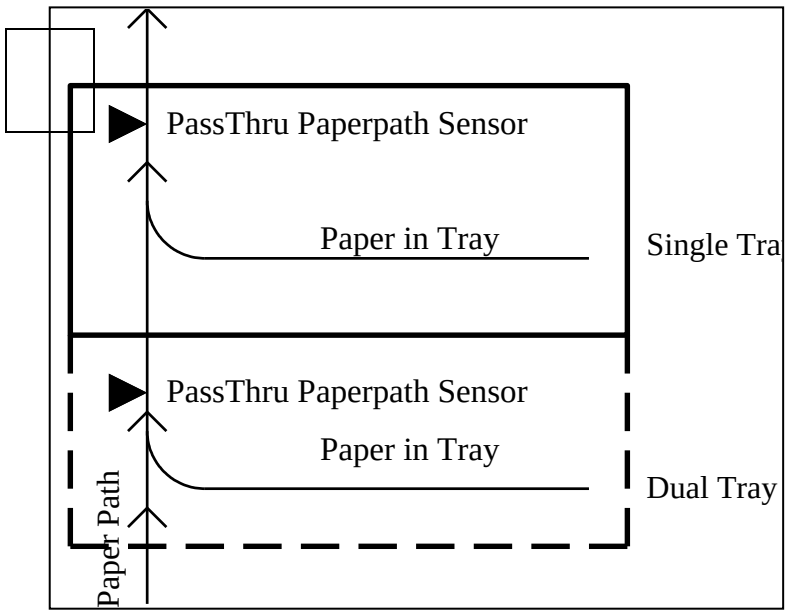
COMMAND SET TABLE by Command Name (alphabetically)					
Command / Notification	**Mnemonic**	**OpCode**	**Command/Response**		**Level**
ASSIGN DEVICE ADDRESS	ADA	0x80	4.2.8		0
BEGIN COMMUNICATIONS SESSION	BCS	0x1C	4.5.10		Resv
BROADCAST COMMUNICATIONS MODE ECHO	BCME	0x8F	4.2.9		0
CANCEL / FLUSH (Logical) PAGE QUEUE	CFPQ	0x07	4.5.15		0
CLEAR / FLUSH DEVICE (Physical) PAPERPATH	CFDP	0x92	4.5.4		0
COMMUNICATIONS ERROR RESPONSE	CER	0xFF	4.6.8		Resv
CONNECT COMMUNICATIONS TO NEXT DEVICE	CCND	0x05	4.2.7		0
DEVICE CONTROLS MEDIA NOTIFICATION	DCMN	0x1A	4.5.8		0
DEVICE MAINTENANCE OPERATIONS	DMO	0x90	4.6.9		Resv
DIAGNOSTIC TEST	DT	0x89	4.7.1		0
DISCONNECT COMMUNICATIONS To NEXT	DCND	0x04	4.2.6		Resv
IMAGE LOCATION NOTIFICATION	ILN	0xB5	4.5.21		
DISTANCE TO END TRACK SPEED	DETS	0xB1	4.3.9	None	
DIVERT MEDIA INTO DEVICE	DMID	0x19	4.5.7		0
DIVERT MEDIA NOT INTO DEVICE	DMNID	0x20	4.6.7		0
DIVICE OPERATION COMMAND	DOC	0x92	4.5.4		0
DOWNLOAD DATA BLOCK	DLDB	0xA2	4.6.20	4.6.20	
DOWNLOAD DATA COMPLETE	DLDC	0xA3	4.6.21	4.6.21	
ENABLE APPLICATION EXECUTION	EAE	0xA0	4.6.22	4.6.22	
END COMMUNICATIONS SESSION	ECS	0x1D	4.511		Resv
HARD RESET	HR	0x01	4.2.2		Resv
INITIATE CALIBRATION PROCESS	ICP	0xB0	4.6.18	None	
INTERVENTION REQUIRED CLEAR	IRC	0x15	4.4.15		0
INVOKE LOW POWER STATE	ILPS	0x1E	4.6.4		0
JOG OUTPUT TRAY	JOT	0x17	4.5.3		Resv
LIGHTING NOTIFICATION	LIN	0xB6	4.6.25		10
LIGHTING PING	LIP	0xB7	4.6.26		10
LOCATE OUTPUT BIN	LOB	0x96	4.6.16		0
MECHANICAL RESET	MR	0x22	4.2.4		0
MEDIA SIZE NOTIFICATION	MSN	0x91	4.6.1		Resv
MEDIA SENSOR A TO D THRESHOLD OFFSET	MSADTO	0xB2	4.3.10	None	
OPERATE DEVICE CONTROL PANEL	ODCP	0x??	4.6.3		Resv
PAGE At EXIT SENSOR NOTIFICATION	PEXSN	0X2C	4.5.18		7
PAGE At PRINTER's INPUT SENSOR	PISN	0x1B	4.5.9		0
PICK PAGE	PP	0x18	4.5.6		0
PICK PAGE AT SET GAP	PPASG	0x23	4.5.13		0
PICK RETRY	PR	0x28	4.5.14		0
PICK TIME ERROR NOTIFICATION	PTEN	0xBA	4.4.48	None	22
PRINTER CHARACTERISTICS NOTIFICATION	PCN	0x81	4.3.7		0,5,7
PRINTER INTERVENTION REQUIRED	PIRN	0x84	4.4.14		0
QUERY AUX POWER STATUS	QAPS	0x08	4.4.17		0
QUERY COVER OPEN LOCATION	QCOL	0x13	4.4.8		0
QUERY DEBUG MESSAGE	QDM	0x16	4.4.16		0
QUERY DEVICE BOOT SOFTWARE REVISION	QDBSR	0X2D	4.3.6		0,5,7
QUERY DEVICE CHARACTERISTICS	QDC	0x8A	4.3.2		0,5,7
QUERY DEVICE INVENTORY	QDI	0x8B	4.3.4		0
QUERY DEVICE MEASURED INTERPAGE GAP	QDMIG	0x2E	4.4.37	4.437	
QUERY DEVICE MAXIMUM SPEED	QDSMAX	0x2B	4.5.17	5.2.26	5

COMMAND SET TABLE by Command Name (alphabetically)					
Command / Notification	Mnemonic	OpCode	Command/Response		Level
QUERY DEVICE MINIMUM SPEED	QDSMIN	0x2A	4.5.16	5.2.25	5
QUERY DEVICE MINIMUM INTERPAGE GAP	QDMG	0x83	4.4.11		0
QUERY DEVICE PICK TIME	QDPT	0x82	4.4.10		0
QUERY DEVICE QUIESCE	QDQ	0x26	4.4.13		0,5
QUERY DEVICE SOFTWARE REVISION	QDSR	0x0A	4.3.5	5.2.4	0,5,7
QUERY DEVICE STATUS 1	QDS1	0x0B	4.4.1	5.2.5	0
QUERY DEVICE TYPE	QDT	0x06	4.3.1	5.2.2	0
QUERY DOT XX ERROR CODE	QDXX	0x99	1.11.22	1.15.27	12
QUERY DIAGNOSTIC TEST COMPLETE	QDTC	0x21	4.7.2		0
QUERY DIAGNOSTIC TEST DATA MESSAGE	QDTDM	0x25	4.7.3	5.2.20	0
QUERY EQUIPMENT CHECK STATUS	QEQCKS	0x10	4.4.5	5.2.9	0
QUERY HEALTH CHECK DATA	QHCD	0xBB	4.4.49		23
QUERY HW IR	QHWIR	0x29	4.4.20		0
QUERY IR ABSOLUTE IMMEDIATE STOP	QIRAIS	0x0E	4.4.4	5.2.8	0
QUERY IR CONDITION - HOMING ALLOWED	QIRCHA	0x0D	4.4.3	5.2.7	0
QUERY MEDIA SENSOR CALIBRATION STATUS	QMSCS	0x40	4.4.21	4.4.21	
QUERY NON-CRITICAL STATUS CHANGE	QNCSC	0x0C	4.4.2	5.2.6	0
QUERY NVRAM READ / WRITE COMPLETE	QNRWC	0x0F	4.4.19		0
QUERY NUMBER TRAYS	QNT	0x11	4.4.6		0
QUERY OPERATION COMPLETE	QOC	0x09	4.4.18		0
QUERY PAGE COMPLETE ID	QPCID	0x14	4.4.12		0
QUERY PAPER PATH CHARACTERISTICS	QPPC	0x8D	4.3.3	5.2.3	0
QUERY PAPER POSITION STATUS	QPPS	0x8E	4.4.7		0
QUERY PICK TIME ERROR	QPTE	0x2F	4.4.44	None	22
QUERY RESET COMPLETE	QRC	0x03	4.2.5	5.2.1	0
QUERRY SERIAL PORT DEBUG DATA	QSPDD	0xA6	4.6.24		8
QUERY TRAY PULLED LOCATION	QTPL	0x27	4.4.9		0
READ DEVICE's NVRAM	RDN	0x93	4.6.12		0
PAGE AT EXIT SENSOR NOTIFICATION	PEXSN	0x2C	4.5.18		7
REQUEST DOWNLOAD DATA	RDLD	0xA1	4.6.19	4.6.19	
RESTORE DEVICE's NVRAM DATA AREA	RDNDA	0x95	4.6.15		0
RESTORE FULL POWER STATE	RFPS	0x1F	4.6.5		0
SELECT INPUT DEVICE	SID	0x85	4.5.1		0
SELECT INPUT DEVICE EXTENDED	SIDX	0xB3	4.5.19	None	
SET INTERPAGE GAP	SIG	0x24	4.5.12		0
SET CONTROL PARAMETERS	SCP	0xB8	4.4.42	None	
SELECT OUTPUT DEVICE	SOD	0x86	4.5.2		0
SELECT PROCESS SPEED	SPS	0x87	4.5.5		0, 5
SOFT RESET	SR	0x02	4.2.3		Resv
SPECIAL DEVELOPMENT COMMAND	SDC	0xFE	4.6.10		Resv
SPECIAL MANUFACTURING COMMAND	SMC	0xFD	4.6.11		Resv
STAPLE MEDIA	SM	0x8C	4.5.4		Resv
SYSTEM CONFIGURATION RESET	SCR	0x00	4.2.1		0
TRACK STAGING	TS	0x97	4.3.8	None	
TRANSMIT REQUESTED READ DATA	TRRD	0x12	4.6.14		0
TRAY LOCK NOTIFICATION	TLN	0x88	4.6.6		Resv
WRITE DEVICE's NVRAM	WDN	0x94	4.6.13		0

APPENDIX C - DEVICE TYPE(S) PHYSICAL MODEL

This appendix documents the physical attributes (ie, model) assumed by the engine for the various device's *device type* returned in the **Query Device Type** response. Device types not shown do not have their physical model defined at this time.

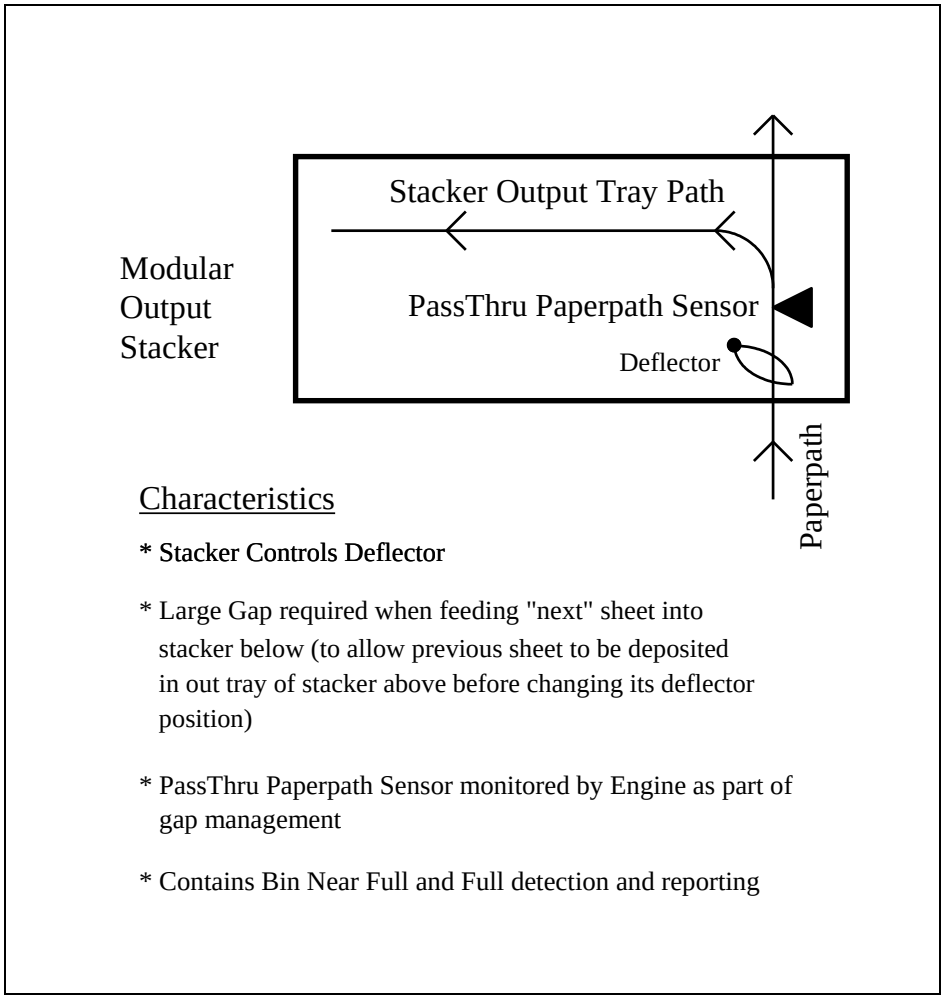
A. TRAY OPTION (Device Type = 0x01)



B. DUPLEXER OPTION (Device Type = 0x04)

Information not included at this time.

C. OUTPUT STACKER (Modular) (Device Type = 0x05)



Characteristics

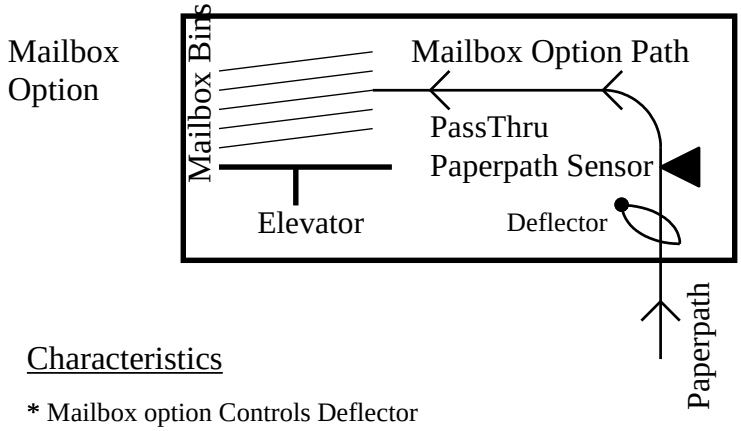
* Stacker Controls Deflector

* Large Gap required when feeding "next" sheet into stacker below (to allow previous sheet to be deposited in out tray of stacker above before changing its deflector position)

* PassThru Paperpath Sensor monitored by Engine as part of gap management

* Contains Bin Near Full and Full detection and reporting

D. MAILBOX OPTION (Device Type = 0x0A)



Characteristics

* Mailbox option Controls Deflector
* Mailbox option May Require QDMG response = 0xFFFF while moving elevator into position
* PassThru Paperpath Sensor monitored by Engine as part of gap management (same as device's paper infeed sensor)
* No PassThru output path to another device
* Contains Bin Near Full and Full detection and reporting
* May or may not contain any "Finishing" features. Must QDC to determine

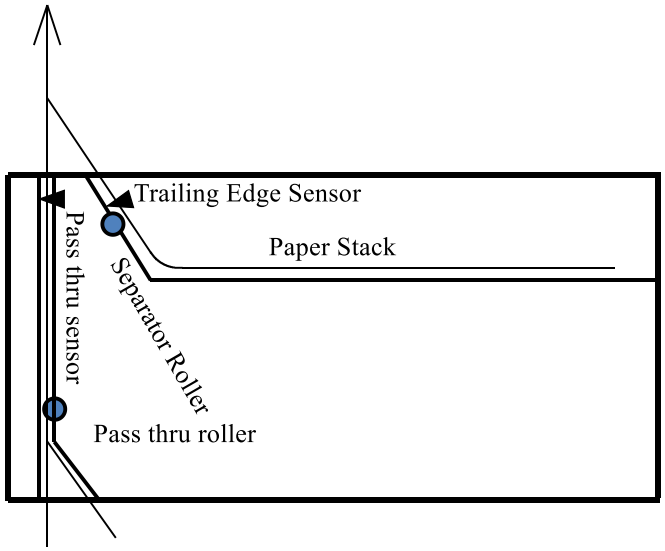
E. HIGH CAPACITY INPUT TRAY (HCIT) (Device Type = 0x32)

Information not included at this time.

F. HIGH CAPACITY OUTPUT FINISHER (HCOF) (Device Type = 0x31)

Information not included at this time.

G. G2 SMART INPUT TRAY (Device Type = 0x50)



Note: For some programs, TE sensor is before separator roller.

Characteristics:

* Elevated paper stack.
* Two distinct paperpaths: pick paper path and feed through paper path.
* Trailing edge sensor and separator roller in the pick path.
* Pass through roller and pass through sensor in the feed through paper path.

14 APPENDIX D – COMPATIBILITY LEVEL DESCRIPTION

This appendix describes the feature set introduced per compatibility level increment

LEVEL	DESCRIPTION
26	Support Query Serial Port Debug Data (QSPDD) command
25	(For devices using the old attach sequence) Support Query Serial Number feature via RDN/WDN/TRRD commands
24	Extended the Device Operation Comand (DOC) • New N2 parameter will contain staple pattern info
23	Support Query Health Check Data (QHCD) command
22	Support Pick Time Error Notification (PTEN) command Support Query Pick Time Error (QPTE) command
21	Support Query Device Measured Interpage Gap (QDMIG)
20	Extended Set Control Parameters (SCP) Command • New N3 parameter used to determine whether the option can use the feed path sensor to determine when to stop the pick
19	Additional value for N1 in Printer Intervention Required Notification (PIRN) command • 0x03 – Option Tray Feed Error
18	Additional value for N1 in Query Device Characteristics (QDC) • 0x80 - Lighting Subcommand
17	Additional value for N1 in Query Device Characteristics (QDC) • 0x93 – Tray Zone Sensing Subcommand
16	Track Speed (TS) reassigned to Op Code of 0xA5
15	
14	Additional value for N1 in Query Paper Path Characteristics (QPPC) command • 0xFE - Speed ratio of Device's 1st Passthru Nip • 0xFF - Distance from Device's Bottom to 1st Passthru Nip
13	Additional value for N1 in Printer Intervention Required Notification (PIRN) • 0x02 – Print Engine Halted, No Jam Release
12	Support Query Dot XX (QDXX) command
11	
10	Support Lighting Ping (LIP) command Support Lighting Notification (LIN) command
9	
8	Extended the Query Device Inventory command • New N5 and N6 parameters contain measured paper length reported by tray device
7	Extended the Query Device Software Revision (QDSR) • New N8 parameter will contain device's Application Code Compatibility level 2 Extended the Query Device Characteristics (QDC) response • New N4 response to indicate Duplexer's physical attachment type
6	
5	Additional value for N1 in Query Device Quiesce (QDQ) response • 0x81 – Device Processing Not Quiesce
4	
3	

2	
1	Extended the Select Input Device eXtended (SIDX) Command • Additional paper type enums for N1 parameter • New N2 parameter to indicate specific tray number in an input device with multiple trays

APPENDIX Y - DOCUMENT CHANGES

This appendix documents the "change history" of this document.

DATE	ENGINEER	REV	CHANGE
11/15/2022	Ivan Caballes	3.56	Clean Up Spec Numbering
9/23/2022	Ivan Caballes	3.55	Modified Staple Patterns names to align with new naming convention
9/7/2022	Ivan Caballes	3.54	Added new Output Device subsystem 0x87 for Device' Fold Finisher, as well as pertinent QDC response specificiation. Repurposed Bit 2 of Operational Capability byte in the Query Device Characteristic response for Output Bins to indicate support for Fold Finishing. Bit was previously used to indicate Auxillary Power support for the device. Added new Staple Patterns in the QDC Stapler Subsystem Response Added new Staple Patterns in SLPQ Staple Pattern Byte
7/13/2022	Ivan Caballes	3.53	Added Compatibility Level 27, indicating support for new Query Number of Covers command
2/11/2019	Bill Brown	3.52	Added notes on hardware interface reflecting some sections being obsolete. Also removed some references to specific OEM partners.
4/20/2018	Ivan Caballes	3.51	Implementation of QSPDD Command, change compatibility level to 26
6/30/2017	Ivan Caballes	3.50	Modified definition of QPPS command to reflect actual usage across different device platforms.
3/10/2017	Ivan Caballes	3.49	Added Appendix showing summary of feature sets added per Compatibility level
7/13/2016	Ivan Caballes	3.48	EV40569 Added support for 2nd staplehead inventory reporting
7/13/2016	Crystal Caneza		Changed SLPQ Staple Pattern values to actual values used
6/09/2016	Del Elliott	3.47	Update Media Types for SIDX
5/27/2016	Del Elliott	3.46	Added RFID_TRAY_TYPE as 0x51
8/06/2015	Rodney Goble	3.45	Added support for QPPS areas.
6/16/2015	Del Elliott	3.44	EV_398833 Staple pattern needs to be added to DOC paperport command
3/3/2015	Chris Redmon	3.43	Added QHCD command for Health Check.
12/10/2014	Junmar Sales II	3.42	Updated QDC command definition for output bin to add support on detecting no accumulate job if early warning level is reported
11/18/2014	Bill Brown	3.41	Updated MSN command definition for when N1=0, and second command byte N2 represents media width in mm (used for universal support in output options)
8/22/2014	Del Elliott	3.40	Add QPTE and PTEN commands for Pick Time Error Correction Algorithm.

7/28/2014	Chris Redmon	3.39	Update QDC lighting to add enumerations for door or cover beacons.
7/28/2014	Chris Redmon	3.38	Correct confusing PCN wording.
04/23/2014	Rodney Goble	3.37	Added support for multiple staple positions. Command affect SLPQ and QDC for staple devices.
04/08/2014	Junmar Sales II	3.36	Added QDC support for Device's without output tray(bins) but has holepunch capabilities
08/15/2013	Shuman	3.35	Added support for Cut Adjust right using SCP command, subcommand 0x02.
10/03/2012	Shuman	3.34	Document support for alternate cut media action in DOC command.
09/25/2012	Shuman	3.33	Added support for Cut Adjust using SCP command, subcommand 0x01.
06/29/2012	Del Elliott	3.32	EV_219472 Report Heavy weight to option
01/05/2011	Yusef Alaas	3.31	Added NA 2-Hole Punch pattern and renamed the exiting patterns to match the industry standards.
11/8/2011	Yusef Alaas	3.30	Added Query Device Measured Interpage Gap (QDMIG)
11/04/2011	Yusef Alaas	3.29	Updated SCP command.
08/04/2011	Yusef Alaas	3.28	Added PIRN: Option Tray Feed Error
07/20/2011	Yusef Alaas	3.27	QDC Lighting get sent to trays with compatibility level of 18 or greater
02/23/2011	Kwame Betnsi-Enchill	3.26	Added support for continuous media input devices
11/8/2010	Bill Brown	3.25	Added support for "zone sensing" of media length in tray options
04/14/2010	Kwame Bentsi-Enchill	3.24	Modified TS command to be usable on all options Fixed an op code conflict
03/30/2010	Yusef Alaas	3.23	Added G2 Input Tray device type. Added Set Control Parameters (SCP). Updated QPPC to support pick path distances. Update QDS1 response to support pick path sensor.
01/14/2010	Kwame Bentsi-Enchill	3.22	Fixed a typo on the QDI command response for hole punch
01/14/2010	Kwame Bentsi-Enchill	3.21	Updated QDQ to support NVRam activity
09/12/2009	Kwame Bentsi-Enchill	3.20	Updated QNCSC command response for Hole punch support
09/12/2009	Kwame Bentsi-Enchill	3.19	Updated QDC command and response for Hole punch support
09/16/2009	Yusef Alaas	3.18	Updated QDC response for lighting to account for beacon LEDs
05/20/2009	Crystal Rose Cabradilla	3.17	Update to PaperPort Compatibility Level 14 for QPPC changes
05/07/09	Ian Quino Fernandez	3.16	Added new option Modular High Capacity Output Stacker device type
03/13/2009	Del Elliott	3.15	Updates for Apollo stackable options
06/12/2008	Kwame Bentsi-Enchill	3.14	Added new SR HTU top option device type
03/21/2008	Bill Brown	3.13	Updated MSN table to be consistent with current RIP enumerations
03/12/2008	Jon Frey	3.12	Added motor error and hardware error values to the QEQCKS response
02/21/2008	Bill Brown	3.11	Added new exception response 0x08 to QOC for Mono
01/28/2008	David Mickan	3.10	Added new command QDXX – Query Dot XX Error Code = 0x99
11/16/2007	Crystal Cabradilla	3.09	TT_277542 - Implement Special Media Tray Mode
11/02/2007	David Mickan	3.08	Added QDC N1=90 (lighting system), 91 (diverter mechanism), and 92 (in-system programming) details. Add device types 0x20 and 0x21 to QDT. Added exception response to QOC 0x07 – Accumulator full
09/14/2007	Dexter Pontañeles	3.07	TT_284252: Add Oficio and A6 to standard media sizes

08/20/2007	Christopher Redmon	3.06	Add LIN and LIP commands to support notifying options how to control lights, and when to sync them up. Also expanded QDC command to support options reporting lighting characteristics.
07/23/2007	Kwame Bentsi-Enchill	3.05	TT_278382 OC: Implement new paperport command to allow the engine to retrieve the options serial debug data
05/23/2007	Del Elliott	3.04	TT_278609: Option needs to report measured media size
03/27/2007	Del Elliott	3.03	TT_275320 Engine needs to be able to configure active state of S2 sensor in the option
07/11/2006	Jon Frey	3.02	Added Sequence Configuration Command
06/08/2006	Bill Brown	3.01	Multiple changes to create unified document for color and mono
05/31/2006	Bill Brown	3.00	New title and rev, Fixed headings and cross-references
04/27/2006	Jon Frey		Correct PTC command length byte
04/24/2006	Jon Frey		TT_251165 Implement Engine Code Support for PTC Command - Stackable Functionality
03/20/2006	Ivan Lacuesta		TT_249357 Define and Implement new paper port command for astros.
03/20/2006	Jon Frey		Add PTC command
12/26/2005	Del Elliott		TT_214422 - Finish Implementing command handler for Query Logical Device Characteristics Updated QDC section.
08/09/2005	Ivan Lacuesta		TT_228253 Change Device Type values for tray and duplexer.
04/26/2005	Leneth Libre		TT_220467 Update PCN for Duplexer.
04/14/2005	Ivan Lacuesta		TT_219887 Support QDPT from Duplexer.
04/14/2005	Leneth Libre		TT_219798 Implement a New Paperport Command gives Boot Software Version
01/21/2002	Bill Brown		Rev 1.1 – Added Folio and Statement to MSN Command
01/16/2002	Bill Brown		Added “Media Mismatch” bit to QIRAIS response, updated clock frequency table, removed “Draft” from background, removed reference to HCOS, Changed revision to 1.0 (from 0.2)
01/02/2002	Matt Boothe		Added FAN command.
			Updated command cross reference tables
12/20/2001	Del Elliott		Turned off Save Version on Close Feature of MS WORD
12/17/2002	Del Elliott		TT_135951 DDTG : Some comm-related 990 Errors should probably be mapped to 982. Updated QEQCKS section.
12/14/2001	Del Elliott		Added RDLD, DLDC, DLDB, DETS, QMSCS, ICP, EAE, MSADTO, SIDX, and TS